

Zurich ServiceNow AI Platform Capabilities

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Common Service Data Model

The CSDM is the data model that admins should follow when they set up ServiceNow products and applications. The standards for defining configuration items (CIs) and relationships between CIs in the CMDB ensure that your data resides in the appropriate CMDB tables. The result is maximum value from ServiceNow AI Platform applications.

About the CSDM

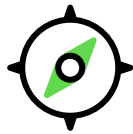
The CSDM data model reflects your service life cycle. The CSDM is the data model standard for all products that use the Configuration Management Database (CMDB).

- ServiceNow AI Platform products, apps, and features are “CSDM-compliant”. That is, they all share identical access to the underlying CSDM data model. This shared model significantly increases the power of your applications.
- The CSDM guidelines ensure unified data access for ServiceNow AI Platform products.
- The CSDM gives you clear direct prescriptive guidelines for service modeling within the CMDB.
- CSDM terms and definitions ensure consistent and accurate service reporting.
- The CSDM data model supports multiple configuration strategies and includes guidelines for using base-system tables and relationships.
- You can use the CMDB query builder to create reports showing CMDB configuration items (CIs) and their relationships.

Expert guidance to assess and improve your CSDM implementation — the CSDM Assessment

The CSDM Assessment provides Impact Customers with leading practices and prescriptive guidance on the CSDM and how it supports processes on the ServiceNow AI Platform. To help your organization plan for and implement CSDM, the assessment includes interactions with ServiceNow CSDM experts and personalized content. See [Common Service Data Model \(CSDM\) Assessment Accelerators](#).

CSDM documentation



Explore

See how the CSDM framework ensures that CIs have the proper relationships with each other and that your data resides in the appropriate CMDB tables. The guidelines help you to get maximum value from your ServiceNow AI Platform applications.

Configure the framework



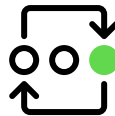
Following the CSDM framework ensures that you meet your primary goal of consistent accuracy in reporting and analytics so you can effectively manage your digital environment.

Implement the framework



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Manage the framework



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Reference

Reference topics provide detailed descriptions of tables, properties,

forms, and roles that are used in the CSDM framework.

Key principles that guided the design and development of the CSDM framework

The framework is intended to help you avoid errors in implementation and to ensure that your ServiceNow AI Platform products generate accurate, complete, and consistent reports. The following principles guided the development of the CSDM framework.

Use simplified concepts

Represent concepts in a simple, direct manner to eliminate duplicates and confusion over data sources.

Design for reporting and analytics

A prime objective of CSDM is to support consistent analysis.

Prescribe the data relationships

Tell users in a clear direct way which relationships and references to use to link CSDM tables.

Share the data model across products

The CSDM identifies a data model that is shared across products to simplify concepts and collaboration. Collaborating with other product teams achieves the best shared design.

Use clear definitions

Use agreed-upon CSDM definitions wherever a table, reference, or attribute is used.

Share base-system tables

zBoot must provide shared base-system CSDM tables by default.

Consistent data integrations

To ensure data integrity, use prescribed technologies when integrating external data sources

Speed adoption

For each new release, provide automation and guidance for CSDM that accelerates upgrading and minimizes issues.

Enable data governance and process

The presence of data within the model provides little value without governance and effective process to manage the truth and validity of the data.

Provide practical user documentation

(The content that you're viewing now) Each product team that references CSDM objects should provide documented guidance on use and/or value of the objects. Links to product guidance appear in [Applying CSDM guidelines to your product — product views](#).

Helpful resources

Some ServiceNow resources that can provide you with helpful information are:

CSDM Community Forum

[Community Forum](#)

Other CSDM resources

See [What is a CSDM?](#) and [CSDM in a nutshell](#).



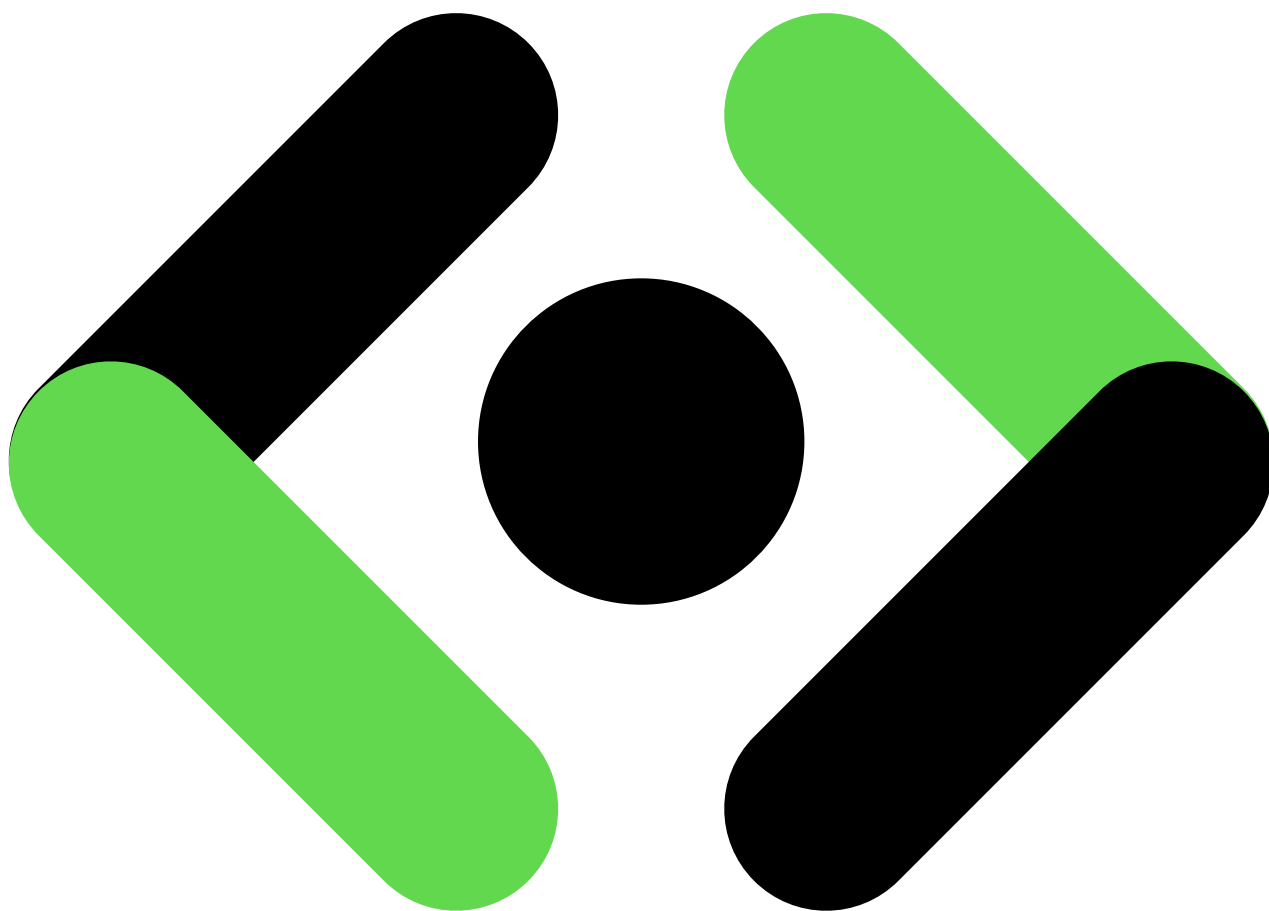
ServiceNow Community

ServiceNow Community



Customer Success Center

Impact



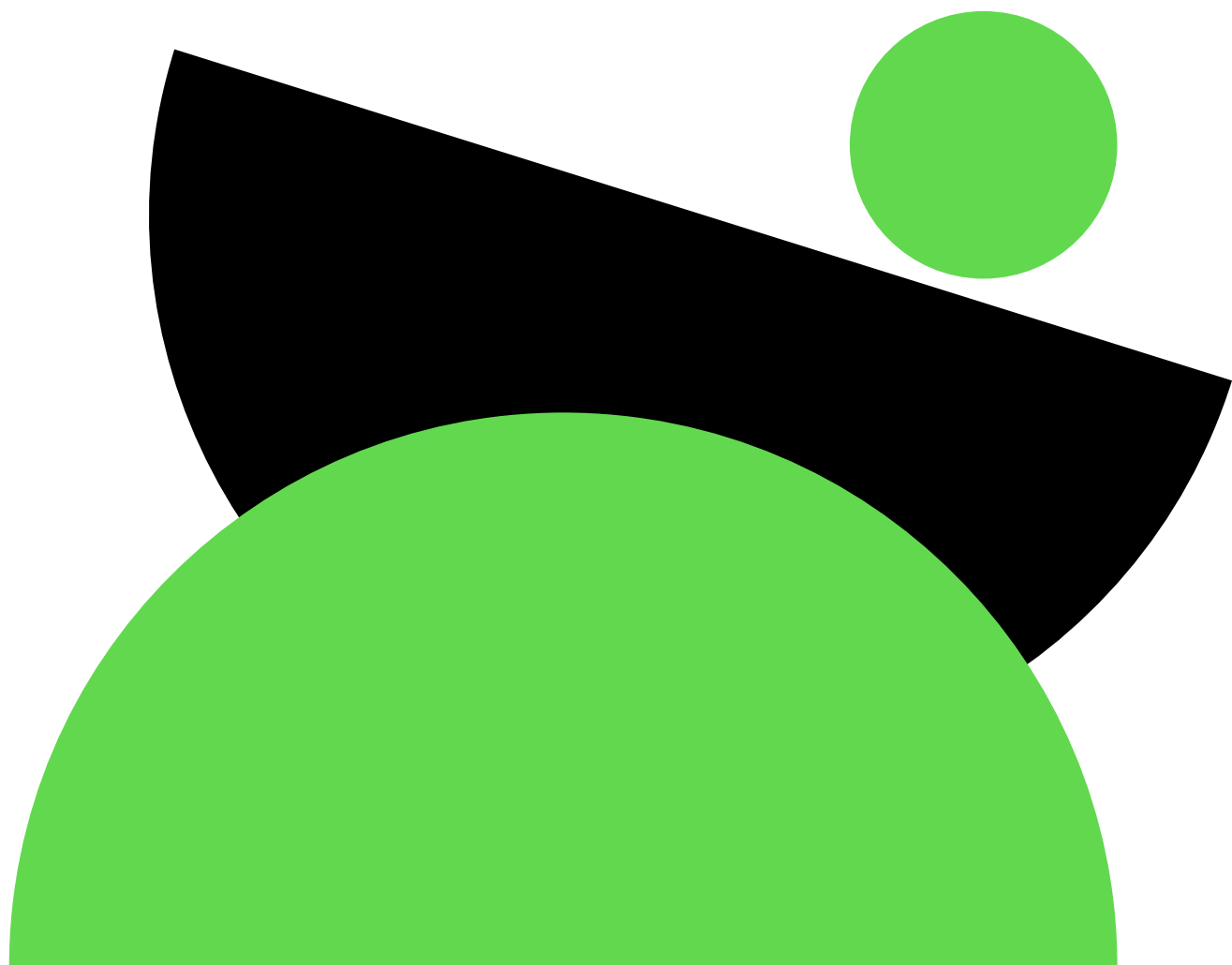
Developer

developer.servicenow.com



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ServiceNow University

ServiceNow University



NowCreate

Best Practices



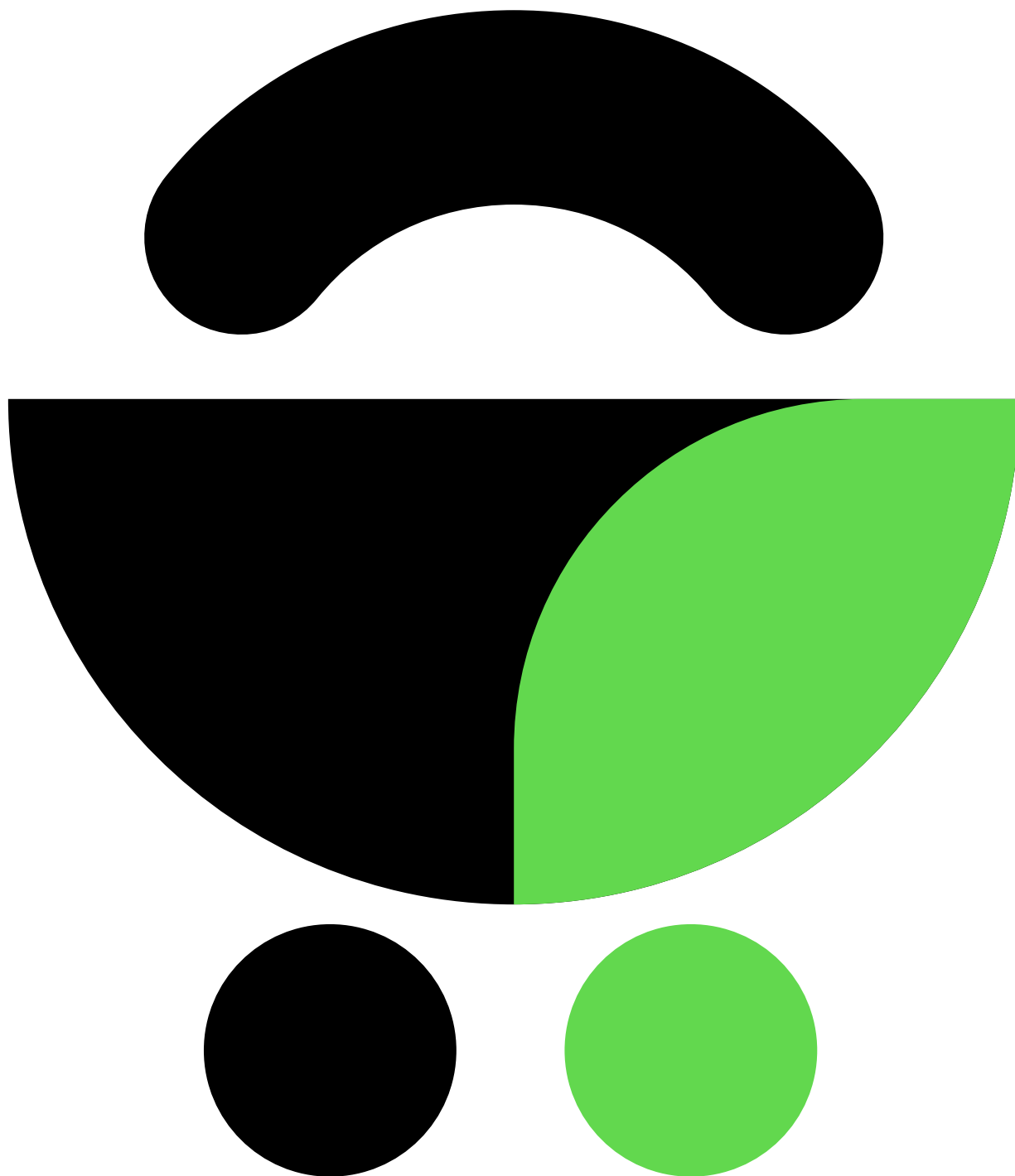
Partner

<https://www.servicenow.com/partners.html>



ServiceNow

<http://servicenow.com>



ServiceNow Store

<http://servicenow.com>



Support

- <https://support.servicenow.com/now>
- [Known Error Portal](#)



Video

Watch Playlist of all CSDM videos.

- [Exploring the CSDM model](#)

The CSDM is the data model that admins should follow when they set up ServiceNow products and applications. The standards for defining configuration items (CIs) and relationships between CIs in the CMDB ensure that your data resides in the appropriate CMDB tables. The result is maximum value from ServiceNow AI Platform applications.

- [Configuring CSDM](#)

Activate and configure CSDM capabilities.

- [Implementing the CSDM framework in stages](#)

Following the CSDM framework ensures that you meet your primary goal of consistent accuracy in reporting and analytics so you can effectively manage your digital environment.

- [Access CSDM features](#)

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- [Applying CSDM guidelines to your product — product views](#)

ServiceNow products are CSDM-aware — they expect that CMDB data is organized according to CSDM guidelines and they benefit from that organization.

- [CSDM reference](#)

Reference topics provide detailed descriptions of tables, properties, forms, and roles that are used in the CSDM framework.

Exploring the CSDM model

The CSDM is the data model that admins should follow when they set up ServiceNow products and applications. The standards for defining configuration items (CIs) and relationships between CIs in the CMDB ensure that your data resides in the appropriate CMDB tables. The result is maximum value from ServiceNow AI Platform applications.

Definitions of CSDM terms

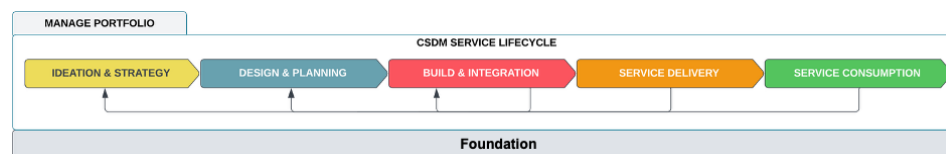
See [CSDM terms](#).

Goal of CSDM

Following the CSDM data model helps you to accurately and completely capture the data objects that help you to run your business.

CSDM provides guidance on where to put data in the platform. Because there are multiple ways of identifying and using data, the CSDM guidelines help you to determine the best data structures for supporting your organization's services.

How tables in CSDM data domains support the service life cycle



1. When you first start working on an instance, your admins install and populate the CMDB data structures. The data is required to enable your platform features and capabilities. This kind of data is kept in tables that are considered foundational to all products on your instance. We therefore refer to this set of tables as forming the Foundation domain — it supports all of the other data domains.

2. You start with an idea — an idea that implements your business model. The idea or concept is saved in data that you work on in the tables of the Ideation & Strategy domain.
3. If your research determines that the idea has merit, you can decide to commit to it and move forward into designing and planning the idea. You might design the products to implement the idea internally or buy something to integrate into your organization's offerings. The supporting data lives in tables in the Design & Planning domain.
4. After the design is complete and acceptable, you can hand it off to the teams that will build or integrate the idea. Designers and developers build the software that implements the idea. The supporting data lives in tables in the Build & Integration domain. Information learned in this stage of the product life cycle is fed back into data in the domains that were involved earlier in the digital product's life cycle.
5. You can now release the idea. It's delivered from the infrastructure and management of the Service Delivery domain to the catalog of business offerings in the Service Consumption domain so that your customers can consume it. Customers can be internal or external. Again, information learned in these stages is fed back.

The Manage Portfolio domain is a layer on top of the CSDM conceptual model that interacts with the other CSDM domains. The typical user, a service owner, might be responsible for services in more than one domain.

Roles that create, maintain, and use the data

Note: The Application Owner, Portfolio Manager (multiple), and Service Owner (multiple) might be the same persons.

Enterprise Architect and Application Owner

- Identify, enter, and maintain data in Business Application records
- Align enterprise applications to capabilities and business strategy
- Manage technology portfolio life cycle

Portfolio Manager and Service Owner

- Own the Service or digital product
- Create Services and service offerings
- Set the SLAs
- Is informed of outages
- Request catalog items to be created for the offerings

Service Desk, Incident Manager, and Change Manager

- Use the CIs and related service-related data to support and maintain services and CIs
- Support Service Operations
- Review and approve changes
- Maintain support processes

App Service Admin and Technical SMEs

- Support, manage, and operate the underpinning technology that services rely on
- Subject matter technical experts that create change requests with data from CMDB

CMDB Manager and CI Analysts

- Maintain the accuracy and integrity of the CMDB
- Work with various SMEs to certify that the data is accurate and aligned to the data model
- Create new CIs and own the SACM processes

Benefits of the CSDM guidelines

Benefit	Value and associated features
Rationalization — Determine how well an enterprise application is performing	Identify insights from operational activities related to services (visualize aggregated information across all your services and supporting operations). Application Portfolio Management (APM) and IT Business Management (ITBM)
Outage reduction — Identify affected business services quickly	Prioritize critical events based on service-related information. IT Service Management (ITSM) and Event Management
Reporting — Consistent and reliable information about your services and digital products	Understand services, their makeups, and their accountabilities. IT Service Management (ITSM)
Alignment — Align business processes to reduce risk and assure compliance	Understand Service Health (availability, CSAT, Performance, Vulnerabilities, IPC stats) Governance, Risk, and Compliance (GRC)

Importance of CSDM to the ServiceNow AI Platform

Value across the enterprise — impact analysis, risk analysis, and business continuity

Product family	Benefits of a mature CMDB that follows the CSDM guidelines	Key metrics
ITSM	• Change management • True service-level reporting • Remove semantic disparity	• Reduced tickets • Reduced MTTR
ITOM	• Visibility across environments • System health and availability • Map issues to business processes for prioritization	• Avoided service outages • Reduced MTTR
ITAM	• Accurate license positions • Visibility across operational and financial data	• Reduced software license fines • Reduced failed audits
Strategic Portfolio Management (SPM)	• Map resources to business processes • Services based on business context	Reduced business risk and outages from outdated technologies

Product family	Benefits of a mature CMDB that follows the CSDM guidelines	Key metrics
	Planning scope and impact of investment / divestment	
SecOps	- Prioritization of vulnerabilities based on business criticality - Faster incident isolation and response - End-to-end risk visibility	Reduced time to remediate priority 1 vulnerabilities
DevOps	- Automated DevOps change management - Link between configuration changes and impacted CIs	Automated change
CSM	- Visibility across customer environment - Accurate response to incidents	Customer satisfaction
IRM (GRC)	- Ensures quality data used for audit - Healthy, trusted data is critical for audit scoping and activities	- Accurately identify audited entities - Reduced audit effort and time

CSDM Community

Join the ServiceNow CSDM Community to grow your CSDM expertise.

CSDM videos in the ServiceNow Community

Playlist of all CSDM videos

- [CSDM data domains](#)

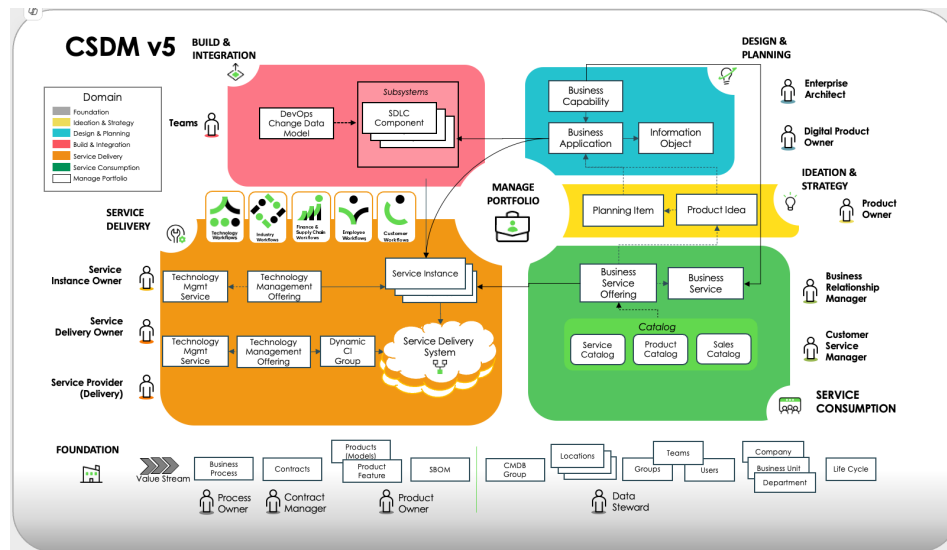
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CSDM data domains

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CSDM domains

As described in [Exploring the CSDM model](#), each domain is associated with one or more products, services, or service types, each of which you can extend as needed. Every box in the diagram (except Catalog Item) represents tables that hold CIs in the CMDB. Each connecting line represents a relationship between CI classes. Roles and user types appear next to their area of responsibility.



BUILD & INTEGRATION

Development details on the enterprise applications (digital products) to be deployed and used by the business. These are not operational CIs.

SERVICE DELIVERY

- Deployed instances of digital products and their related and discoverable components.
- Documentation of the services that provide and support the deployed instances.
- Tables used by ITOM products.

DESIGN & PLANNING

Details on the design and planning of digital products. These are not operational CIs.

IDEATION & STRATEGY

- Ideas, concepts, and considerations for creating:
- New or additional services
 - Improvements and enhancements to existing services

SERVICE CONSUMPTION

- The portfolio and request catalog of business service offerings that depend on the deployed digital products.
- Tables used by Service Portfolio Management (SPM) and Customer Service Management (CSM).

FOUNDATION

Tables that contain base data that is referenced from or to objects in the other CSDM domains. Foundation data is required before you can use ServiceNow products or add data to the CMDB.

Foundation domain

Tables in the Foundation domain contain base data that is referenced from or to objects in the other CSDM domains. Before you can use ServiceNow products, you must populate foundational data. For more information, see [Foundation domain in the CSDM model](#)

Ideation & Strategy domain

The Ideation & Strategy domain represents the ideas, concepts, and considerations for both the creation of new or additional services as well as improvements and enhancements to existing services. These capabilities are part of Strategic Portfolio Management (SPM). For more information, see [Ideation & Strategy domain in the CSDM model](#).

Design & Planning domain

Enterprise architects and digital product owners work on the design and planning of digital products that your organization can either buy or build. For more information, see [Design & Planning domain in the CSDM model](#).

Build & Integration domain

Development teams use the tables in the Build & Integration domain during the build effort (systems development life cycle — SDLC or Agile development) of digital products. The tables represent the logical development details of the enterprise applications (digital products) to be deployed and used by your organization. These are not operational CIs. For more information, see [Build & Integration domain in the CSDM model](#).

Service Delivery domain

The Service Delivery domain represents the overall end-to-end service delivery system that includes the infrastructure, technologies, integration patterns (infrastructure, systems, data, processes, dependency models), service delivery networks, and operational models. Together, these items deliver the CSDM-compliant services to internal and external users and/or organizations. For more information, see [Service Delivery domain in the CSDM model](#).

Service Consumption domain

Internal or external consumers can request business services through the request catalogs in the Service Consumption domain. The business service offerings in the request catalogs are the deployed digital products in the Service Delivery domain. Business relationship managers and customer service managers might sell or even consume elements of the Service Delivery domain. For more information, see [Service Consumption domain in the CSDM model](#).

Manage Portfolio domain

The Manage Portfolio domain is a layer on top of the CSDM conceptual model that interacts with the other CSDM domains. The typical user, a service owner, might be responsible for services in more than one domain. For more information, see [Manage Portfolio domain in the CSDM model](#).

Services and service types

A service enables you to achieve the outcomes that you want with minimal risks and without ownership of specific costs and risks. (This definition is consistent with the base definition of "service" in ITIL v3 and IT4IT.) Services typically have three components: the interaction, the offering, and the service system.

The ServiceNow AI Platform includes the following base-system service types that you can extend to align with the service types in your organization.

Technology management service [cmdb_ci_service_technical] table (formerly Technical service)

Technology management services are the systems associated with the admins of CIs in the Service Delivery domain. Service instance owners, Technology management service owners, and Technology service owners. Technology management services are typically lower-level leaf nodes of one or more business services or business service instances in a structured hierarchy.

- Technology management services are operational CIs.
- A technology management service must be a one-level service and not a hierarchy of technology management services.
- Technology management services are used for impact in Incident Management, Problem Management, and Change Management. Also used for approvals for Change Management.
- Technology management services must be provider-focused and include the technology provided for the business to consume or sell.

For more information, see [Manage Portfolio domain in the CSDM model](#).

Service Instances — Service Instance (called application service before CSDM v5) [cmdb_ci_service_auto] table

A service instance is a logical representation of a deployed application stack. Because service instances are logical in nature, they should use the Logical life-cycle value pairs. Service instances follow the same life-cycle guidance as any other logical CI.

- A service instance is an operational CI and a unique instance of an application.
- Used in Incident Management, Problem Management, and Change Management.
- Can be created for each region and each environment (Development, QA, and Production).
- Can be created via manual mapping, service mapping with entry point, and dynamic query.

For more information about leaf nodes and structured hierarchies, see [Design & Planning domain in the CSDM model](#).

You can monitor service instances (called application service before CSDM v5) on the Application Services dashboard. For more information, see [Use Application Services dashboard to monitor health](#).

Business services (cmdb_ci_service_business)

A business service is a service type that is published to business users. A business service typically implements one or more business capabilities.

Usually, business users order business services. Business users can select the desired offering and service commitment levels via the Service Catalog. For example, procurement, shipping, and finance.

- A business service is an operational CI.
- A business service must be a one-level service and not a hierarchy of business services.
- A business service can be used for impact in Incident Management, Problem Management, and Change Management and for approvals for Change.

- A business service must be focused on the consumer or seller.

For more information, see [Service Consumption domain in the CSDM model](#).

CSDM videos in the ServiceNow Community

[Playlist of all CSDM videos](#)

- [Foundation domain in the CSDM model](#)

Tables in the Foundation domain contain base data that is referenced from or to objects in the other CSDM domains. Before you can use ServiceNow products, you must populate foundational data.

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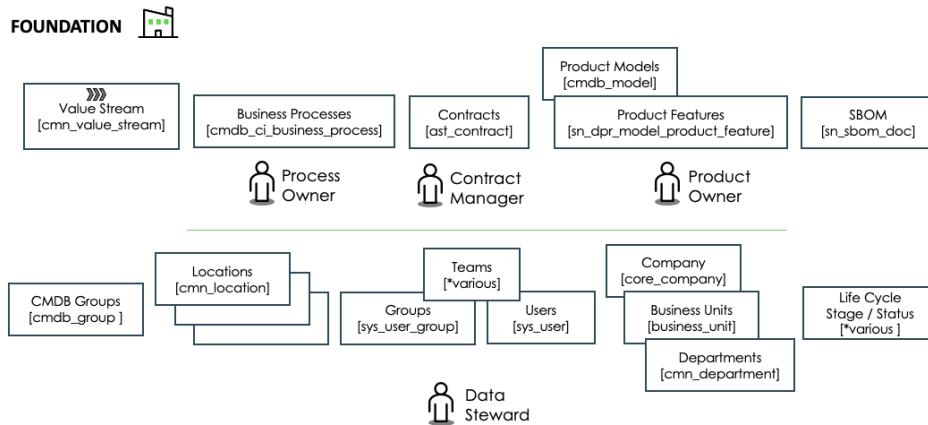
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Foundation domain in the CSDM model

Tables in the Foundation domain contain base data that is referenced from or to objects in the other CSDM domains. Before you can use ServiceNow products, you must populate foundational data.

The tables in the Foundation domain aren't used in CMDB relationships. Instead, the tables contain critical referential data. Typical users of the domain are process owners, data stewards, product owners, and contract managers.



Note: For an introductory walk-through of the tables and attributes that you should populate for any domain, see the videos listed in [CSDM resources](#).

Foundation domain tables used during the service life cycle

Individual Foundation domain tables are accessed as needed during each phase of the service life cycle. Each topic that describes the life-cycle phase of a domain identifies the foundation tables that are active during that phase. See the following diagrams:

- Tables used during the Ideation & Strategy phase of the service life cycle
- Tables used during the Design & Planning phase of the service life cycle
- Tables used during the Build & Integration phase of the service life cycle
- Tables used during the Service Delivery phase of the service life cycle
- Tables used during the Service Consumption phase of the service life cycle

Data managed by the chief strategist: Value stream

Value stream: A value stream is the set of actions that take place to add value to a customer from the initial request through realization of value by the customer. The value stream begins with the initial concept, moves through various stages of development and on through delivery

and support. A value stream always begins and ends with a customer. The value stream is aligned with your organization's business processes.

Data managed by the process owner: Business processes

A business process has a well-defined start and finish. Examples of business processes in the banking industry are the customer onboarding process and the credit check process. Each business process can have levels of criticality and impact. Business processes are stored in the `cmdb_ci_business_process` table.

In a parent-child relationship, business processes can be identified by using the parent attribute as a reference to a parent business process.

The business process is a manually-maintained CI that can identify declared and determined criticality as well as impact to confidentiality, integrity, and availability. Business processes can be reviewed monthly, quarterly, semi-annually, or annually. In addition, the next review date can be recorded. For further information, see [Business process management](#) and [Create a business process](#).

Data managed by the contract manager: Contracts

A contract is a binding agreement between two parties. In the ServiceNow AI Platform, contracts contain detailed information such as the contract number, start and end dates, active status, terms and conditions statements, documents, renewal information, and financial terms.

- A contract is not a CI. Contracts use contract model types from the [Product Models](#) module. Contracts are stored in the `[ast_contract]` table.
- Use the Contract Management application to manage and track contracts. See [Contract Management application](#).
- In the Service Level Management application, contracts group together SLAs that relate to a single vendor or customer, as well as the CIs, locations, groups, users, and child contracts that are related to the contract. For more information, see [Define a service contract](#).
- Service contracts used by Vendor Management Workspace can support tangible/physical CIs as part of an SLA.

- In the Customer Service Management product, service contracts define the type of support that customers receive. A contract can include an account and contact or a consumer and the specific assets that are covered. A contract can also include multiple service entitlements and SLAs. See [Define a service contract in Customer Service Management](#).

For more information, see [Definitions of life-cycle values for document and contract entities](#).

Data managed by the product owner

Products and product models

Products might be bundled to create a collection or group of products, for example a FlashBlade server (hardware model), or a 24/7 support service (service model). Additionally, you can identify the products reaching end-of-life as defined by third-party providers or internal product owners. You can also bundle other products as components to represent the set of products that your organization develops, sells, or uses.

A product model is a specific version or configuration of a product used to manage and track applications on the ServiceNow AI Platform. The Product Model tables are not CIs. CIs reference Product Models using the **Model ID** attribute (available on all CMDB tables). For example, a Service Offering CI may reference a Service Offering Model, while a Windows Server may reference a Hardware Model.

Product models identify the product owner, team, product status, compatibility with other products, reference to product catalog, and reference objects in the various stages of a product's life cycle.

Product Models are recorded in the [cmdb_model] table through the following extended tables, known as product model classes.

- Application Model [cmdb_application_product_model] (version-agnostic)
- Software Model [cmdb_software_product_model] (version-specific)
- Contract Model [cmdb_contract_product_model]
- Facility Model [cmdb_facility_product_model]

- Hardware Model [cmdb_hardware_product_model] (tangible/physical devices)
- Consumable Model [cmdb_consumable_product_model]
- Service Model [cmdb_service_product_model]

Application, service, and software class instance CIs aren't created through Discovery, so their **Model ID [model_id]** values might not refer to product model records. To help you to migrate to a product-centric management paradigm, each instance of a logical CI should be associated with a product model. For recommendations, see [Auto-generate product models for logical CIs](#).

Product features

In digital product release, a feature (what you're developing toward) exists in a particular version of software or service.

Software bills of material (SBOMs)

An SBOM is the overall collection of components of a piece of software.

Cyclone DX is supported.

Data managed by the data steward

CMDB groups

A CMDB group is a collection of CIs (but is not, itself, a CI). A group is based on the results of saved Query Builder queries, encoded queries, or manual entries. You can apply an action to all members of a group at one time.

- You can work with a CMDB group across the ServiceNow AI Platform.
- For the CSDM, the Dynamic CI Group references a CMDB group to provide a list of CIs based on a common criteria.
- CMDB groups are stored in the Group [cmdb_group] table.
- The CMDB group can potentially replace the spreadsheets that you might be using to group your CIs.

For more information, see [CMDB groups](#).

Locations

Data that comes from multiple sources and federated integrations is difficult to maintain. The following attributes have been added to the location [cmn_location] table to simplify management:

- **Source:** The origin of the location record.
- **Location type:** The position of the location record in the hierarchy of locations. You can use the following options to create a hierarchy of location data to suit your requirements: Region, County, State/Province, City, Site, Building/Structure, Floor, and Room.
- **Managed by group:** The group that governs or manages this location record.
- **Validation** (duplicate and primary): Flag duplicate records and manually filter locations that are not be displayed.
- **Life cycle stage** and **Life cycle stage status**.

Life-cycle values

life-cycle value pairs track the life cycles for products, assets, contracts, CIs, locations, and other objects. Using the standard CSDM life-cycle values consistently helps you to effectively track objects through their transitions over time. Reporting can therefore accurately reflect the actual states of CIs: usage, availability, end of support, and so on.

When you enable the CSDM framework, you can start using the Life Cycle Stage and Life Cycle Stage Status values to track an asset's life cycle. To use the fields, follow the procedure described in [Activate the CSDM plugin](#). The following processes can use the life-cycle value pairs:

- [Life cycle of product entities](#)
- [Life cycle of tangible/physical CIs](#)
- [Life cycle of intangible/logical entities](#)
- [Life cycle of document and contract entities](#)
- [Life cycle of location entities](#)

See the ServiceNow Community video: [CSDM V4 product and life cycle discussion](#)

Common data managed by the data steward

Common data is stored in the following tables:

- Company: [core_company]
- Business unit: [business_unit]
- Department: [cmn_department]
- Location: [cmn_location]
- Groups [sys_user_group] and users [sys_user]. When the full definition of a CI requires that you identify the contact groups that are associated with the CI, you can add that data in the **Teams** related list for the CI.

Common data elements are not configuration items. Common data is shared and used throughout the ServiceNow AI Platform. Common data includes organizational structure (Company, Business Unit, Department), locations, groups, and users. Many ServiceNow AI Platform products depend on common data to provide business value.

Planning your common data is essential to the effective implementation of ServiceNow AI Platform products and features. Consider the following issues:

- Do you have a trusted source for the data?
- Do you have multiple data sources?
- How often does the data change?
- Do you have the depth of data that the CIs require?
- Who maintains the data?

CSDM videos in the ServiceNow Community

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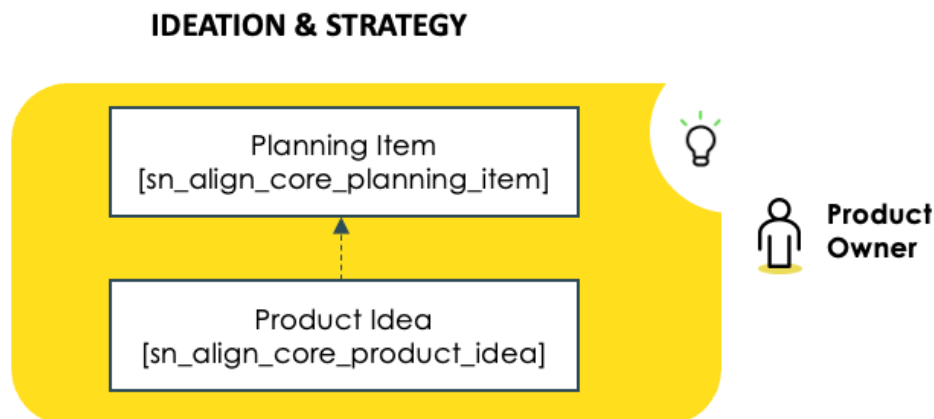
Ideation & Strategy domain in the CSDM model

The Ideation & Strategy domain represents the ideas, concepts, and considerations for both the creation of new or additional services as well as improvements and enhancements to existing services. These capabilities are part of Strategic Portfolio Management (SPM).

Tables used in the Ideation & Strategy domain

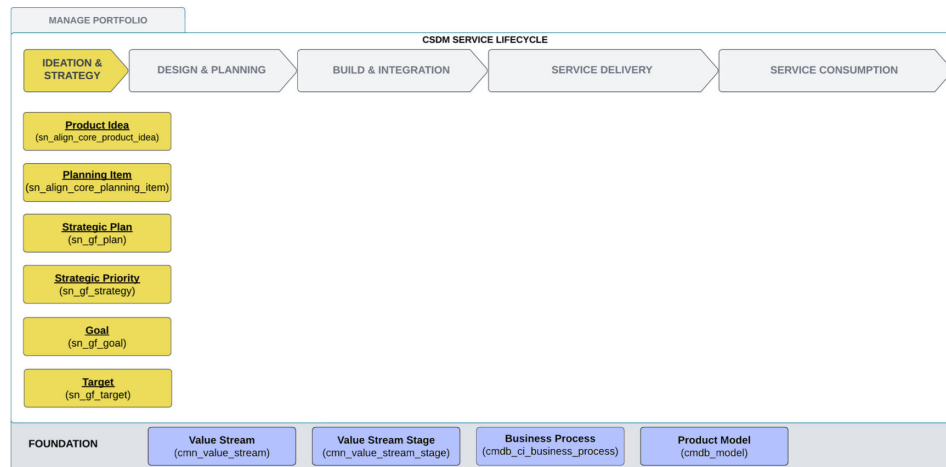
The Ideation & Strategy domain supports the efforts of product owners during the initial stages of working out the ideas for digital products.

- Planning Item [sn_align_core_planning_item] table: A planning item is a type of work item that planning organizations use to visualize planning data. The planning data supports enterprise-level planning, project-level planning, and portfolio planning.
- Product Idea [sn_align_core_product_idea] table: An idea for a complete product, feature, enhancement change proposal, or suggestion that can be curated and/or promoted into demand, project, epic, or story.



Note: CIs in the Ideation & Strategy domain aren't operational, so you can't select them for Incident Management, Problem Management, or Change Management.

Tables used during the Ideation & Strategy phase of the service life cycle



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Design & Planning domain in the CSDM model

Enterprise architects and digital product owners work on the design and planning of digital products that your organization can either buy or build.

Tables in the Design & Planning domain

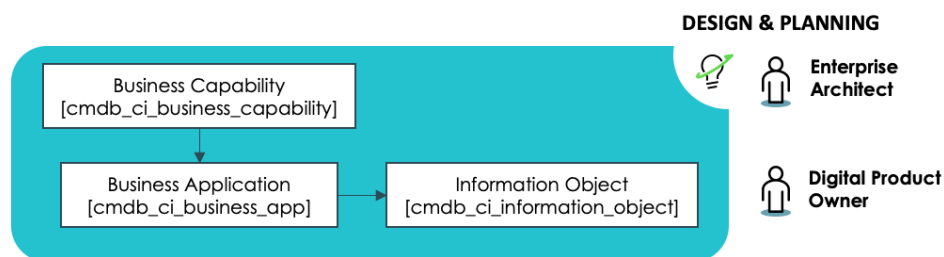
CIs in the Design & Planning domain aren't operational, so you can't select them for Incident Management, Problem Management, or Change Management.

The Design & Planning domain includes the tables used by Application Portfolio Management (APM). You use APM to rationalize and manage your business applications, but you're not required to use APM to benefit from the data in these tables. The Design & Planning domain includes the following tables:

- Business Capability [cmdb_ci_business_capability] table. Business Capability: A business capability is a high-level capability required by

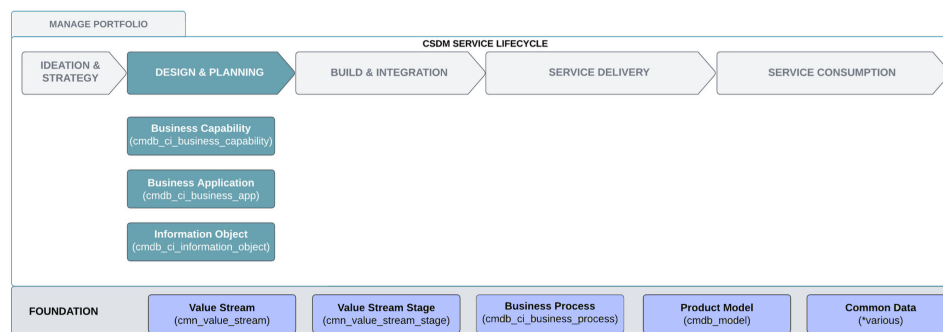
the organization to execute its business model. A business capability provides design guidance for implementing applications.

- Business Application [cmdb_ci_business_app] table. A business application is a purchased or internally-developed application that supports a business capability. A Business Application, ultimately, relates to an Application Service table and not any other type of Service Instance. For more information, see [CI relationships in the CSDM](#).
- Information Object [cmdb_ci_information_object] table. An information object describes the type of data that is being interchanged between the business application and the database that serves it.



Note: CIs in the Design & Planning domain aren't operational, so you can't select them for Incident Management, Problem Management, or Change Management.

Tables used during the Design & Planning phase of the service life cycle



Relationships between CIs that support decision making

An accurate service model that includes the following relationships can serve as the foundation for strategically aligned architectural decisions. The relationships enable you to determine the risks involved in using particular business capabilities. In addition, they enable you to assess services based on their relationships to business capabilities and business applications.

- Business capability Provided By::Provides Business application
- Business capability Provided By::Provides Business service (Service Consumption domain)
- Business application Uses::Used By Information object
- Business application (reference attribute) Business application
- Business application Contains::Contained By SDLC component (optional) (Build & Integration domain)
- Business application Consumes::Consumed By Service instance (Service Delivery domain)

Relationship between a business capability and its supporting business applications

This relationship supports visualization and reporting.

- A business capability has the Provided By::Provides relationship with its supporting business applications

-

A business capability is a high-level capability that supports a business model or fulfills a mission for your organization.

A business capability typically describes a specific task that achieves one or more business outcomes. Business capabilities are often listed as verbs (for example, Manage HR, Manage financial data, or Provide IT support services). You can use business capabilities to rationalize and prioritize the cost of business applications and business services.

- A business application represents the software and infrastructure that provides a business function (for example, the titles catalog).

Business applications are not strictly required, but they are strongly recommended because they increase productivity and perform other business functions such as accounts payables, accounts receivables, and general ledger. You can use APM to add any business application for which you must track costs, usage, business value, functionality, and risks.

Relationship between a business application and the service instances

A business application has Consumes::Consumed By relationships with service instances.

The relationship connects the record of the business application that is used in planning and design with where and how it's realized operationally, represented by service instances. The relationship accounts for each use of a business application in the development, test, and production environments (dev, test, and prod service instances). Often there are multiple production deployments. For example, a large retailer uses a business application that runs a cash register in each of its 1,000 stores. There are therefore 1,000 production instances of the service instance — one per store — for that one business application. [See the "CSDM in a nutshell" video](#) for additional discussion of the relationship.

Business capabilities represented in a hierarchy

You can represent business capabilities in a hierarchy of a parent business capability and one or more lower-level (child) capabilities. Child capabilities (leaf nodes) are represented by numeric values: 1.0 for the parent and 2.0 through 6.0 for the leaf nodes. If you add, update, or delete a capability at a leaf node, be sure to update the levels of all the capabilities for the leaf nodes in that hierarchy, as applicable. If a business capability hierarchy requires more than six levels, divide the structure into multiple business capabilities.

Use the Business Capability form to create, modify, and extend business capabilities.

<

≡

Business Capability
Manage supplier recovery

🔗

⚙️

⋮

Dashboard

Form

Update

Delete

* Name

Manage supplier recovery

Parent

Service products after sales

🔍

ⓘ

Level

2

Business Unit

🔍

Department

🔍

Description

Related Items

🔍

Search for CI

+

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Update

Delete

Related Links

[Multisource Data Preview](#)
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🔍

Capabilities

Order ▾

Search

⚙️

—

New

Parent = Manage supplier recovery

🔍	Name	Description	Level	Order ▲
No records to display				

Related Links

[Update Capability Level and HierarchyID](#)

To update capabilities, select the **Update Capability Level and HierarchyID** related link. Follow these guidelines when you update capabilities to ensure that the capability map reflects the change:

- If the parent capability is updated in the hierarchy, the levels of all its leaf node capabilities are recalculated.
- The total number of leaf node levels in a hierarchy can't exceed six.
- When adding a capability, the hierarchy level is automatically assigned based on the parent capability level.
- You can delete only leaf node-level capabilities or capabilities without leaf node levels.
- Don't create circular relationships. For instance, when creating a parent capability, a leaf node capability can't be its parent.

Adding a Business application

A business application is a manually managed CI class that has a Consumes :: Consumed By relationship with a service instance. You must therefore manually create required relationships to CIs (for example, with instances of the service instances in use). Creating relationships also enables you to relate business applications to infrastructure CIs such as databases and web servers. If needed, you can integrate or connect two or more business applications to establish their relationship.

A business application can span the following entities:

- Environments (for example, Development, Test, or Production).
- Geographies (for example, the Americas or the Asia Pacific Japan [APJ])
- Regions (for example, Europe, the Middle East, and Africa [EMEA]).

Use either of the following methods to add a business application:

-

Import the list of applications from a spreadsheet or third-party tool. To import data, define a data source and a transform map and then run or schedule an import.

- Use the Business Application form.

Business Application
New record [Business Application ReadOnly view]

Name: Compliance Training Status: In Production

Business process: Learning Path Technology stack: Java

Application type: COTS User base: 1000+

Architecture type: Web Based Platform: Oracle

Install type: Cloud Last change applied date: 2018-10-01

Business Unit: Legal

Department: Legal

Description: The compliance training business application focuses on security, SEC rules, trade policies and other regulatory training for all employees.

Owners: Compliance

Business owner: Fred Luddy

IT Application owner: Abel Tuter

Last updated by:

Submit

Information object

- An information object is a CI that displays and describes the information (or type of data) that the application receives from the database. Information objects are part of the information portfolio and are referenced by the business application. An information object has a Used By::Uses with a business application.
- Information objects are mapped to the information object [cmdb_ci_information_object] table.
- You can use the information object table to identify the types of data that a business application uses, including highly sensitive data such as:
 - Personally Identifiable Information (PII)
 - Payment Card Industry Data Security Standard (PCI DSS) data
 - Health Insurance Portability and Accountability Act (HIPAA) data

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Related concepts

- [CI relationships in the CMDB](#)

Build & Integration domain in the CSDM model

Development teams use the tables in the Build & Integration domain during the build effort (systems development life cycle — SDLC or Agile development) of digital products. The tables represent the logical development details of the enterprise applications (digital products) to be deployed and used by your organization. These are not operational CIs.

Tables and relationships in the Build & Integration domain

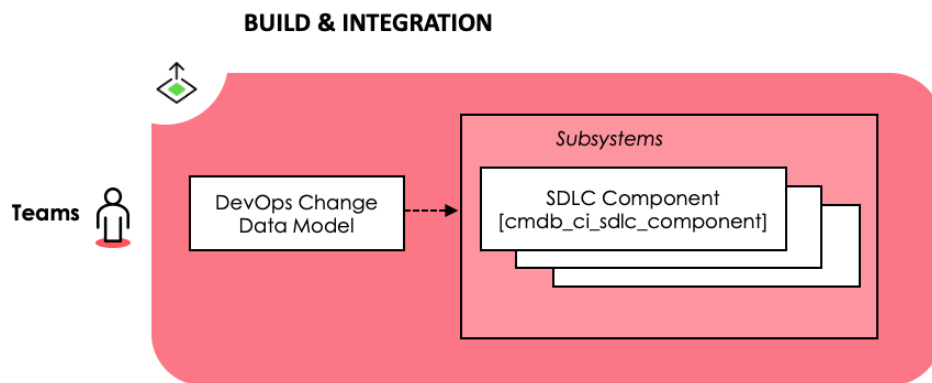
An SDLC component is a CI that represents a unique development effort of code. It represents parts of a larger business application or digital product broken down into its individually developed components. In other words, the SDLC component is a software element of a larger application or technology. SDLC component CI records in the SDLC Component [cmdb_ci_sdlic_component] table enable the DevOps product to provide enhanced capabilities for visualizing and managing your application development pipeline.

Records in the table are not operational and are not direct targets of the ITSM Incident Management, Problem Management, or Change Management processes. You therefore are not required to configure SDLC component records. The SDLC Component table is available through the CMDB schema version 1.33.

The DevOps Change Data Model is built on a powerful data model that connects your DevOps toolchain to the work and data already in the platform. The model provides the ability to accelerate changes while ensuring effective and transparent risk management.

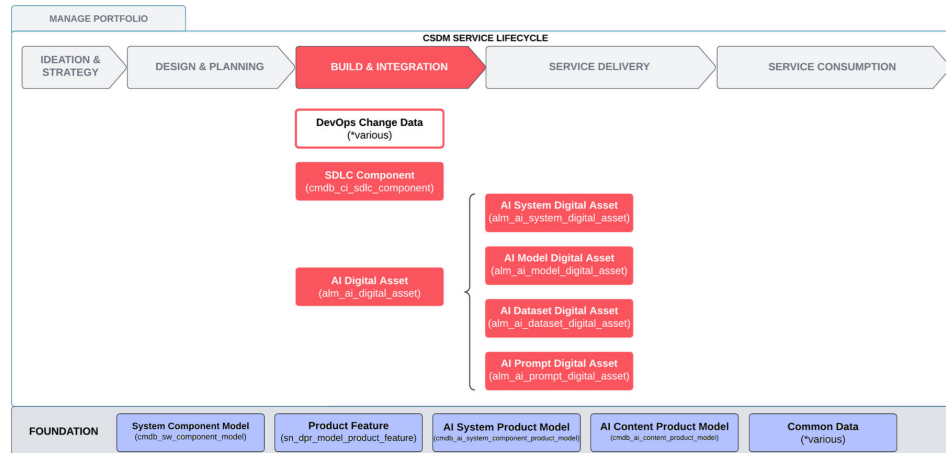
The Build & Integration domain includes the following tables and relationships:

- Business application (Service Consumption domain)
Contains::Contained By SDLC component
- SDLC component has a Contains::Contained By relationship with the Service instance table in the Service Delivery domain.



Tables used during the Build & Integration phase of the service life cycle

The AI System Digital Asset represents the Software/ Application components for AI and Gen AI services. For the purposes of build, integration, and AI governance, the AI System Digital Asset (ServiceNow, third party, or custom) represents the deployable software source, binaries, and components that feed into the Build & Integration pipelines. An asset can be deployed or not currently deployed.



Types of SDLC component

- **Application:** A service instance is a deployed instance of the SDLC application component. Examples include micro services and APIs. The build team typically builds service instances on behalf of the Service Owner (as described in [Manage Portfolio domain in the CSDM model](#)).

See [Use Application Services dashboard to monitor health](#).

- **Infrastructure:** Any infrastructure CI that represents a snapshot of its configuration details is a deployed instance of the SDLC infrastructure component. Examples include database and security configurations.

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Service Delivery domain in the CSDM model

The Service Delivery domain represents the overall end-to-end service delivery system that includes the infrastructure, technologies, integration patterns (infrastructure, systems, data, processes, dependency models), service delivery networks, and operational models. Together, these items deliver the CSDM-compliant services to internal and external users and/or organizations.

Tables in the Service Delivery domain

Users of tables in the Service Delivery domain provide and manage the operational environments and infrastructure that support your organization's business applications (as developed in the Build & Integration domain). Typical users are service instance owners (for application and platform) and service delivery owners or service providers (for infrastructure and delivery).

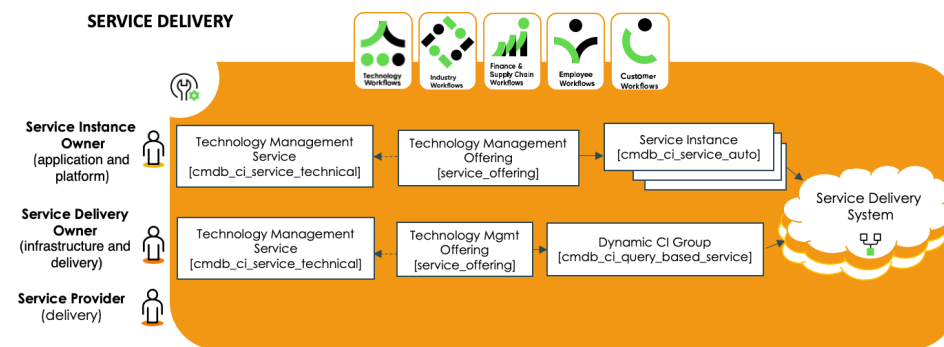
The tables in the Service Delivery domain represent the technology that your business sells or consumes in the provider view. Service Mapping and Discovery populate the tables. The tables in the domain also enable you to manage CIs and their relationships. The domain includes the following tables:

- Technology management service (formerly Technical service) [cmdb_ci_service_technical] table. The **Service Classification** is Technical Service.
- Technology management offering (formerly Technical service offering) [service_offering] table. The **Service Classification** is Technical Service.
- Request catalog. Technology consumers can request technology management offerings through a request catalog. Catalogs are described in detail in [Service Catalog](#).
- Dynamic CI group [cmdb_ci_query_based_service] table. **Service Classification** is Technical Service. Technology management services in Event Management use the cmdb_ci_query_based_service table.
- Service instance (called application service before CSDM v5) [cmdb_ci_service_auto] tables. The **Service Classification** is Application Service.
- For manually-created and Service Mapping: Mapped Application Service [cmdb_ci_service_discovered] table (included in the base system).
- When query-based: [cmdb_ci_query_based_service].
- When tag-based: [cmdb_ci_service_by_tags].

The CIs in this domain are deployed instances of digital products and their related and discoverable components (such as installed applications, servers, and network components) plus documentation

of the services that provide and support the deployed instances. The domain also represents the portfolio of technology management services in use. For more information on life cycles, see [Definitions of life-cycle values for tangible/physical CIs](#).

The services are operational, which means that you can select them for ITSM Incident Management, Problem Management, or Change Management.



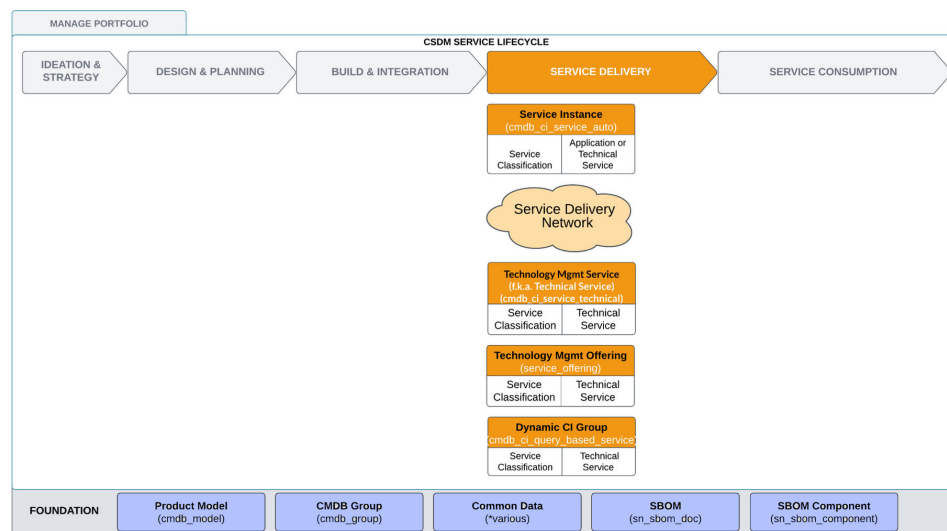
Relationships between Service Delivery tables

- SDLC component Contains::Contained by Service instance (optional to use this item).
- Business application Consumes::Consumed by Service instance.
- Service instance (a logical representation of a deployed system or application stack) Depends on::Used by /Sends Data To:: Service instance.
- Service instance Depends on::Used by Application.
- Application Runs on::Runs Infrastructure CIs.
- Technology management service uses a reference attribute to specify the relationship with Infrastructure Technology management offering. Published to service owners and typically underpins one or more business services. A Technology management service may have an operational view made up of one or more Technology management offerings.

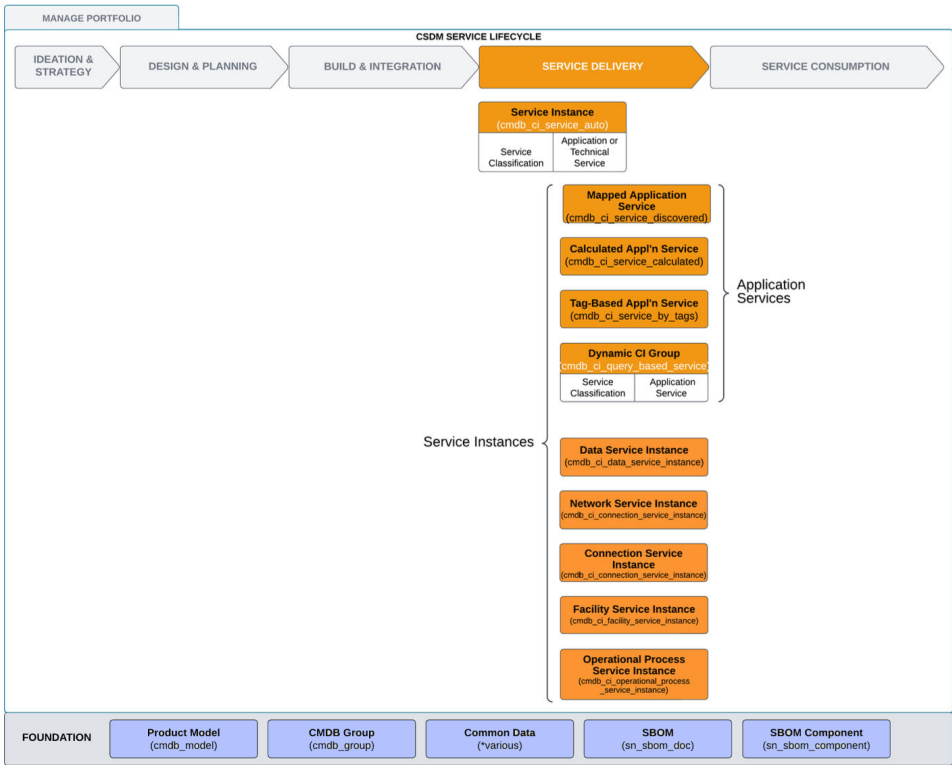
- Technology management offering Contains::Contained by Service instance. Stratification of the Technical Service into options, including localization/geography, environment, pricing, availability, capability, support group (INCIDENT), technical approval group (CHANGE) and packaging options.
- Technology management offering Contains::Contained by Dynamic CI Group.
- Dynamic CI Group (a dynamic grouping of CIs, based on results of CMDB Groups queries) uses a related list to specify the relationship with Infrastructure CIs.
- Business Service Offering Depends on::Used by Service instance.

Note: Business services and technology management services connect to the `spm_service_portfolio` through the `spm_taxonomy_node`. See [Service Portfolio Management taxonomy](#).

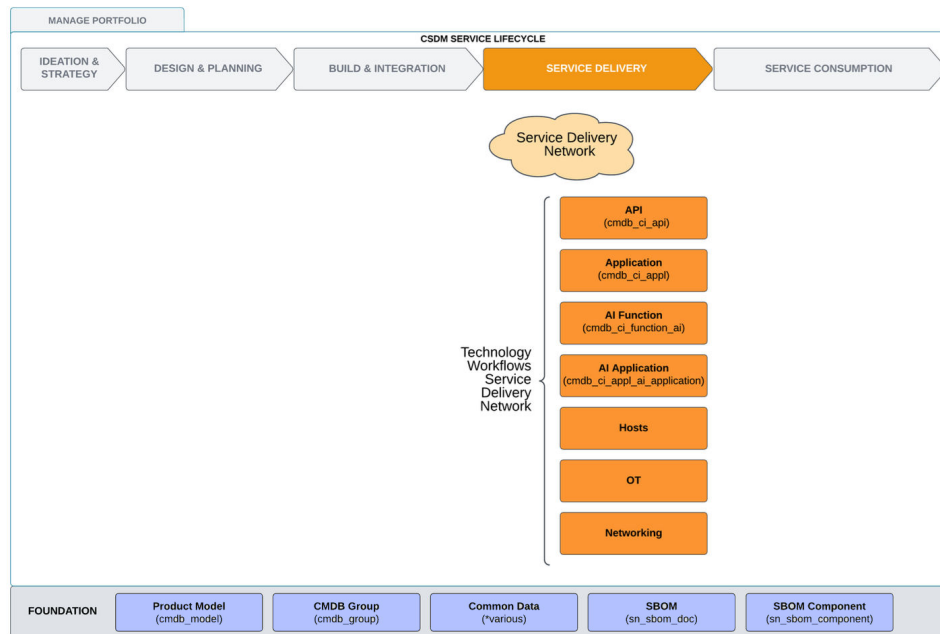
Tables used during the Service Delivery phase of the service life cycle



Tables used during the Service Delivery phase of the service life cycle



AI component tables used during the Service Delivery phase of the service life cycle



API: The API data model is made available within the CMDB to assist in managing your API data. As part of API Insights, you can centralize your management of APIs including, but not limited to, viewing API details, comparing APIs, identify and resolve data gaps, and manage service relationships. **API Function:** from a virtual perspective. **API Application:** From an on-premises perspective.

AI Function: AI SaaS applications deployed on public cloud platforms that offer scalable, on-demand services for machine learning, data processing, and AI-driven tasks. The applications provide flexible solutions without the need for on-premises infrastructure management.

AI Application: AI software applications that can run on various platforms such as Linux, Windows, Docker containers, or Kubernetes (K8) clusters. The platforms support diverse AI workloads, including machine learning models, data analytics, and intelligent services or AI-enabled applications.

Technology management services

Technology management services are associated with service owners and are typically layered under one or more business services or service instances. A technology management service may have one or more technology management offerings.

Users of technology management services can view and manage the technologies that you provide to the business. Event Management enables you to monitor service performance. You can also use Event Management to identify health issues for related infrastructure CIs and service instances.

Technology management services can be managed as part of the Service Portfolio in the Service Consumption domain (that is, a Service Portfolio hierarchy can be referenced from a technology management service). This allows for a more complete hierarchy and management of both technology management services and business services within the Service Portfolio Management workspace and related workspaces. You can make better decisions when you know how spend on technology management services can improve performance and reliability of your business services.

Note: Business services and technology management services connect to the `spm_service_portfolio` through the `spm_taxonomy_node`. See [Service Portfolio Management taxonomy](#).

Technology management offerings

Technology consumers can request technology management offerings (TMO) through the request catalog. Catalogs are described in [Service Catalog](#). The consumer can typically select the following features and options:

- Level of performance
- Location or geography
- Environment
- Pricing
- Availability

- Capability
- Support group (for incident)
- Technical approval group (for change)
- Packaging options (commitments)

Tip: Use dynamic CI groups to significantly improve management. For more information, see [Synchronize user groups for a technology management offering](#).

Technology management offerings typically have the following components:

One or more service commitments

A service commitment defines the service delivery obligations agreed to between the consumer and the provider. Service commitments uniquely define the level of service in terms of availability, criticality, scope, pricing, and other factors. For example, an organization may offer two levels of support for a service instance:

- Support for a production-level offering: Provides a high level of availability and criticality for production instances. Includes a 24/7, 5-minute response time guarantee (24 hours per day seven days per week).
- Support for a non-production-level offering: Limited availability and criticality for non-production instances. Includes a 60-minute response time guarantee between 8:00 a.m. and 5:00 p.m., Monday through Friday.

A service offering subscription that records which users have access to an offering

A technology management offering references the **Service Categorization** attribute in service tables to denote whether a technology management offering or offering is related to a business service or to a technology management service. Technology management offerings that are mapped to the [service_offering] table are classified as "technology management service" and are derived from the service. The technology management offering is based on how the parent serves a specific technical need. Every technology

management service should have at least one technology management offering.

Important: Each CI associated through a Dynamic CI Group can be related to only one technology management service or technology management offering. Conflicts can result when one service includes multiple offerings with different SLAs, OLAs, Support Groups, and commitments.

Dynamic CI groups

A dynamic CI group is comprised of CIs that result from a CMDB Groups query. For example, you can create a dynamic CI group based on location: "all web servers in Detroit" or "all Oracle databases in Mumbai".

Note: Dynamic CI groups contain only CIs and can't contain other CI groups.

Dynamic CI groups are mapped to the [cmdb_ci_query_based_service] table and are classified as either service instance or technology management service, as applicable. You might want to use dynamic CI groups in the following situations:

Query-based service instance

You don't have Service Mapping enabled yet, but you have 12 servers and three database instances in MyAppServiceProd. You can replace your spreadsheets with a dynamic CI group as a service instance.

See [Use the Dynamic CI Group method to populate application services](#).

Managed group of Infrastructure CIs

The web servers in Detroit are managed by the DetroitRockCity technology management offering. Instead of manually creating relationships from technology management offerings to Infrastructure CIs, use a dynamic CI group. A single relationship from your technology management offering CI (DetroitRockCity) to your dynamic CI Group (web servers in Detroit) gives you the visibility you need.

A way to manage patches for your CIs

In Change Management, you can select the dynamic CI group for the CIs you need to update and use a business rule to auto-populate the **Affected CI** field.

Service instances (called application services before CSDM v5)

Service instance a relabel of [cmdb_ci_service_auto], which previously was called application service.

Application services still exist. They were always there as extended tables depending on how you managed that application service.

For manual creation or service mapping, you go into the mapped application service.

For tag-based, it would live in the tag-based application service table. The tables still exist and they are still application services and you are still gonna go to the application service wizard to create application services.

New types of service instances are extensions of [cmdb_ci_service_auto]:

- Data service instance [cmdb_ci_data_service_instance]
- Network service instance [cmdb_ci_network_service_instance]
- Operational Process service instance [cmdb_ci_operational_process_service_instance]
- Connection service instance [cmdb_ci_connection_service_instance]
- Facility service instance [cmdb_ci_facility_service_instance]

The Service Instance (called application service before CSDM v5) [cmdb_ci_service_auto] table supports service instances. A service instance is a set of interconnected applications and hosts that are configured to offer a service. A service instance is a logical representation of a deployed system or an application stack. Using service instances, you can view maps and change history for services. For example, the Event Management application can monitor service performance and identify health issues for service instances.

Service instances can be internal, like an organization's email system or customer-facing, like an organization's website. For example, creating financial reports through a web-based application requires a computer, web server, application server, databases, middleware, and network infrastructure. The applications and hosts are configured to offer the service of financial reporting. A service instance represents an instance of such a business application or system in the development, test, or production environment.

Service instances are the entry points for the Service Mapping feature. Service instances underpin a business or technology management service and are mapped to the CMDB Service Instance (called application service before CSDM v5) [cmdb_ci_service_auto] table for common reporting.

Service instances are key relationship entities for IT Service Management (ITSM), IT Operations Management (ITOM), Strategic Portfolio Management (SPM), and Customer Service Management (CSM).

Service instances include relationships between business applications, business services, technology management services, applications, and infrastructure CIs. You can expose a service instance by using the related business or technology management offering. For more information, see [Use Application Services dashboard to monitor health](#).

Types of service instance in the Service Instance data model:

- Data service instance
- Network service instance
- Operational Process service instance
- Connection service instance
- Facility service instance

The table that a service instance maps to depends on the method used to create it:

Methods mapped to tables

Method used to create the service instance	Mapped to table
Top Down Discovery (Service Mapping)	cmdb_ci_service_discovered
Dynamic CI Group (Query-based)	cmdb_ci_query_based_service
Tags	cmdb_ci_service_tags
Manual entry, using the Create a Service Instance form (formerly, the Create an Application Service form)	cmdb_ci_service_discovered

- For more information about service instances and the methods you can use to create them, see [Application services](#) and [Create an application service](#).
- You can specify required attributes for service instances. For more information, see [Modify the attributes and relationships required for application services](#).
- You can set a relationship between a service instance and the components of other CSDM domains. See [Service Mapping](#).

Applications

An application is any program or module that defines behavior and performs a specific function. Applications are typically discoverable instances and provide a specific set of functions for one or more services.

- The application table and extended tables contain uniquely discovered instances of code in use on the host.
- Applications are considered infrastructure CIs.
- The instance is limited to the applications on a single host. This limitation ensures that applications are uniquely identified during discovery.

- There's a one-to-many (and not a one-to-one) relationship between the application and the service instance. A single installed application, such as a database instance, may support multiple service instances depending on the configuration and the use of the applications.

Note: The application table [cmdb_ci_appl] isn't an inventory or portfolio of your applications. Don't make the mistake of storing managed application details in the application table. Those details (inventory or application portfolio objects) belong in the business application table (as documented in [Design & Planning domain in the CSDM model](#)).

Infrastructure CIs

Infrastructure CIs are managed physical and logical components. A CI can be a single module, such as a server, database, or a router or a complete system such as a web server, database, or infrastructure.

The underlying infrastructure components or CIs can be complicated. The complexity increases as data structures are layered on top of physical CIs. For that reason, you should work with a business relationship manager or enterprise architect to define your business capabilities and business applications.

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Related tasks

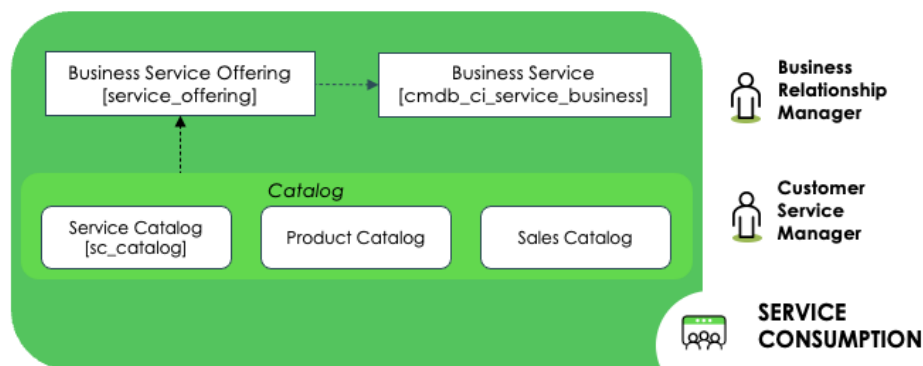
- [Synchronize user groups for a technology management offering](#)

Service Consumption domain in the CSDM model

Internal or external consumers can request business services through the request catalogs in the Service Consumption domain. The business service offerings in the request catalogs are the deployed digital products in the Service Delivery domain. Business relationship managers and customer service managers might sell or even consume elements of the Service Delivery domain.

Tables in the Service Consumption domain

Tables in the Service Consumption domain are used by Service Portfolio Management and Customer Service Management (CSM). You can select the tables in the Service Consumption domain to use with Incident Management and Change Management.



The Service Consumption domain includes the following tables:

- The Business Service [cmdb_ci_service_business] table (**Service Classification** is Business Service) extends the core Service [cmdb_ci_service] table. For example, a service might be "Manage HR".

Note: Before the Business Service table was added, all Business Services existed in the Service table. In the future, all Business Services might migrate from core cmdb_ci_service to cmdb_ci_service_business. Until then, both tables operate identically.

- The Business Service offering [service_offering] table holds the offerings that consumers can select in the catalog. Taken together, the business service offerings make up the business service. In our HR example, offerings might be "Manage Onboarding" and "Manage Interviews". Each offering has a Depends on::Used by relationship with a service instance in the Service Delivery domain.
- Service Catalog [sc_catalog] table. The Service Catalog tables hold the business service offerings that consumers can select.

- Business capability Provided By::Provides Business service
- Service portfolio (reference attribute) Business service
- Business service (reference attribute) Business service offering
- Business service offering Depends on::Used by Service instance

You're not required to use Service Portfolio Management or CSM to use the referenced tables, but those products enable you to manage workflows and report service-related data.

Tables used during the Service Consumption phase of the service life cycle



Business service offerings

Business service offerings (**Service Classification** is Business Service) are the starting point for configuring Service Portfolio Management. Business service offerings inherit from Business Services. Business service offerings consist of one or more service commitments that define the level of service in terms of availability, scope, pricing, and other factors. For example, an organization might offer two levels of desktop support:

- A silver offering of upgrades and virus protection.
- A gold offering with the silver commitments plus a response time guarantee of 30 minutes between the hours of 8:00 a.m. and 5:00 p.m., Monday through Friday.

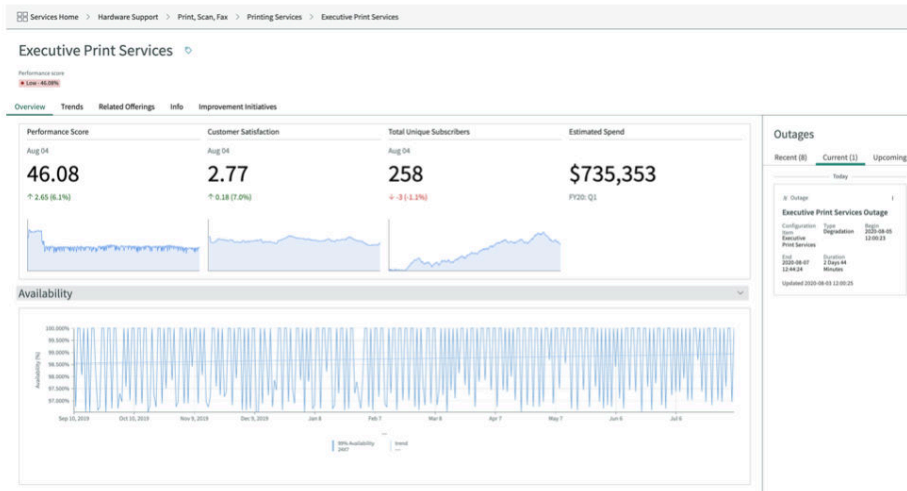
Note: Business services and technology management services connect to the `spm_service_portfolio` through the `spm_taxonomy_node`. See [Service Portfolio Management taxonomy](#).

Business service offerings have the following characteristics:

- Business service offerings tailor the service by capability, availability, pricing, and packaging options. You can use the service offering to set different levels of performance and features for a particular service.
- Business service offering commitments define the agreed-upon service delivery obligations.
- Business service offering subscriptions record which users have access to an offering.
- Business service offerings are the CMDB records that identify the specific business area and the entity where the service is delivered. Some business services and service offerings depend on the service instance.
- Business service offerings are derived from the service and are refined depending on how the parent serves a particular business need.

Note: You should configure at least one business service offering for each operational business service or technology management service.

You can view your business service offerings in the Digital Portfolio Management (DPM).



Business service offerings typically have different service-level agreements (SLAs) depending on their commitments. Without a business service offering, SLAs remain at a process level only. For example, the SLA stays at a P1 incident or a minor change, and doesn't refer to the affected business service offering.

You can represent business services and business service offerings as catalog items in the service catalog to make them available for consumers.

Business services

A business service is associated with business users and is typically layered beneath one or more business capabilities. A business service can contain one or more business service offerings.

Business consumers can use the request catalog to order business services, business service offerings, and service commitment levels. Catalogs are described in detail in [Service Catalog](#). Business services are mapped to the `cmdb_ci_service_business` table and are classified as "business services."

Service portfolios

A service portfolio is a hierarchical collection of business services, products, projects, or applications. A portfolio can represent a strategic business objective and enables you to manage all included items as a

group (for example, life cycles). Items are organized into the following categories:

- Objective (business intent)
- Capability
- Organization (for example, enterprise resource planning [ERP] or financial management)
- Geography (location)

Catalogs

A request catalog enables consumers to order and manage business and technical products, services, service commitment options, and offerings (for example, the Human Resources [HR] service catalog). Catalogs contain catalog items and are the starting point for consumers to access available services. Catalogs are described in [Service Catalog](#).

Catalog Item

A catalog item is an item or a service that a consumer can request from the catalog. A service can contain multiple catalog items (for example, the employee onboarding catalog). Catalog items are listed on the service portal and are available to the users that need them (either through subscription or job responsibility). Each catalog item is linked to one business service offering.

CSDM videos in the ServiceNow Community

[Playlist of all CSDM videos](#)

Manage Portfolio domain in the CSDM model

The Manage Portfolio domain is a layer on top of the CSDM conceptual model that interacts with the other CSDM domains. The typical user, a service owner, might be responsible for services in more than one domain.



For example, in the Service Consumption domain, the service owner for Human Resources (HR) might be financially responsible for the business application that provides HR services. The service owner might also need to manage the HR application instances (known as service instances or systems) and might also be accountable for the impact the application has on the business.

Because of these additional responsibilities, CSDM enables service owners to oversee business applications and their deployed instances. This visibility enables service owners to perform their duties and meet their responsibilities.

CSDM videos in the ServiceNow Community

[Playlist of all CSDM videos](#)

Configuring CSDM

Activate and configure CSDM capabilities.

Configuration overview

- [Activate the CSDM plugin](#)

Activate the CSDM plugin so you can begin implementing the CSDM data model.

- [Configure the CSDM Data Foundations dashboard](#)

Use the CSDM Data Foundations dashboard to monitor and evaluate key foundational metrics of the CSDM framework.

- [Migrate and sync existing data to the CSDM framework](#)

You complete several tasks to ensure that your existing application data migrates successfully to the required tables in the CMDB.

Activate the CSDM plugin

Activate the CSDM plugin so you can begin implementing the CSDM data model.

Before you begin

Important: Before you activate the CSDM plugin, you must map your existing life-cycle data to standard CSDM attributes. The mapping enables you to track assets effectively through their life cycle transitions with the [CMDB Data Manager](#). For instructions, see [Migrate to CSDM life cycle standards](#).

- Role required: itil and itil_admin

Procedure

Activate the com.snc.cmdb.csdm.activation plugin as described in [Activate a plugin](#).

Configure the CSDM Data Foundations dashboard

Use the CSDM Data Foundations dashboard to monitor and evaluate key foundational metrics of the CSDM framework.

Before you begin

For an introduction to the dashboard, see [CSDM Data Foundations dashboard](#).

- Before you use the dashboard for the first time, populate the CSDM metrics: Navigate to **All > System Scheduler > Scheduled Jobs** and run the CSDM Get Well Metric Collection job.
- The CSDM Data Foundations dashboard adds the following scheduled jobs that must be running:

- **CSDM Get Well Metric Collection:** Calculates and stores details for compliant CIs associated with metrics. Data appears on the dashboard only after the first run of this scheduled job. metrics' scores are stored in the CSDM Data Foundations Metric Scores [sn_getwell_csdm_score] table. The job runs daily by default.
- **CSDM Data Foundations PA Metric Collection:** Calculates the total count of non-compliant CIs that are associated with each metric. It also provides trending data over time for the non-compliant CIs associated with metrics.
- **CMDB Get Well CIs Processed Via IRE Metric Batch Collection:** Invoked once a day (by default) by the CMDB Get Well Metric Collection job. Repeatedly executes a 10-minute run/one-minute pause cycle to process as many batch files as possible.
- **Role required:** app_service_admin, app_service_user, asset, cmdb_read, itil_admin, portfolio_admin, service_viewer, or technology_service_owner

Procedure

1. Navigate to **All > CSDM > Configuration > CSDM Data Foundations Dashboard**.
2. Select a tab.

The reports on each tab display the data metrics that are appropriate for the maturity of your organization's CSDM implementation stage (foundation, crawl, walk, run, and fly).

3. Review the reports.
Note the percentages and color-coding in the **Result** column for each metric.
 - If the percentage is 100%, the CSDM framework has the information it needs. You don't need to do anything else.
 - Otherwise, required information is missing and additional actions are required. Continue with [step 5](#).
 Note the metrics on the **Foundation** tab:

Named Product Models with Product Owners

Shows cmdb_model records that meet the following conditions:

- **Status = In production**
- **Name** and **Owner** is not empty

Configuration Item Status Values

Shows the percentage of CIs with default status values.

- 80%: At least 1 to 5 CIs have custom status values.
- 60%: 6 to 10 CIs have custom status values.
- 40%: 11 to 15 CIs have custom status values.
- 20%: 16 or more CIs have custom status values.

To view the default base-system status values, enter `sn_getwell_oob_status_table_field.list` in the navigator **Filter** text box. The Configuration Item Status Values form displays the list of elements and associated tables. Select a table name to see the list of default labels and values.

Configuration Item Status Values			
		Search	Label ▼ Search
Table field = cmdb_ci			
		Label	Value
☐		<u>Retired</u>	7
☐		<u>Pending Install</u>	4
☐		<u>Installed</u>	1
☐		<u>Stolen</u>	8
☐		<u>Absent</u>	100
☐		<u>In Stock</u>	6
☐		<u>Pending Repair</u>	5
☐		<u>On Order</u>	2
☐		<u>In Maintenance</u>	3

Business Units with Companies

Shows business unit records where the **Company** field is not empty.

Locations with Parents.

Shows cmdb_ci records that meet the following conditions:

- **Status** = installed
- **Operational status** = operational
- **Location** and **Location.parent** is not empty

4. Select the tiles associated with the foundational metrics to access Performance Analytics widgets.

Performance Analytics widgets are provided by the CSDM PA Metric Collection scheduled job. These widgets provide trending data over time for the non-compliant CIs associated with the metric.



- Ensure that the real-time option is selected () and then select **Show Records** to view the list of CIs.
 - Select the **Breakdowns** context menu to view available breakdowns.
5. Scroll to the list of CIs in the **Custom Status Values** related list.

The charts show the number of custom values that have been defined for each element. Select a chart to view custom values that have been defined for the element. This example shows custom label-value combinations for the `install_status` element.

Custom Status Values New Search Table Search					
All > To delete = false > Element = install_status					
		Table	Element	Label	Value
☐		<u>cmdb_ci</u>	install_status	Absent2	100
☐		<u>cmdb_ci</u>	install_status	Absent	101
☐		<u>service_offering</u>	install_status	Installed by me	1
☐	Actions on selected rows...				

6. Select a CI to drill down to the form view.

The form view provides the required information. If you don't see the form, you might not have sufficient access privileges. Contact your ServiceNow administrator.

7. When you're finished using the form, select **Update** or **Delete** to return to the list view.
8. Navigate to return to the CSDM Data Foundations dashboard.

Result

The key foundational metric results are available for you to review and analyze.

Migrate and sync existing data to the CSDM framework

You complete several tasks to ensure that your existing application data migrates successfully to the required tables in the CMDB.

Before you begin

Role required: admin

About this task

Keep the following points in mind:

- Some CSDM tables have been introduced recently so you might not be familiar with them. See the documentation for your product to learn about unfamiliar tables.
- You can continue to use customized or non-conforming CMDB tables. If you do so, however, you might not get the full benefit of your products.
- Be sure to use the migration tools that are described in [Assistance with the CSDM life-cycle sync process](#).

Manage the attributes that you are using. Rationalize your custom attributes. Use the following guidelines to decide whether you really need to keep all customizations:

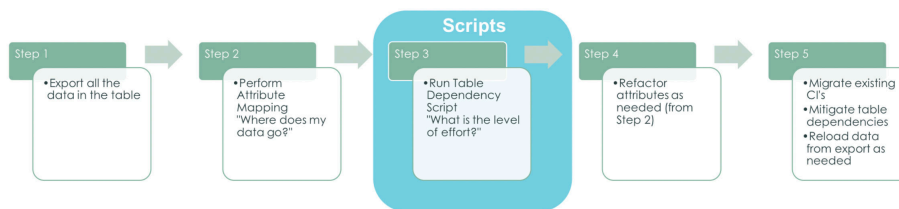
- Preferred: The custom attribute doesn't have a related base-system attribute but you need to use it.
- Keep: The custom attribute doesn't have a related base-system attribute but it's required for a unique use case.
- Refactor: The custom attribute does have a base-system attribute or a capability that can be migrated.
- Do Not Need: The customization is no longer needed. Delete the attributes that you don't use or use only rarely. Consider deleting attributes if there's a better way to address a use case.

Consider related dependencies. Moving configuration items (CIs) to a new table doesn't automatically move related dependencies. To identify related dependencies, use the scripts described in [Migrating into CSDM identifying table dependencies](#) (available on the [ServiceNow Community](#)).

Important: The script doesn't move your data or their dependencies. It only identifies the dependencies. You refactor data and dependencies as part of the migration.

After running the scripts and evaluating the data, you will have a better idea of the effort required to migrate your data. Decide whether you need all referenced reports, rules, and scripts. Then decide what you want to migrate and make a migration plan.

Migration workflow



Procedure

1. Back up your data.
Export your data (with all attributes) to Excel and keep the file in a secure location. Have a contingency plan in case issues arise.

2. Map the attributes.
Identify the table where the data should go. Make sure that destination table has the required base-system attributes. Rationalize your custom attributes. Decide which customizations you will keep.

3. Move CIs from existing classes to CMDB classes.

Note: Don't forget about non-conforming tables and their dependencies. You could have hundreds of reports, business rules, scripts, table references, and more that need the data in your non-conforming tables.

Moving CIs to a new table does not automatically move reports, business rules, and so on. As described in the following steps, the fix script identifies dependencies that you need to refactor. You can [download the fix script](#) from the ServiceNow Community.

4. Refactor the attributes.
Solidify the data model and get the data ready for migration.

Make sure you have completed the attribute mapping-related tasks described in earlier steps. Follow the guidelines and refactor your data as needed.

5. Migrate the data.

Note: Make sure that you have a valid and recent backup. Make another backup if necessary. During migration, you lose all customized or base-system attributes that are not in the same table hierarchy.

Keep these points in mind as you proceed:

- Migrate your CIs to the new class to move the CI and all its related objects, incidents, and changes to the new table.
- Start with a few CIs and increase the number when you feel comfortable.

6. Remediate your table dependencies:
 - a. Modify reports to use the new table.
 - b. Migrate business rules and scripts as needed.
 - c. Update table references as needed.

7. Reload data into the new attributes using the backup that you made earlier.
8. Validate all data and dependencies.

Result

You've successfully migrated your application to the CSDM framework, and your data is in the required CMDB locations.

- [Assistance with the CSDM life-cycle sync process](#)

Several features help you to ensure accuracy and completeness for the process of syncing legacy data with life-cycle data in the CSDM framework.

- [Legacy status values that are auto-updated](#)

Several legacy statuses are automatically mapped to the Life Cycle Stage and Life Cycle Stage Status fields.

Assistance with the CSDM life-cycle sync process

Several features help you to ensure accuracy and completeness for the process of syncing legacy data with life-cycle data in the CSDM framework.

Before you perform the life-cycle sync process

Legacy status values that are auto-updated

Several legacy statuses are automatically mapped to the Life Cycle Stage and Life Cycle Stage Status fields. For more information, see [Legacy status values that are auto-updated](#).

Data migration script

Use the fix script from the ServiceNow Community to identify table dependencies. See [Migrating into CSDM identifying table dependencies](#).

CMDB Data Foundations dashboard

Navigate to **All > Configuration** and then select **CMDB Data Foundations Dashboard**.

- CMDB dashboard: Each tab provides key metrics that evaluate configuration and customizations in the CMDB.
- The metrics provide visible results of evaluation. Color codes and weighted priority help with planning.
- Each metric includes a link to a remediation playbook with background and plays for remediation.

See [Monitor health in CSDM and CMDB Data Foundations Dashboards](#).

CSDM Data Foundations dashboard

Navigate to **All > Configuration** and then select **CSDM Data Foundations Dashboard**.

The CSDM dashboard focuses on key data elements to support you in implementing the CSDM framework. The reports on each tab display the data metrics that are appropriate for the maturity of your organization's CSDM implementation stage (foundation, crawl, walk, run, and fly). See [CSDM Data Foundations dashboard](#). In particular, the **Crawl** tab presents a report titled **CIs with Mismatched Operational Status and Life Cycle Stage** that shows a summary of life-cycle value mismatches among CMDB, asset, and IBI and enables you to view mismatch details. See ['Crawl' stage reports on the CSDM Data Foundations dashboard](#).

View the contents of life-cycle tables

To view how life-cycle value pairs are mapped in the mapping tables, select **All**, and then enter `life_cycle_stage.list` in the search text box.

Migrating nonconforming CIs into the correct CSDM tables

If you already have CSDM data but not in the recommended tables, you can run a script to identify dependencies that you may have created on the nonconforming tables. The results of the script identify referenced dependencies such as reports, UI scripts, business rules, and other references for a specified table (editable in the script). See this [post on the Now Community site](#)

The life-cycle sync process presents the discrepancy report

When you start the sync process, the discrepancy report [Enabling life-cycle sync from legacy-to-asset](#) helps you to isolate and remediate mapping issues or missing data issues before proceeding with migration and syncing. The report identifies issues and presents links that take you to the mappings that must be resolved before you can sync the data.

Related concepts

- [Enabling life-cycle sync from legacy-to-asset](#)

Legacy status values that are auto-updated

Several legacy statuses are automatically mapped to the Life Cycle Stage and Life Cycle Stage Status fields.

Values that can be auto-updated

When you follow the procedure described in [Activate the CSDM plugin](#), the following legacy statuses are automatically mapped to the Life Cycle Stage and Life Cycle Stage Status fields:

Important: Legacy field values are not deleted after you map them to Life Cycle Stage and Life Cycle Stage Status values.

- Product Model Status
- Asset State
- Asset Substate
- Contract Status
- CI Install Status
- CI Operational Status
- CI Hardware Status (tangible/physical CIs)
- CI Hardware Substatus (tangible/physical CIs)

Note: Mappings are inherited from parent tables to child tables.

Implementing the CSDM framework in stages

Following the CSDM framework ensures that you meet your primary goal of consistent accuracy in reporting and analytics so you can effectively manage your digital environment.

The CSDM data framework

The CSDM is the data model that admins should follow when they set up ServiceNow products and applications. The standards for defining configuration items (CIs) and relationships between CIs in the CMDB ensure that your data resides in the appropriate CMDB tables. The result is maximum value from ServiceNow AI Platform applications.

Activating CSDM

Activate the CSDM plugin

Activate the CSDM plugin so you can begin implementing the CSDM data model.

Enabling life-cycle sync from legacy-to-asset

You can align life-cycle values for each product instance on the asset, CI, and IBI tables. A one-time process moves legacy status values for asset and CI across the platform to standard CSDM life-cycle value pairs (life cycle stage and life cycle stage status). Business rules then run regularly to ensure identical IBI, asset, and CI life-cycle data for each product instance.

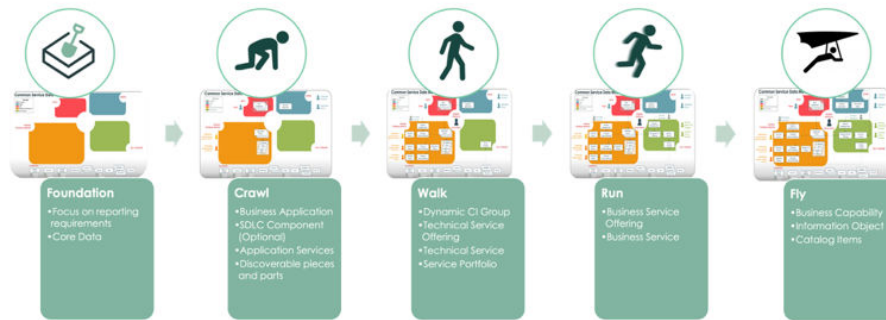
Migrate and sync existing data to the CSDM framework

You complete several tasks to ensure that your existing application data migrates successfully to the required tables in the CMDB.

Implementing the CSDM framework

It's best to use a staged approach when you implement the CSDM framework. Each implementation stage involves particular information types and provides specific benefits. Because each stage builds on the preceding stage, we use an analogy to the way a person develops:

foundation, crawl, walk, run, and, eventually, fly.



Note: Business applications reference information objects in the information portfolio. You might need to implement the Information Object [cmdb_ci_information_object] table earlier than the Fly stage. Your business requirements determine the right stage for implementing the table.

CSDM implementation stage — Foundation

In the Foundation stage of implementing the CSDM framework, you prepare the referential data that enables accurate reporting to support good business decisions. Use the base-system tables when you begin implementing the CSDM to derive the highest value from your ServiceNow products and the ServiceNow AI Platform.

CSDM implementation stage — Crawl

In the Crawl stage, you work on base-system CMDB tables that are associated with IT Service Management (ITSM).

CSDM implementation stage — Walk

In the Walk stage, you identify and populate the network infrastructure CIs and applications that your organization's technical teams support.

CSDM implementation stage — Run

In the Run stage, you set up the relationship between a technology and the business that sells and/or consumes the technology.

CSDM implementation stage — Fly

When you reach the Fly stage, you've accomplished all or most of the process of implementing the CSDM framework. The fly stage completes the process.

Key guidelines for you to follow

- When linking CSDM tables, use only the relationships that are designed in the model.
- Collaborate on the shared data model with other product teams. Also, when you extend CSDM and related functionality, be sure to follow the provided guidance. Following the guidance and collaborating with other product teams helps you achieve the best design.
- Use agreed-upon CSDM definitions whenever you use a table, reference, or attribute.
- Use the provided CSDM base tables.
- Use the recommended technologies when you integrate external data sources. The specified process ensures data integrity and integration consistency.
- Follow the provided guidance for setting up and using ServiceNow AI Platform products.

Improve your CSDM implementation

The CSDM Assessment provides Impact Customers with leading practices and prescriptive guidance on the CSDM and how it supports processes on the ServiceNow AI Platform. To help your organization plan for and implement CSDM, the assessment includes interactions with ServiceNow CSDM experts and personalized content. See [Common Service Data Model \(CSDM\) Assessment Accelerators](#).

- [Enabling life-cycle sync from legacy-to-asset](#)

You can align life-cycle values for each product instance on the asset, CI, and IBI tables. A one-time process moves legacy status values for asset and CI across the platform to standard CSDM life-cycle value pairs (life cycle stage and life cycle stage status). Business rules then run regularly to ensure identical IBI, asset, and CI life-cycle data for each product instance.

- Matching the usage of dynamic CI groups to service type

The type of service determines how you use dynamic CI groups.

- CSDM implementation stage — Foundation

In the Foundation stage of implementing the CSDM framework, you prepare the referential data that enables accurate reporting to support good business decisions. Use the base-system tables when you begin implementing the CSDM to derive the highest value from your ServiceNow products and the ServiceNow AI Platform.

- CSDM implementation stage — Crawl

In the Crawl stage, you work on base-system CMDB tables that are associated with IT Service Management (ITSM).

- CSDM implementation stage — Walk

In the Walk stage, you identify and populate the network infrastructure CIs and applications that your organization's technical teams support.

- CSDM implementation stage — Run

In the Run stage, you set up the relationship between a technology and the business that sells and/or consumes the technology.

- CSDM implementation stage — Fly

When you reach the Fly stage, you've accomplished all or most of the process of implementing the CSDM framework. The fly stage completes the process.

Enabling life-cycle sync from legacy-to-asset

You can align life-cycle values for each product instance on the asset, CI, and IBI tables. A one-time process moves legacy status values for asset and CI across the platform to standard CSDM life-cycle value pairs (life cycle stage and life cycle stage status). Business rules then run regularly to ensure identical IBI, asset, and CI life-cycle data for each product instance.

This topic provides an overview of the entire align-and-sync process.

- Detailed instructions for running the align-and-sync process appears in [Migrate legacy asset and CI settings to CSDM life-cycle values](#).
- A detailed reference to all sync options appears in [How life-cycle values for Asset, CI, and IBI are synced](#).

Tip: Terms used in this document are defined in [CSDM life-cycle terms](#).

Why enable CSDM life-cycle sync between CI and asset entities

When the one-time data sync operation finishes, CSDM life-cycle values for each product instance as represented in the asset and CI tables will be identical. Business rules then sync life-cycle value pairs among the asset, CI, and IBI tables. The business rules then run on regular schedules to ensure that life-cycle values for the asset, CI, and IBI tables remain aligned.

See [How life-cycle values for Asset, CI, and IBI are synced](#).

The following products, for example, benefit from the standardized values:

- To handle alerts appropriately, the Event Management and Operational Intelligence features need to know whether a CI is in the maintenance life cycle stage status.
- To report cost data effectively, the Cloud Cost Management feature needs to know the CSDM life-cycle values of a CI.
- To generate consistent tasks and workflows, the Audit Management and Compliance features need to use standard CSDM life-cycle values.

How the mapping between legacy status values and CSDM CSDM life-cycle values is specified

A life-cycle value pair is the combination life cycle stage and life cycle stage status values for a CI, asset, or IBI over the life cycle of a product instance. The base system includes the life-cycle mapping [life_cycle_mapping] table that holds mapping rules. The default rules specify how to align common legacy status values to a CSDM life-cycle

value pair. You can update and add rules as needed. (The table is described more fully in [Life cycle mapping table](#).)

Life cycle mapping
cmdb_ci_hardware

* Mapping for table: Hardware [cmdb_ci_hardware] Application:

* Priority: 10 Active: ☒

Fields

Legacy value		CSDM life cycle values	
* Legacy field name	Install Status	Life cycle control	Hardware - Dep
* Legacy field value	Pending Install	Table	Hardware
Legacy subfield name		Life Cycle Stage	Deploy
Legacy subfield value		Life Cycle Stage Status	Test

Note: If you are activating the CSDM (csdm.lifecycle.migration.activated) plugin for the first time, and you have custom states and substates for assets or CIs, then you will create and configure mapping between the custom legacy values and the CSDM life-cycle values.

Alignment/sync process

To start the align and sync process, you select **Enable life cycle sync** on the Life cycle mapping table list page. That action sets the csdm.lifecycle.sync.between.ci.and.asset.activated system property to true and starts the following activities:

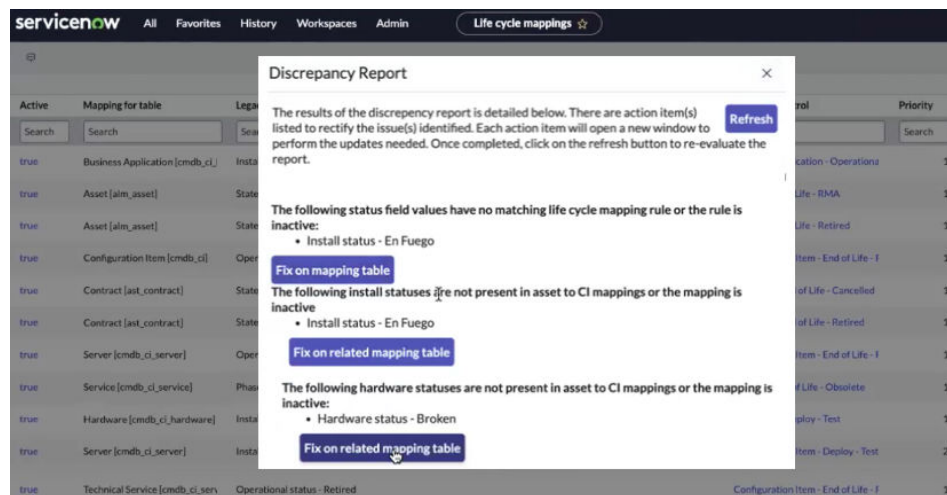
Phase 1: Run the discrepancy check and correct all mappings

The system runs the discrepancy check

The discrepancy check script checks all mapping rules:

- Verify CSDM life-cycle mapping rules for asset tables.
- Verify CSDM life-cycle mapping rules for CMDB CI tables.
- Verify legacy asset-to-CI mapping rules that appear in the alm_asset_state_mapping and alm_asset_ci_state_mapping tables.

Note: For each CI that is not associated with an asset (non-asset CI), use CSDM mapping tables to derive the life-cycle value pair for the CI from the legacy status.



The script generates a discrepancy report. The report displays the list of missing or inactive mappings that you need to repair before you can perform bulk alignment of life-cycle values.

- Legacy status values that aren't yet mapped to life-cycle value pairs.
- Legacy status values that aren't included in a mapping rule or that are included in an inactive rule.
- Custom legacy status values. For each custom value, the system adds a record to the life_cycle_mapping table. Those mapping records, however, are inactive because some required field values are not yet set. You will edit and activate those records so all custom legacy status values will be aligned.

- After the one-time alignment, legacy status values between CI and asset will be auto-synced. The report therefore identifies hardware statuses or install statuses that are not present or that are included in an inactive mapping rule between asset and CI.

Manual process: If there are mapping errors, add or update mappings

Add missing mapping rules and ensure that current rules are correct and active. The buttons open mapping tables where you can update mapping rules. Instructions for updating individual rules appear in [Map legacy status values to CSDM life-cycle values](#).

Run the discrepancy check again

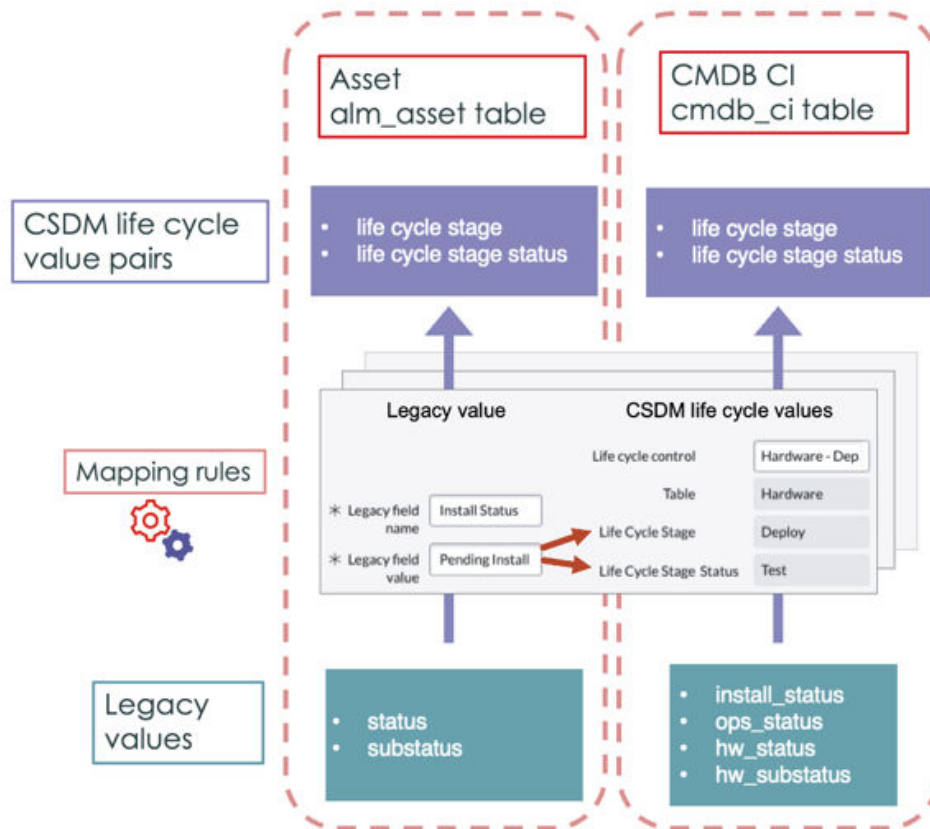
Run the discrepancy check and fix errors until all errors are fixed.

Phase 2: Automated bulk alignment: Align all legacy CMDB CI and asset data to CSDM life-cycle value pairs

When all mapping errors are fixed, a scheduled job in the batch process performs one-time-only bulk alignment to life cycle stage and life cycle stage status life-cycle value pairs.

During bulk alignment, the process uses the mapping rules to derive the CSDM life-cycle value pair from legacy status values for both asset and CMDB CI records. During the process, the **Activate** button is disabled and you are blocked from updating mapping rules.

Note: IBI records do not change. Active sync between legacy and life-cycle values for IBI records is already in place.



Step 1: For all assets that do not have life-cycle value pairs, generate the values

Use the CSDM life-cycle mapping rules for asset tables to derive the life-cycle value pairs from the legacy status values.

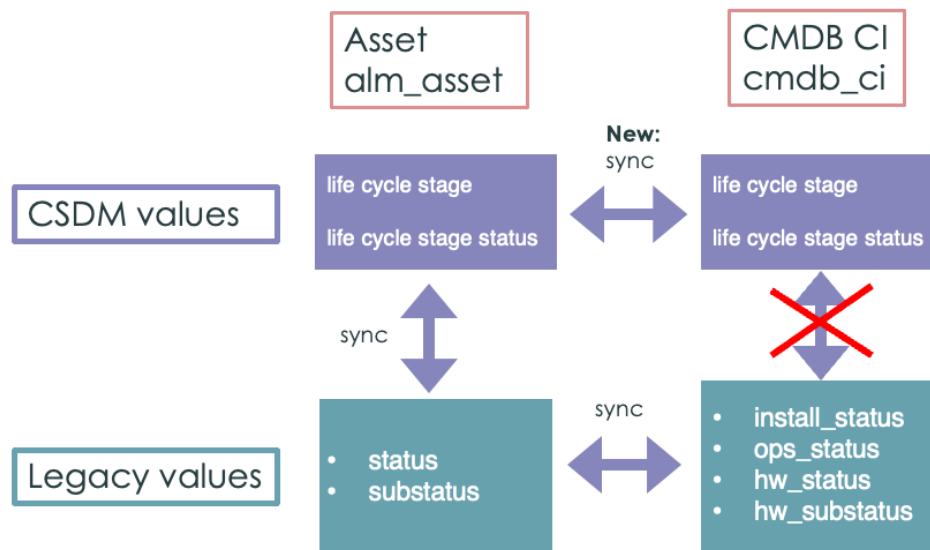
Step 2: One-time sync: For each CI whose CSDM life-cycle values do not match the asset values, update the CI with the asset values

This one-time sync will not be repeated because in one of the following phases, bi-directional sync is implemented between CI life-cycle value pairs and asset life-cycle value pairs.

Step 3: One-time sync: If a CI is not associated with an asset, use CSDM mapping tables to derive the CI's life-cycle values from the legacy status.

This one-time sync will not be repeated because bi-directional sync continues between CI legacy values and asset legacy values. Future CI values will derive from the new bi-directional sync between asset and CI life-cycle value pairs.

Phase 3: Automated update to sync operations



New: Activate sync of life-cycle data between asset and CMDB CI

CSDM life-cycle value pairs are bi-directionally synced between asset and CI.

New: Deactivate Bi-directional sync between CMDB CI legacy values and CSDM life-cycle values

As a result, changes to CI legacy values will not update CSDM life-cycle value pairs (the values will diverge over time). Instead, CSDM life-cycle value pairs are updated by the bi-directionally sync with asset life-cycle value pairs. A report on the CSDM dashboard lists CIs whose life-cycle values diverge from legacy values.

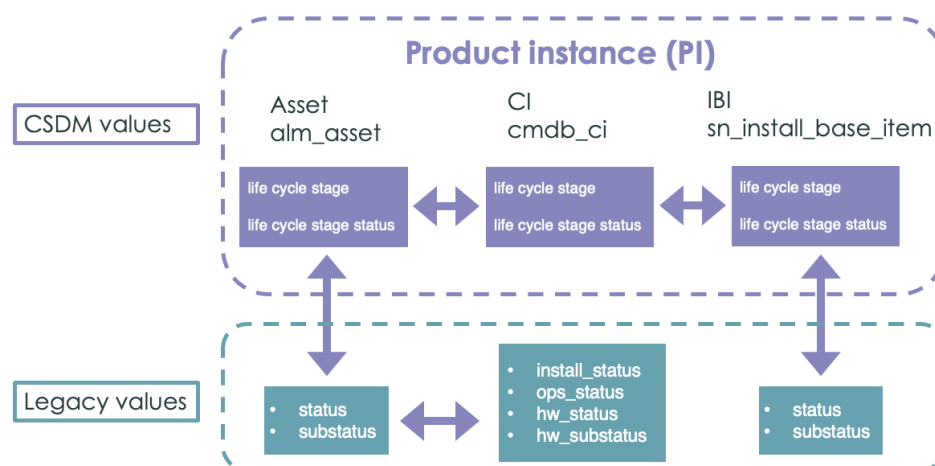
No change: Sync between asset legacy values and CI legacy values

The existing bi-directional sync between asset legacy values and CI legacy values continues.

(No change if sync is currently active): Activate sync between asset legacy and asset life-cycle values.

Bi-directional sync between asset legacy values and life-cycle value pairs continues.

Phase 4: Ongoing automated sync operations



Immediately after sync operations are completed, business rules run to align life-cycle value pairs in the asset, CI, and IBI tables for each product instance. Business rules then run on a regular schedule to incorporate changes to legacy status values as updates to life-cycle value pairs and to maintain alignment among Asset, CI, and IBI.

Activate life-cycle alignment and sync

Instructions [Migrate legacy asset and CI settings to CSDM life-cycle values](#)

Map legacy status values to CSDM life-cycle values

Use the Life Cycle Mapping module to specify how existing legacy status values should be converted to CSDM life-cycle value pairs (life cycle

stage and life cycle stage status). You map both asset and CI legacy status values to life-cycle value pairs.

Before you begin

Role required: `itil_admin` or `asset_admin`

About this task

Several legacy statuses are automatically mapped to the Life Cycle Stage and Life Cycle Stage Status fields. For more information, see [Legacy status values that are auto-updated](#).

Because the base system includes a large number of default mappings, you might not have to create many custom mappings. To view the list of default mappings, navigate to **All > CSDM > Life Cycle Mapping**.

Tip: Terms used in this document are defined in [CSDM life-cycle terms](#).

In the following example, your existing data uses a status attribute named **Install Status** for hardware CIs. You configure the Life cycle mapping form to map the existing **Pending Install** value of the **Install Status** attribute to the **Deploy/Test** life-cycle value pair in the CMDB.

Life cycle mapping
cmdb_ci_hardware

* Mapping for table: Hardware [cmdb_ci_hardware] Application:

* Priority: Active: ☒

Fields

* Legacy field name: Install Status Life cycle control: Hardware - Dep

* Legacy field value: Pending Install Table: Hardware

Legacy subfield name: Life Cycle Stage: Deploy

Legacy subfield value: Life Cycle Stage Status: Test

Procedure

1. Navigate to **All > CSDM > Life Cycle Mapping**.
2. On the Life cycle mappings list view, select **New** and then fill in the Life cycle mapping form as described in [Life cycle mapping form](#).
3. Select **Submit**.
4. Repeat the process and, when you have mapped all legacy status values, migrate them to the CMDB.
For instructions, see [Migrate legacy asset and CI settings to CSDM life-cycle values](#).

Migrate legacy asset and CI settings to CSDM life-cycle values

Activate life-cycle sync to migrate your legacy asset and CI status settings to the CSDM standard life-cycle value pairs. After the one-time migration process, business rules maintain sync among IBI, asset, and CI life-cycle values.

Before you begin

Tip: Terms used in this document are defined in [CSDM life-cycle terms](#).

Before you enable life-cycle sync, navigate to **CSDM > Life Cycle Mapping**. Review the prepopulated mappings in the **Life Cycle Mappings** list view:

- Adjust and add any mappings as needed for your environment.
- Review mappings for any custom status values. Those mappings are incomplete and you must provide the desired standard life cycle control to map to.
- Ensure that all mappings are configured with a life cycle control.
- Ensure that all mappings are activated.

Activation can proceed only if the following conditions apply:

- All mapping records are set to active and are configured with a life cycle control
- All mapping records for custom legacy values are fully configured.

Role required: itil_admin or asset_admin

Procedure

1. Navigate to **CSDM > Life Cycle Mapping**.
2. On the **Life Cycle Mappings** list view, select **Enable life cycle sync** on the Life cycle mapping table list page.
That action sets the `csdm.lifecycle.sync.between.ci.and.asset.activated` system property to `true` and runs a script that validates mappings. The script generates a discrepancy report. The report displays the list of missing or inactive mappings that you need to repair before you can perform bulk alignment of life-cycle values.
3. Select an item on the report to access the mappings that need attention, correct the mappings, and then select **Refresh** to run the report again.

Result

One-time sync is done. All cmdb_ci and asset records should have life cycle stage and life cycle stage status values, either coming directly from what the asset had (if you have an asset/CI combo), or from the life cycle mapping table directly for non-asset CIs.

Business rules now perform ongoing alignment of life-cycle data as described in [Enabling life-cycle sync from legacy-to-asset](#).

What to do next

After the data has migrated successfully, you can start managing data following the CSDM model:

1. [Activate the CSDM Activation \(com.snc.cmdb.csdm.activation\) plugin](#).
2. Use the [CMDB Data Manager](#) to centrally govern the life cycle of CIs in bulk and in a standard and consistent way.

Matching the usage of dynamic CI groups to service type

The type of service determines how you use dynamic CI groups.

Dynamic CI groups

- A dynamic CI group is a user-defined collection of CIs that is used as a group CI in the CI field of the record for Incident Management, Problem Management, and Change Management.
- You set up dynamic CI groups based on common attributes or criteria.
- A dynamic CI group cannot contain other groups.

Dynamic CI groups for technology management services (formerly Technical services)

You can group up to 10,000 similar items that are associated with a technology management service and its associated technology management offering. In this usage, a dynamic CI group functions as a collection of CIs that are managed individually. Because the CIs aren't

necessarily related to each other, service associations are not needed. Impact analysis (through Incident Management, Problem Management, or Change Management) is also not meaningful or expected for unrelated CIs. Here are some examples:

- Servers with a particular operating system
- Network gear based on location
- Application servers based on a running process
- CIs that make up a service instance
- CIs that share a particular support group

The Service Classification attribute identifies the service as a technology management service. The ownership sync process assigns CI ownership through the technology management service, as described in [Synchronizing group assignment attributes](#).

Dynamic CI groups for service instances

For a service instance (called application service before CSDM v5), impact analysis outcomes are expected because when any individual CI is impacted, the entire dynamic CI group is affected. All related business services or technology management services are also affected. Because of this interdependence, you must establish service associations and impact analysis is important.

The Service Classification attribute identifies the service as a service instance.

See [Use the Dynamic CI Group method to populate application services](#).

- [Synchronizing group assignment attributes](#)

To empower a particular user group to manage a collection of CIs or CI classes, set group assignment attributes through the Technology management offering or the CI Class Manager. The operation synchronizes the group attribute data across all CIs that belong to the specified CI class or groups of CIs.

- [Set the group for a CI or an entire class of CIs](#)

Synchronize group assignment attributes on entire CI classes and individual CIs using the CI Class Manager.

- [Synchronize user groups for a technology management offering](#)

Synchronize group assignment attributes on entire CI classes and individual CIs by creating a dynamic CI group for use by a technology management offering.

Synchronizing group assignment attributes

To empower a particular user group to manage a collection of CIs or CI classes, set group assignment attributes through the Technology management offering or the CI Class Manager. The operation synchronizes the group attribute data across all CIs that belong to the specified CI class or groups of CIs.

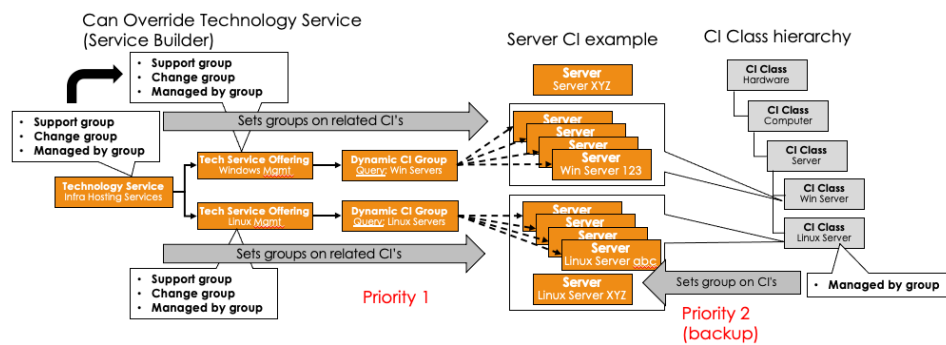
Methods for synchronizing group assignment attributes

Note: The Managed by Group setting identifies the group that manages a CI class (ensuring that it is complete and correct). It might or might not be the same group as repairs an individual CI.

- [Synchronize user groups for a technology management offering:](#) Directly set the **Support group**, **Change group**, or **Managed by group** attributes in a technology management offering. The settings are applied to all CIs that are associated with the technology management offering.
- [Set the group for a CI or an entire class of CIs:](#) Set the **Managed by group** attribute for a specific class in the CI Class Manager. All CIs within the class will have their **Managed by group** field populated based on the value specified in the CI Class Manager. With this method, the **Managed by group** setting is applied only to the CIs that aren't associated with a technology management offering. For CIs that are managed by a technology management offering, the **Managed by group** field is first synchronized with its dynamic CI group. The field is then synchronized with the CIs in the dynamic CI group, overwriting the entry from the CI Class Manager. For instructions, see [Synchronize user groups for a technology management offering](#).
- **Use Support Group and Change Group:** By using dynamic CI groups, data synchronization enables you to manage data that cannot be

discovered. The values in the **Support Group** and **Change Group** (previously labeled **Assignment Group**) fields in the cmdb_ci table are synchronized with their related dynamic CI groups and with all the CIs that are contained as part of that dynamic CI group object.

Team data synchronization



Related tasks

- [Synchronize user groups for a technology management offering](#)

Related reference

- [CI Class Manager](#)

Set the group for a CI or an entire class of CIs

Synchronize group assignment attributes on entire CI classes and individual CIs using the CI Class Manager.

Before you begin

Role required: itil and itil_admin

About this task

Note: The Managed by Group setting identifies the group that manages a CI class (ensuring that it is complete and correct). It might or might not be the same group as repairs an individual CI.

You can streamline approval of CMDB Data Manager policies by populating the Manage by Group attribute of CIs that you plan to target in policies. Use the CI Class Manager to populate the attribute for an entire class in a single synchronization operation. If the Managed by Group attribute isn't populated for a CI, then the approval process is directed to the administrator.

Set the **Managed by group** attribute for a specific class in the CI Class Manager. All CIs within the class will have their **Managed by group** field populated based on the value specified in the CI Class Manager. With this method, the **Managed by group** setting is applied only to the CIs that aren't associated with a technology management offering. For CIs that are managed by a technology management offering, the **Managed by group** field is first synchronized with its dynamic CI group. The field is then synchronized with the CIs in the dynamic CI group, overwriting the entry from the CI Class Manager.

Procedure

1. Navigate to **All > Configuration > CI Class Manager**.
2. Select **Hierarchy** to expand the CI Classes list, select a class to display details for, and then select **Basic Info**.
3. Specify the value in the **Managed by Group** attribute and select **Save**.
4. To verify the change, select **CI List**.
The change should be applied to the **Managed by Group** attribute of all CIs of the class unless they are associated with a technology management offering. The change is applied only to the CIs in the class and not to the sub classes under a CI.

Example

In this example, you define a user group that can manage all CIs belonging to the **Linux Server** class:

1. Navigate to the CI Class Manager and select the **Linux Server** class in the list.
2. Select **Basic Info** and, in the **Managed By Group** attribute, select **sys_user_group** and then select **Save**.

3. To verify that the attribute was updated, select **CI List** and navigate to the Linux Server class. You will see that the **Managed By Group** attribute has been updated to **sys_user_group**.

Related tasks

- [Synchronize user groups for a technology management offering](#)

Related concepts

- [Synchronizing group assignment attributes](#)

Synchronize user groups for a technology management offering

Synchronize group assignment attributes on entire CI classes and individual CIs by creating a dynamic CI group for use by a technology management offering.

Before you begin

Role required: itil and itil_admin

About this task

Directly set the **Support group**, **Change group**, or **Managed by group** attributes in a technology management offering. The settings are applied to all CIs that are associated with the technology management offering.

In addition to the method described in this procedure, you can use the CI Class Manager to set any CI property value. If the **Support group**, **Change group**, or **Managed by group** value is set in the technology management offering, the value specified by the technology management offering takes precedence.

Note: The Managed by Group setting identifies the group that manages a CI class (ensuring that it is complete and correct). It might or might not be the same group as repairs an individual CI.

Procedure

1. Navigate to **All > Configuration > CMDB Groups** and create a new CMDB group.
See [CMDB groups](#) for details.
2. Navigate to **All > Configuration > Dynamic CI Groups**.
3. Create a new dynamic CI group and associate it with the CMDB group that you created.

See [Service Delivery domain in the CSDM model](#) for more information on dynamic CI groups.

4. Navigate to **All > CSDM > Technical Service Offering** and create a new technology management offering.
See [Service Delivery domain in the CSDM model](#) for more information on technology management offerings.

5. Navigate to the **CI Relationships** table, select **New**, and then enter the following values:

- **Parent:** Select the technology management offering that you created.
- **Child:** Select the dynamic CI group that you created.

6. Select **Submit**.

You have created a relationship between the technology management offering and the dynamic CI group.

7. Navigate to **All > CSDM > Technical Service Offering** and open the technology management offering that you created.

8. Enter values in the following fields:

- Support group
- Change group
- Managed by group

The screenshot shows the 'Offering' form in ServiceNow. The 'Name' field is 'tso1'. The 'Support group' field is highlighted with a red box and contains the value 'Database San Diego'. Other fields include 'Model ID', 'Owned by', 'Parent' (SAP Materials Management), 'Used for' (Production), 'Operational status' (Operational), 'Version', 'Approval group', 'Change group' (Hardware), 'Managed by group' (Help Desk), 'Managed by', 'Business criticality' (2 - somewhat critical), 'SLA', and 'Location'. The form also has sections for 'Short Description', 'Description', and 'Comments'.

9. Right-click the header and select **Save**.
A message confirms that data synchronization has been enabled for the fields.
10. The values that you specified are applied to the related dynamic CI group and all associated CIs.

Note: If a new CI is added to the class, the data will be synchronized only after the scheduled CSDM Data Sync job is completed. If you need to synchronize the data immediately, navigate to **Scheduled Jobs > CSDM Data Sync** and select **Execute**.

CSDM implementation stage — Foundation

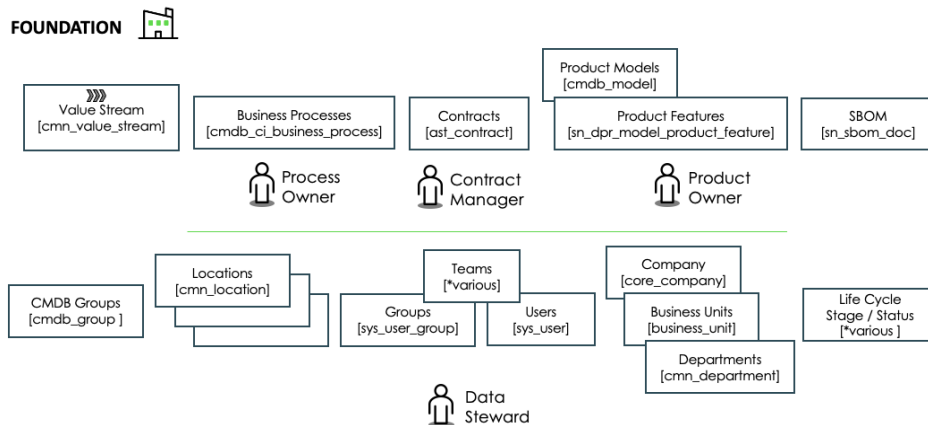
In the Foundation stage of implementing the CSDM framework, you prepare the referential data that enables accurate reporting to support good business decisions. Use the base-system tables when you begin implementing the CSDM to derive the highest value from your ServiceNow products and the ServiceNow AI Platform.

Benefits of preparing the data in the Foundation stage

The basis of any good data model is the foundational data that is referenced throughout the model.

- The base-system tables in the CMDB act as the foundation for many ServiceNow AI Platform products.
- The tables help your organization align with reporting requirements early to expedite the value you get from ServiceNow AI Platform products. You can reduce or eliminate costly rework tasks needed to align with reporting requirements.

Tables that you work on during the Foundation stage



Business Process [cmdb_ci_business_process] table

A business process has a well-defined start and finish. Examples of business processes in the banking industry are the customer onboarding process and the credit check process. Each business process can have levels of criticality and impact. Business processes are stored in the cmdb_ci_business_process table.

Contract [ast_contract] table

The Contract table identifies binding agreements between two parties. When you populate services provided by vendors into the CMDB, consider the role that contracts play when evaluating service level agreements (SLAs).

Product model [model_id] table

The Product model [model_id] table identifies the unique types of products your organization develops or consumes. When you group assets and CIs by product model, you unify and relate CIs that are part of the same digital product and portfolios of products. Grouping assets and CIs by product model can help you plan projects, monitor costs, and rationalize your data. Discovery can populate tangible/physical product models after they're operational, but other types of product models require planning from product owners.

Use the CSDM Product Model Assignment job to auto-generate a product model record (application model, service model, or software model) for each logical CI that is not yet associated with a product model. Product models are ideal for associating CIs that are parts of a single digital product. See [Auto-generate product models for logical CIs](#).

CMDB Group [cmdb_ci_query_based_service] table

The CMDB Group table identifies a collection of CIs based on the results of saved Query Builder queries, encoded queries, or manual entries.

CMDB groups are critical elements of dynamic CI groups and the strategic management of CIs. Decide early how you want to report CI information and how you want to monitor CIs. These decisions affect how you create CMDB groups. For Change and Incident processes, there are two distinct impact analysis behaviors for dynamic CI groups. See [Matching the usage of dynamic CI groups to service type](#).

Software bills of material (SBOM)

An SBOM is the overall collection of components of a piece of software. Cyclone DX is supported.

Location [cmn_location] table

The Location table uniquely identifies geographic locations. You can create a hierarchy of location data using the Parent attribute. The hierarchy might include entries that match your reporting requirements. For example, you could populate the location table as follows
Your organization's location attributes



To include more detail in reports, you could extend the Location table to include floors, rooms, and even datacenters. With hierarchy capabilities, trusted source data, and your requirements in hand, you can create locations that support your future reporting needs.

Group [sys_user_group] table

The Group table identifies sets of users that share a common purpose. Groups may perform tasks such as approving change requests, resolving incidents, receiving email notifications, or performing work order tasks. Groups also use the referential data in the CMDB to identify how CIs are managed (for example, the Managed by Group) and supported (for example, the Support Group). Any business rules, assignment rules, system roles, or attributes that refer to a group automatically apply to all group members.

Note: The Managed by Group setting identifies the group that manages a CI class (ensuring that it is complete and correct). It might or might not be the same group as repairs an individual CI.

User [sys_user] table

The User table identifies the persons and applications that have access to your ServiceNow instance. You can organize users into groups that are associated with the Company, Business Unit, and Department tables.

Organizational structure

Organization structure tables identify internal business structures and external customers, manufacturers, and vendors.

Company [core_company] table

The Company table is populated with the legal entities of companies. Entities can be either internal (your organization) or external. You can use the Parent attribute to build a hierarchy. Consider the legal entities that you need for reporting when the CMDB is populated.

- Internal entries should focus on a hierarchy of legal entities rather than a hierarchy of business units within a legal entity.
- External entries are identified by a True or False flag. The Customer flag identifies your external customers.

The Manufacturer flag identifies companies that create products that you consume. An internal organization might be a manufacturer.

- The Vendor flag identifies organizations that provide products that you purchase. An internal organization might be a vendor.

Business Unit [business_unit] table

The hierarchy of your business is populated in the Business Unit table with a reference to the parent company. A business unit is a part of your organization that is responsible for specific operations, such as finance, human resources (HR), or IT. A hierarchy within a business unit is common. For large multinational organizations, you might have business units that identify independent regional operations and the specific operations within the region.

Department [cmn_department] table

The Department table includes a finer level of detail about a business unit. The Department table gives you another way to categorize users, groups, assets, and CIs.

Life cycle tables

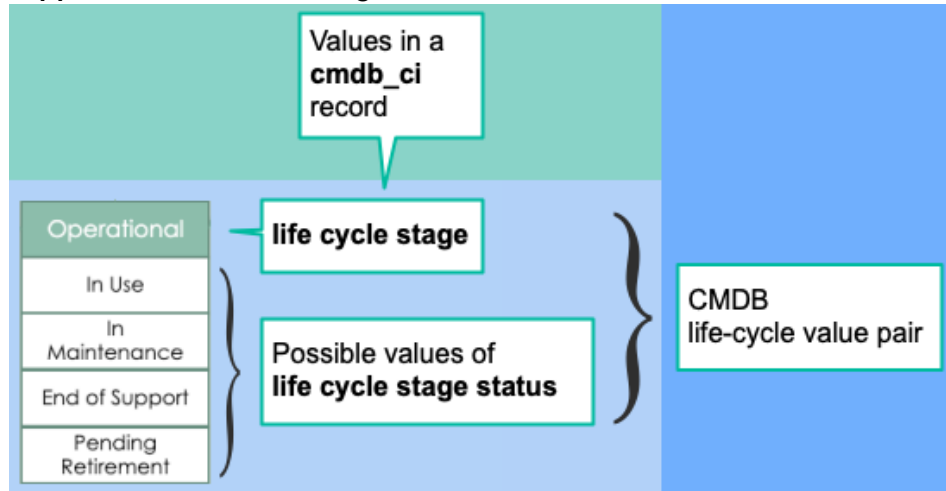
CSDM life-cycle value pairs track the life cycles for products, assets, contracts, CIs, locations, and other objects. Using the standard values consistently helps you to track objects through their transitions over time. Reporting can therefore accurately reflect the actual states of CIs: usage, availability, end of support, and so on.

The standard CSDM life-cycle value pair covers all phases of a product instance life cycle.

- A life cycle stage is one of the broad phases that a CI moves through, for example from inception or procurement to operation and then perhaps to end of life.
- life cycle stage status is the particular status of a CI within its current life cycle stage.

For example, a tangible/physical CI in the **Operational** stage might change stage status over time from **In Use** to **In Maintenance** to **End of Support**. A different tangible/physical CI might go from **In Use** to **End of**

Support without ever having been in **In Maintenance** status.



Note: The [life_cycle_control] table uses the type of CI (tangible/physical, document and contract, location and so on) to determine which life cycle stage status values are available for each life cycle stage.

To take full advantage of the CSDM life-cycle standards, you can map legacy status data to the life-cycle value pairs. See [Enabling life-cycle sync from legacy-to-asset](#).

The following assets can use life-cycle value pairs:

- Life cycle of product entities
- Life cycle of tangible/physical CIs
- Life cycle of intangible/logical entities
- Life cycle of document and contract entities
- Life cycle of location entities

Watch the ServiceNow Community video: [CSDM V4 product and life cycle discussion](#)

CSDM implementation stage — Crawl

In the Crawl stage, you work on base-system CMDB tables that are associated with IT Service Management (ITSM).

Benefits of the operations that you perform in the Crawl stage

- The operations provide the minimum CMDB support requirements for Incident Management and Change Management.
- Setting up APM is faster because your business application data is in the right place in the CMDB.
- The operations build the foundation for using DevOps because your SDLC component data is populated and ready to relate to your applications.
- Service Mapping is ready to use for mapping entry points because your service instance data is populated.
- The operations build the foundation for using TPM risk details, a capability of APM.

The operations prepare you to manage and monitor the life cycles and versions of the underlying technologies of the business applications in your enterprise.

The data enables you to identify outdated or at-risk software using APM, Service Mapping and Software Asset Management (SAM) Professional.

Tables that you work on during the Crawl stage

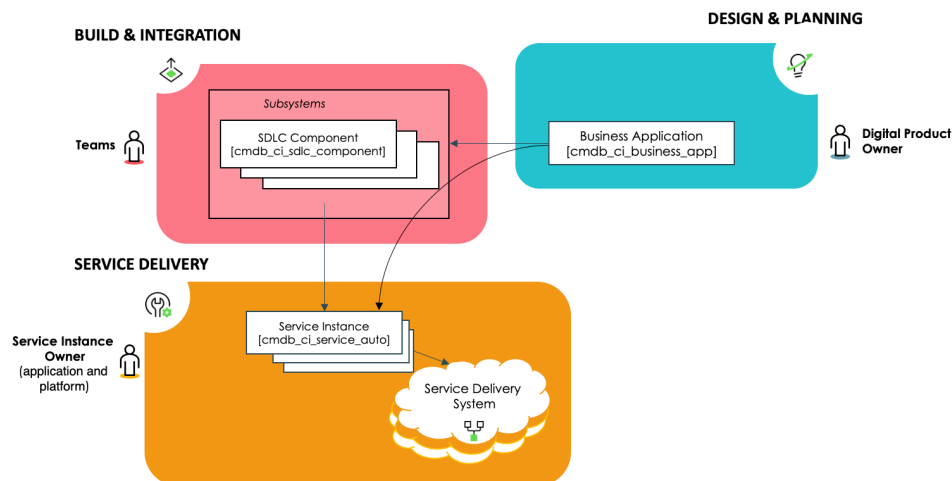
Important: Future products and product enhancements depend on the data that you prepare in each of the tables.

During this stage, you work on the following base-system CMDB tables:

- Business Application [cmdb_ci_business_app] table
- Mapped Application Service [cmdb_ci_service_discovered] table.
- Application [cmdb_ci_appl] table (discoverable)

- Server/host (discoverable)

Note: Some of the classes that you implement in this stage are logical CIs. Logical CIs aren't created through Discovery, so their **Model ID** values might not refer to product model (application model, service model, or software model) records. To help you to migrate to a product-centric management paradigm, each instance of a logical CI should be associated with a product model. See [Auto-generate product models for logical CIs](#).



Start by focusing on applications and the application-related data in these areas and tables:

Business Application [cmdb_ci_business_app] table

A business application is a base-system CMDB table that stores your inventory, application portfolio, and their metadata.

Note: Because this table holds conceptual data, not operational CIs, it is not used by ITSM Incident Management, Problem Management, or Change Management processes.

SDLC Component [cmdb_ci_sdlc_component] table

An SDLC component is a CI that represents a unique development effort of code. It represents parts of a larger business application or digital product broken down into its individually developed components. In other words, the SDLC component is a software element of a larger application or technology. SDLC component CI records in the SDLC Component [cmdb_ci_sdlc_component] table enable the DevOps product to provide enhanced capabilities for visualizing and managing your application development pipeline.

This table represents the software part or element of a larger whole for applications and infrastructure. Related material may serve as representative of developmental details. It can be used if you need to identify the stratification of a business application or digital product.

Note: Because this table holds logical data, not operational CIs, it is not used by ITSM Incident Management, Problem Management, or Change Management processes.

Service Instance (called application service before CSDM v5) [cmdb_ci_service_auto] table

The service instance is typically the system that a caller identifies when they report an issue with an application.

A mapped service instance is a base-system CMDB table that identifies the related business application in use. The service instance ties all the elements of the CSDM together where applications are present.

You might have several service instances representing each deployment based on the environment (development, QA, production) and location or geography (North America, Asia Pacific).

Because service instances are logical in nature, they should use the Logical life-cycle value pairs. Service instances follow the same life-cycle guidance as any other logical CI.

See [Use Application Services dashboard to monitor health](#).

Application [cmdb_ci_appl] table

An application is a base-system CMDB table that represents the discoverable instance of an application: code related to a process in

use on a host. This table isn't an inventory of your applications. Because of the high level of complexity involved, don't try to manually populate the application table. Discovery creates and maintains this table.

Important: The application table [cmdb_ci_appl] isn't an inventory or portfolio of your applications. Don't make the mistake of storing managed application details in the application table. Those details (inventory or application portfolio objects) belong in the business application table (as documented in [Design & Planning domain in the CSDM model](#)).

The application might be identified as the root cause of an incident. However, if you're not using Event Management, the application might not be the initial cause.

If you're using Discovery, applications are automatically related to their host, which provides an impact hierarchy from server-to-host applications.

CSDM implementation stage — Walk

In the Walk stage, you identify and populate the network infrastructure CIs and applications that your organization's technical teams support.

Benefits of the operations that you perform in the Walk stage

Managing discovered infrastructure CIs

The operations facilitate managing the discovered infrastructure CIs. You might manually managing the metadata on these CIs, such as support group and technical approval group. By identifying the technology management offering that manages the CIs, you can perform the following operations:

- Configure the instance to populate and synchronize this metadata onto the related child objects.
- Eliminate the manual effort of maintaining the metadata on thousands of CIs.

View supported CIs

The operations establish a view of the CIs that your organization's technical teams support.

You can see the specific support assignments, which you can change as needed based on your support structure, operational-level agreements (OLAs), and commitments.

Also, this view enables you to formalize for your process for supporting applications and technology owners.

Prepare for Service Portfolio Management

The operations build the foundation for using Service Portfolio Management (Service Portfolio Management).

You can start using Service Portfolio Management more quickly because your service data are in the right place.

Use the request catalog

The operations enable you to order technology service offerings through the request catalog. You can also automate ordering some offerings to enhance the request workflow and update or create related CIs. Catalogs are described in detail in [Service Catalog](#).

Note: The request catalog is not a CMDB table.

Prepare for ITOM products

The operations build the foundation for Information Technology Operations Management (ITOM) products, such as Service Mapping and Discovery.

Enable Automation

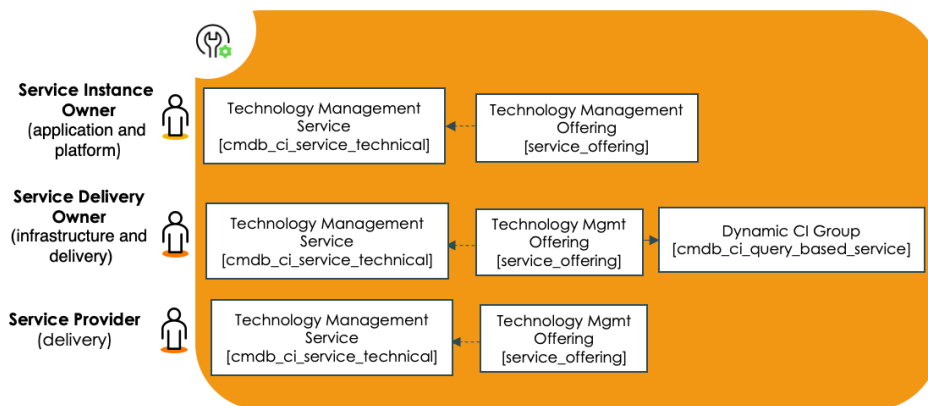
The operations enable more automated methods of grouping CIs for identification and management by Technology management offerings.

Tables that you work on during the Walk stage

The walk stage includes base-system CMDB tables that identify the technology provider.

Note: Some of the classes that you implement in this stage are logical CIs. Logical CIs aren't created through Discovery, so their **Model ID** values might not refer to product model (application model, service model, or software model) records. To help you to migrate to a product-centric management paradigm, each instance of a logical CI should be associated with a product model. See [Auto-generate product models for logical CIs](#).

SERVICE DELIVERY



Technology management service [cmdb_ci_service_technical] table (formerly Technical service), or [cmdb_query_based_services] for Event Management

The cmdb_ci_service_technical table has a service classification of "technical service". This base-system CMDB table identifies the provider of the technology that your business consumes.

Technology management offering [service_offering] table

A technology management offering is a service offering with a service classification of "technical service". Technology management offerings may be further divided as follows:

- Location and geography
- Environment (production or non-production)
- Pricing

- Availability
- Support group (for Incident Management)
- Technical approval group (for Change Management)
- Packaging options (commitments)

The technology management offering comes from the service, based on how the parent serves a specific technical need. Every operational technology management service must be associated with at least one technology management offering.

Note: Not all technology management offerings have to be related to applications or infrastructure CIs. Managed Service Providers can provide technology management offerings.

Dynamic CI group [cmdb_ci_query_based_service] table

A dynamic CI group is a collection of CIs based on the results of saved Query Builder queries, encoded queries, or manual entries. Query Builder is described in [Exploring CMDB Query Builder](#). For more information about dynamic CI groups and how you can use them, see [Manage Portfolio domain in the CSDM model](#).

CSDM implementation stage — Run

In the Run stage, you set up the relationship between a technology and the business that sells and/or consumes the technology.

ITSM considerations during the Run stage

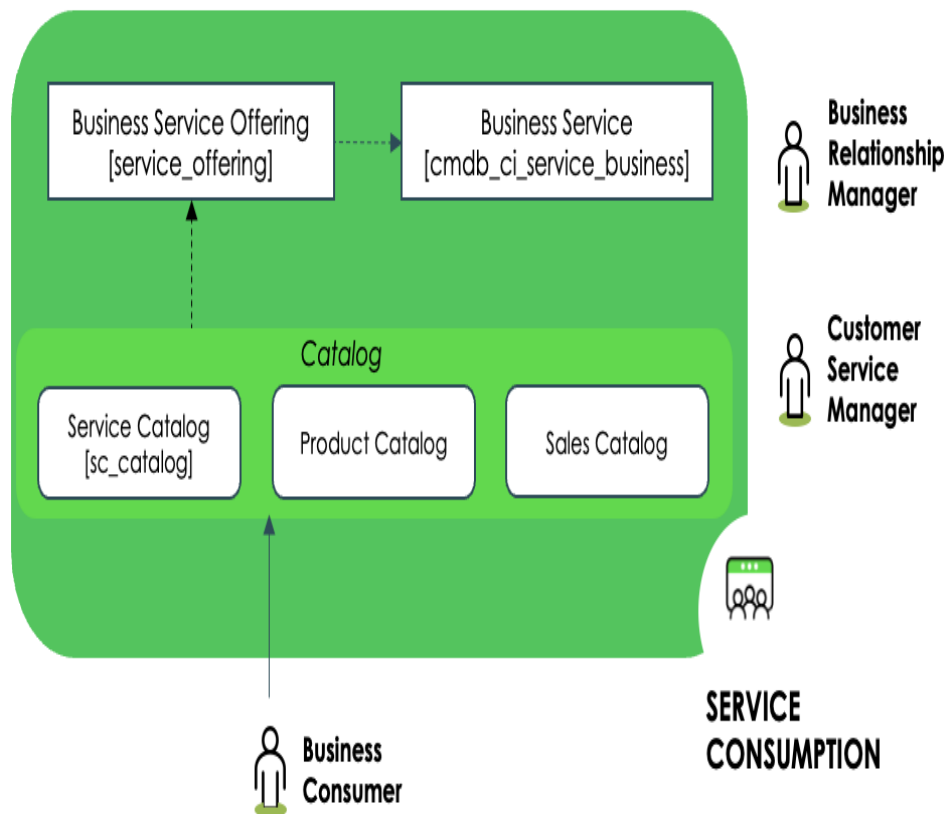
When you use ITSM, you must understand the impact that a technology can have on your business. For example, your business might perform the following actions:

- Consume the technology.
- Sell the technology (as is the case with Customer Service Management).
- Both sell the technology and consume it.

Benefits of the operations that you perform in the Run stage

- Run-stage operations ensure impact assessment for Incident Management and Change Management. Within an incident or change, you can identify the impacted business, assuming relationships exist between the selected CI and the impacted businesses.
- Run-stage operations provide a foundation for using Service Portfolio Management in the Digital Portfolio Management (DPM). Service owners can monitor service portfolios and understand service-related information including service trends, improvement initiatives, service performance, and outage monitoring.
- Run-stage operations provide a foundation for ITSM capabilities. This foundation populates the related “Subscribe by” table on a service offering to identify the business and subscribers affected. Business service offerings can identify subscribers by user, company, location, department, and group.

Tables that you work on during the Run stage



Note: Some of the classes that you implement in this stage are logical CIs. Logical CIs aren't created through Discovery, so their **Model ID** values might not refer to product model (application model, service model, or software model) records. To help you to migrate to a product-centric management paradigm, each instance of a logical CI should be associated with a product model. See [Auto-generate product models for logical CIs](#).

Business service portfolio [service_portfolio] table

Note: The Business service portfolio is not a CMDB table.

A business service portfolio is not a CMDB table. A business service portfolio is a hierarchical collection of business services (products and services) that define a business objective.

Business service [cmdb_ci_service_business] table

The business service table is a base-system CMDB table. This table identifies a business objective that uses (and depends on) the infrastructure that technology uses.

This dependency means that the business service must sell or consume that infrastructure.

Business service offering [service_offering] table (service offering classified as a "business service")

Business service offerings are the starting point for configuring Service Portfolio Management. Business service offerings consist of one or more service commitments. These service commitments uniquely define the level of service in terms of availability, scope, pricing, and other factors.

The business service offering comes from the service. The business service offering is fine-tuned based on how the parent serves a specific technical need.

Every business service should have at least one business service offering.

Request catalog

A catalog (sometimes called a request catalog, service catalog, or service request catalog) is a set of business and technical products, services, service commitment options, and offerings that users can order on a self-service basis. You can manage a catalog to present your available products and services to users as catalog items. Catalogs are described in detail in [Service Catalog](#).

CSDM implementation stage — Fly

When you reach the Fly stage, you've accomplished all or most of the process of implementing the CSDM framework. The fly stage completes the process.

Benefits of the operations that you perform in the Fly stage

The Fly stage is a foundation for using APM capabilities

You can use Application Portfolio Management (APM) capabilities to rationalize your business applications. Ask questions such as the following:

- Are you spending too much on your business capabilities?
- Are you spending too little on your business capabilities?
- Should you increase the amount you spend on emerging business capabilities?

The Fly stage is a foundation for using APM with Service Portfolio Management capabilities

You can use APM with Service Portfolio Management capabilities to rationalize your business services and related offerings. Ask questions such as the following:

- Are you spending too much or too little on services?
- Are you spending too little or too little on services?
- Are they the right services compared to emerging capabilities?

The Fly stage is a foundation for using ITSM capabilities

Use a request catalog to relate a service offering to a catalog item. You can also enhance the request workflow to automatically populate the "Subscribe by" table. Service catalogs are described in detail in [Service Catalog](#).

Note: The request catalog is not a CMDB table.

The Fly stage can help you to manage business services

If your environment has a combination of CIs from each of the CSDM domains, this stage provides away to manage business services.

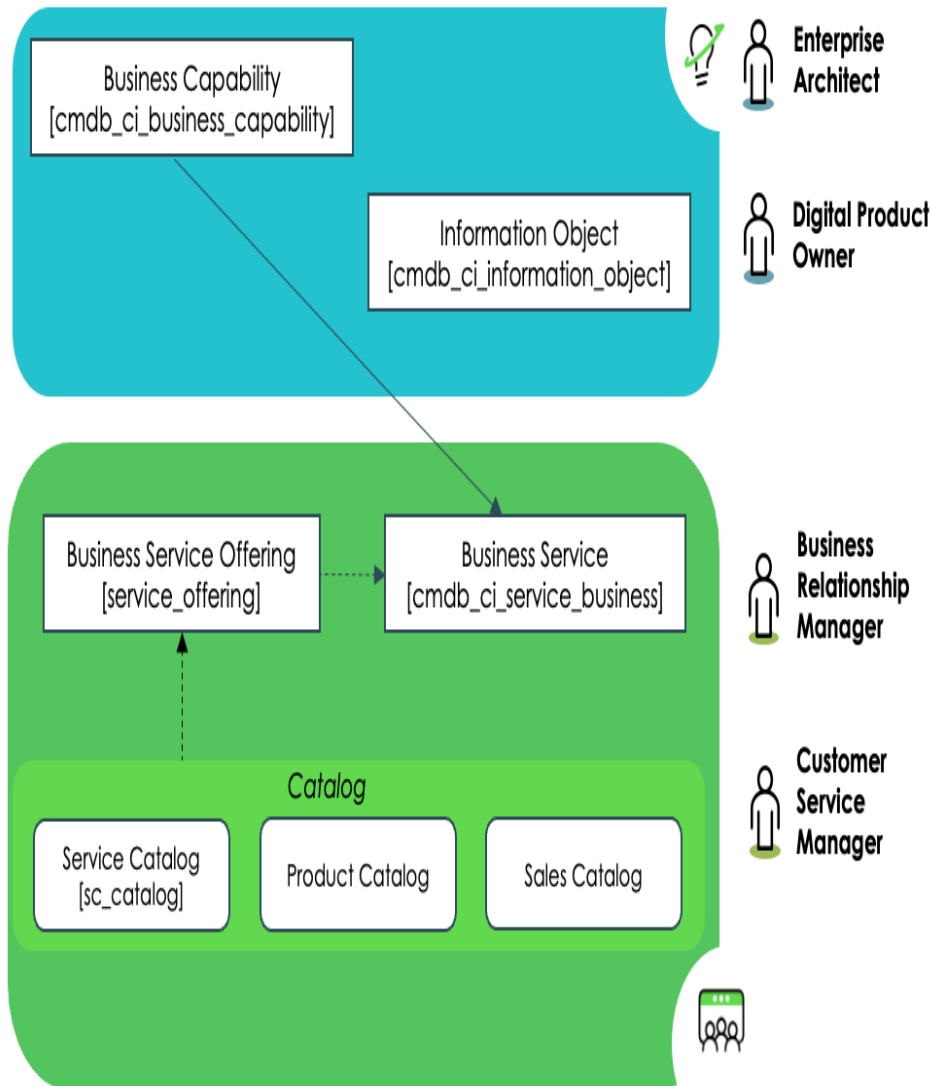
The Fly stage is a way to identify the types of data that may be contained in or used by your business applications

The information object table helps you see what's in your information portfolio.

Tables that you work on during the Fly stage

You reach the fly stage after you have accomplished all or most of the earlier stages.

DESIGN & PLANNING



SERVICE CONSUMPTION

Note: Some of the classes that you implement in this stage are logical CIs. Logical CIs aren't created through Discovery, so their **Model ID** values might not refer to product model (application model, service model, or software model) records. To help you to migrate to a product-centric management paradigm, each instance of a logical CI should be associated with a product model. See [Auto-generate product models for logical CIs](#).

The fly stage completes the remaining aspects of CSDM framework:

Business capability [cmdb_ci_business_capability] table

A business capability is a high-level capability that supports a business model or fulfills a mission for your organization.

Information object [cmdb_ci_information_object] table

You capture asset information as information objects. You can connect the information objects to your business applications to create an application portfolio that you can use at any time.

You can use the Information Object table to identify the types of data that a business application uses, including highly sensitive data such as:

- Personally Identifiable Information (PII)
- Payment Card Industry Data Security Standard (PCI DSS) data
- Health Insurance Portability and Accountability Act (HIPAA) data

Information objects are part of the information portfolio. The information portfolio links to the following data:

- Data Domains: Total number of records in the Data Domain[sn_apm_data_domain]. table
- Information Objects: Total number of records in the Information Object [cmdb_ci_information_object] table.
- Database Instances: Total number of records in the Database Instance [cmdb_ci_db_instance] table.
- Database Catalogs: Total number of records in the Database Catalog [cmdb_ci_db_catalog] table.

Important: You might need to implement the Information object [cmdb_ci_information_object] table as part of an earlier stage. Your business requirements determine the right stage for implementing the table.

Request catalog

Users request services through the request catalog. The request catalog is not a CMDB table. Catalogs are described in detail in [Service Catalog](#).

The fly stage includes these components:

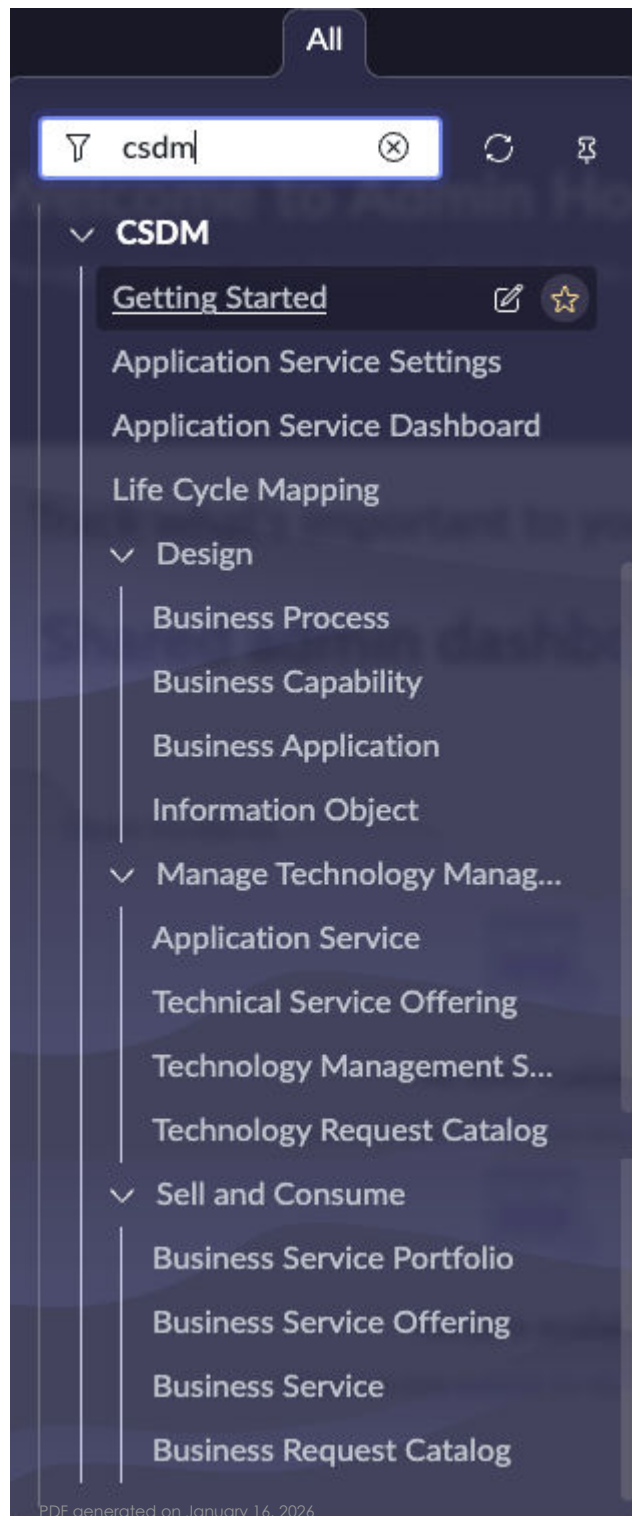
- Business service portfolio [service_portfolio] table
- Business service [cmdb_ci_service_business] table; (service classified as a "business service")
- Business service offering [service_offering] table; (service offering classified as a "business service")

Access CSDM features

The CSDM is the data model that admins should follow when they set up ServiceNow products and applications. The standards for defining configuration items (CIs) and relationships between CIs in the CMDB ensure that your data resides in the appropriate CMDB tables. The result is maximum value from ServiceNow AI Platform applications.

CSDM modules

Navigate to the modules that assist you in implementing and managing the CSDM domains and their components.



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CSDM Data Foundations Dashboard

The CSDM Data Foundations dashboard displays key CSDM indicators on a single page to help you get the full benefit from your ServiceNow AI Platform products. See [CSDM Data Foundations dashboard](#).

Getting Started

Select **Getting Started** to open the library of CSDM user documentation — the documentation you are viewing now.

Service Instance Settings

Use the Service Instance Settings module to specify the attributes and relationships that are required when a user creates a service instance. For more information, see [Modify the attributes and relationships required for application services](#).

See [Use Application Services dashboard to monitor health](#).

Service instances Dashboard

The Application Services dashboard enables you to monitor and manage service instances (service instances were formerly called application services) to ensure that they are fully configured and are populated in the CMDB. See [Monitoring and managing Configuring service instance settings](#).

Life Cycle Mapping

Use the Life Cycle Mapping module to specify how existing legacy status values should be converted to CSDM life-cycle value pairs (life cycle stage and life cycle stage status). You map both asset and CI legacy status values to life-cycle value pairs. See [Enabling life-cycle sync from legacy-to-asset](#).

Design

Work in the tables that are referenced in the Design & Planning domain of the CSDM. See [Design & Planning domain in the CSDM model](#).

Service Delivery

Work in the tables that are referenced in the Service Delivery domain of the CSDM. See [Service Delivery domain in the CSDM model](#).

Sell and Consume

Work in the tables that are referenced in the Service Consumption domain of the CSDM. See [Service Consumption domain in the CSDM model](#).

Synchronize data for 'Managed by' and 'Change' groups

[Synchronizing group assignment attributes](#)

[Set the group for a CI or an entire class of CIs](#)

[Synchronize user groups for a technology management offering](#)

- [CSDM Data Foundations dashboard](#)

The CSDM Data Foundations dashboard displays key CSDM indicators on a single page to help you get the full benefit from your ServiceNow AI Platform products.

- [Monitoring and managing Configuring service instance settings](#)

The Application Services dashboard enables you to monitor and manage service instances (service instances were formerly called application services) to ensure that they are fully configured and are populated in the CMDB.

CSDM Data Foundations dashboard

The CSDM Data Foundations dashboard displays key CSDM indicators on a single page to help you get the full benefit from your ServiceNow AI Platform products.

Access the dashboard

Use either of the following methods to open the dashboard:

Navigate to **All > Configuration**, and then select **CSDM Data Foundations Dashboard**.

1. In CMDB Workspace, navigate to **Workspaces > CMDB Workspace**.
2. Select **Management** in the CMDB Workspace menu bar.

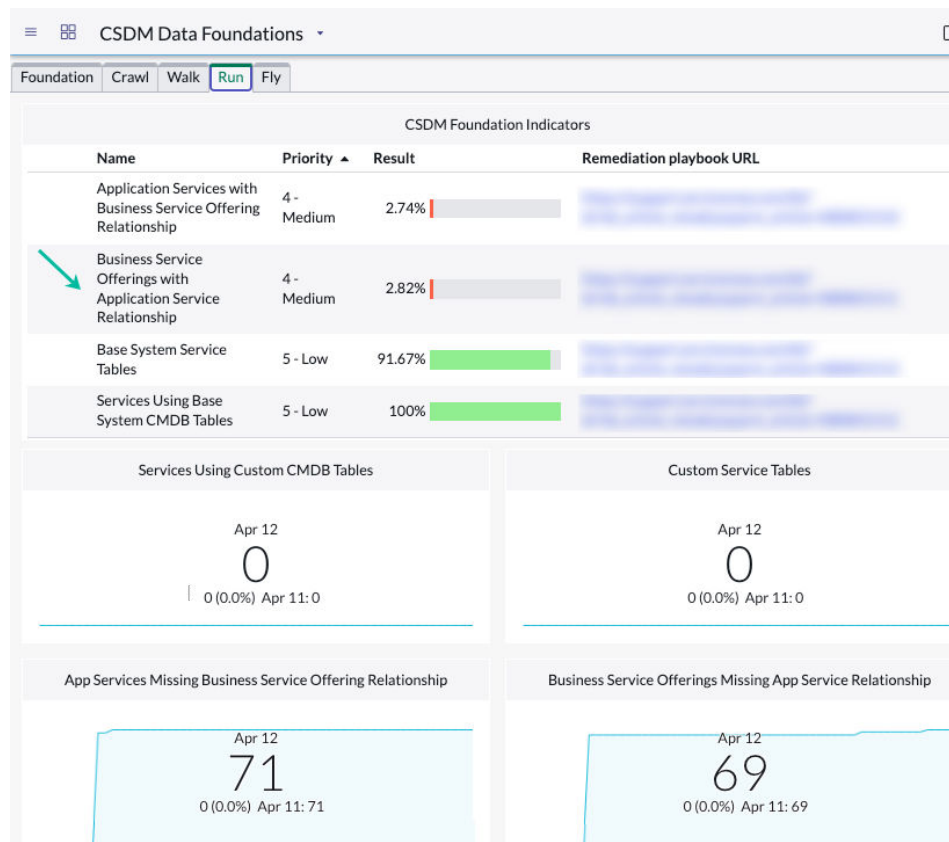
3. Select **CSDM Data Foundations Dashboard** in the Optimize section of the Management tools.

Select the change dashboard icon () to toggle between the CSDM and CMDB dashboards.

Example

The reports on each tab display the data metrics that are appropriate for the maturity of your organization's CSDM implementation stage (foundation, crawl, walk, run, and fly). A label on each report identifies the metric being displayed.

In this example, a report on the **Run** tab indicates that several business service offerings don't have the required relationships to service instances . With this knowledge, Service owners can add the relationships to ensure that customer service agents get complete information on the impact of applications that are down.



Tabs on the dashboard

- Tables in the Foundation domain contain base data that is referenced from or to objects in the other CSDM domains. Before you can use ServiceNow products, you must populate foundational data. See ['Foundation' stage reports on the CSDM Data Foundations dashboard](#).
- In the Crawl stage, you work on base-system CMDB tables that are associated with IT Service Management (ITSM). See ['Crawl' stage reports on the CSDM Data Foundations dashboard](#).
- In the Walk stage, you identify and populate the network infrastructure CIs and applications that your organization's technical teams support. See ['Walk' stage reports on the CSDM Data Foundations dashboard](#).

- In the Run stage, you set up the relationship between a technology and the business that sells and/or consumes the technology. See ['Run' stage reports on the CSDM Data Foundations dashboard](#).
- When you reach the Fly stage, you've accomplished all or most of the process of implementing the CSDM framework. The fly stage completes the process. See ['Fly' stage reports on the CSDM Data Foundations dashboard](#).

Managing performance of data collection

The system runs scheduled jobs that collect data for three CSDM metrics:

- App services that are missing a business app relationship
- Business apps that are missing an app service relationship
- App services where the relationship to business apps is not Consumed by

For information on managing data collection for a metric that isn't needed or that affects performance of the dashboard, see [Manage performance](#).

Related tasks

- [Configure the CSDM Data Foundations dashboard](#)

Related concepts

- [Monitor health in CSDM and CMDB Data Foundations Dashboards](#)

Monitoring and managing Configuring service instance settings

The Application Services dashboard enables you to monitor and manage service instances (service instances were formerly called application services) to ensure that they are fully configured and are populated in the CMDB.

Working in the dashboard

A report on the dashboard, for example, can direct you to a service instance that is not configured with a service population method so you can repair it. For information on working in the dashboard, see [Use Application Services dashboard to monitor health](#).

Applying CSDM guidelines to your product — product views

ServiceNow products are CSDM-aware — they expect that CMDB data is organized according to CSDM guidelines and they benefit from that organization.

The topics in the next section identify the specific CMDB tables that the products update and obtain data from. The topics provide practical instructions for implementing the CSDM guidelines.

- Brief overview of the function and benefits of the product.
- CMDB tables managed by the product.
- CMDB tables used by the product.
- Mention of other ServiceNow products that add value to the product.
- Mention of other ServiceNow products that benefit from the product.
- Use case and examples.

The topics do not address the following information:

- Implementing the ServiceNow product.
- Configuring systems that are not included in the base system. Such systems might be referred to in this product view, but they are out-of-scope.
- Using other ServiceNow products, such as IT Service Management (ITSM), Strategic Portfolio Management (SPM), and IT Operations Management (ITOM).

Note: A CSDM use case might have one or more data types. Other ServiceNow products may also manage those data types. The use cases don't describe all possible product dependencies.

Products and features that benefit when following the CSDM guidelines

Change Management

Applying CSDM guidelines to Change Management

Customer Service Management

Customer Service Management and the CSDM guidelines

Digital Portfolio Management

Apply the CSDM guidelines to Digital Portfolio Management

Enterprise Architecture (EA) [formerly Application Portfolio Management]

Applying the CSDM guidelines to Enterprise Architecture

HR Service Delivery for Healthcare

HR Service Delivery and CSDM tables

Incident Management

Applying CSDM guidelines to Incident Management

Install Base Management

Common Service Data Model framework for Install Base Management

ITOM Health

Applying the CSDM guidelines to ITOM AIOps

ITOM Visibility

Applying the CSDM guidelines to ITOM Visibility

Operational Technology Manager

Implementing the CSDM framework for Operational Technology

Problem Management

[Applying CSDM guidelines to Problem Management](#)

RPA Hub

[Applying the CSDM guidelines to RPA Hub](#)

Service Catalog

[Service Catalog and CSDM](#)

Service Portfolio Management

[Applying the CSDM guidelines to Service Portfolio Management](#)

Vulnerability Response

[Vulnerability Response applications and CSDM tables](#)

CSDM videos in the ServiceNow Community

[Playlist of all CSDM videos](#)

CSDM reference

Reference topics provide detailed descriptions of tables, properties, forms, and roles that are used in the CSDM framework.

Basics

[CSDM terms](#)

[CI relationships in the CSDM](#)

[How CSDM concepts map to CMDB tables](#)

Life cycles

[Life cycle of document and contract entities](#)

[Life cycle of tangible/physical CIs](#)

[Life cycle of location entities](#)

Life cycle of intangible/logical entities

Life cycle of product entities

- [CSDM resources](#)

Resources and videos that complement the documentation.

- [CSDM terms](#)

Most ServiceNow products and ServiceNow AI Platform applications align closely with the Common Service Data Model. This table defines terms as they are used across the ServiceNow AI Platform.

- [How CSDM concepts map to CMDB tables](#)

The objects in the conceptual CSDM framework must map to the physical model objects (CIs and CI class tables) in the CMDB. You can navigate in the platform to see dictionary attributes.

- [CI relationships in the CSDM](#)

For configuration management to be most effective, establish relationships between the objects and CIs in the conceptual CSDM.

- [How life-cycle values for Asset, CI, and IBI are synced](#)

Your organization gains significant value from your ServiceNow AI Platform applications when you take advantage of the option to directly map legacy status values to CSDM life cycle stage and life cycle stage status values and auto-sync the values going forward.

- [Life cycle mapping form](#)

Use the Life Cycle Mapping module to specify how existing legacy status values should be converted to CSDM life-cycle value pairs (life cycle stage and life cycle stage status). You map both asset and CI legacy status values to life-cycle value pairs.

- ['Foundation' stage reports on the CSDM Data Foundations dashboard](#)

The **Foundation** tab on the CSDM Data Foundations dashboard identifies data that might be missing from the CMDB.

- ['Crawl' stage reports on the CSDM Data Foundations dashboard](#)

Crawl stage reports on the CSDM Data Foundations dashboard

- ['Walk' stage reports on the CSDM Data Foundations dashboard](#)

Walk stage reports on the CSDM Data Foundations dashboard

- ['Run' stage reports on the CSDM Data Foundations dashboard](#)

Run stage reports on the CSDM Data Foundations dashboard

- ['Fly' stage reports on the CSDM Data Foundations dashboard](#)

Fly stage reports on the CSDM Data Foundations dashboard

- [CSDM life-cycle terms](#)

Most ServiceNow products and ServiceNow AI Platform applications align closely with the Common Service Data Model. This table defines terms as they are used across the ServiceNow AI Platform.

- [Life cycle of document and contract entities](#)

The document and contract life-cycle value pairs represent the overall life cycle of document assets (contracts) and CIs (business process documentation) as related to their products. The life-cycle values for the document and contract life-cycle process are visible only in tables related to document entities in Contracts and CMDB.

- [Life cycle of tangible/physical CIs](#)

The tangible/physical life-cycle states represent the overall life cycle of physical assets and CIs as related to their products. Tangible/physical assets are physical items that are stocked, for example computers, monitors, and keyboards. The stages and statuses for the tangible/physical life-cycle process are visible only in hardware-related tables in Asset Management and the CMDB.

- [Life cycle of location entities](#)

The values for the location life-cycle process reflect the locations used by your organization and are visible only in the common data locations table.

- [Life cycle of intangible/logical entities](#)

The intangible/logical life-cycle value pairs represent the overall life cycle of logical assets and CIs as related to their products. A logical or software asset includes items like applications, services, and licenses. The life cycle stage and life cycle stage status values of logical items are visible only in tables related to intangible/logical items in Asset Management and the CMDB.

- [Life cycle of product entities](#)

The product life-cycle value pairs represent the overall life cycle of a product model, a specific version, or a product configuration. The life-cycle values for the product life-cycle process are visible only in Product (Models) tables.

CSDM resources

Resources and videos that complement the documentation.

Important resources

- [Download the CSDM White paper from the Now Create site](#)
- [Join the ServiceNow CSDM Community to grow your CSDM expertise.](#)

Videos for getting started

- CSDM 101: Everything you need to know: <https://youtu.be/hANONH1c1vQ>
- Ask the experts: Getting started with CSDM: <https://youtu.be/F1Qznf7kgnE>

CSDM terms

Most ServiceNow products and ServiceNow AI Platform applications align closely with the Common Service Data Model. This table defines terms as they are used across the ServiceNow AI Platform.

Common CSDM terms

Term	Definition	Notes
Application	Any deployed program, module, or group of programs that is designed to provide specific functionality on a computer infrastructure.	Defines behavior and has specific functionality associated with it. Applications are typically discoverable functionality, like Apache Web Server.
Asset	An item whose financial value is tracked.	Many assets are CIs and vice versa, but that is not always the case. Assets have a life cycle with financial considerations, for example, Microsoft 365.
Business Application	Represents all software and infrastructure environments (dev, test, prod) that are configured to provide functionality.	Used to increase productivity and perform other business functions accurately. For example, Dell Online.
Business capability	High-level capability that an organization requires to execute its business model or fulfill its mission.	Typically described in the context of performing one or more specific tasks to achieve business outcomes. For example, demand management or financial planning.
Business service	A business service is a service type that is published to business users.	Usually, business users order business services. Business users can select the desired

Term	Definition	Notes
	A business service typically implements one or more business capabilities.	offering and service commitment levels via the Service Catalog. For example, procurement, shipping, and finance. • A business service is an operational CI. • A business service must be a one-level service and not a hierarchy of business services. • A business service can be used for impact in Incident Management, Problem Management, and Change Management and for approvals for Change. • A business service must be focused on the consumer or seller.
Business service offering	Business service offerings are the starting point for configuring Service Portfolio Management. Business service offerings inherit from Business Services. Business service	See Service Consumption domain in the CSDM model for details.

Term	Definition	Notes
	offerings consist of one or more service commitments that define the level of service in terms of availability, scope, pricing, and other factors. For example, an organization might offer two levels of desktop support: • A silver offering of upgrades and virus protection. • A gold offering with the silver commitments plus a response time guarantee of 30 minutes between the hours of 8:00 a.m. and 5:00 p.m., Monday through Friday.	
Configuration item (CI)	Physical and logical components of an infrastructure that are currently or soon will be under configuration management.	Might be a single module such as a server, database, or router or a more complex item, such as a complete system. For example, a web server, database, or infrastructure.
Operating model	An abstract and ideally	Typically represents the various elements

Term	Definition	Notes
	visual representation (model) of how an organization delivers value to its customers or beneficiaries.	of how an organization operates. It usually incorporates strategy positions such as the innovation model, degree of intelligent automation, industry alignment, provider delivery models, and the business expectations of IT.
Portfolio	Collection of services, products, projects, or applications.	Used to manage like items together for a business. Portfolios may be grouped by objective, capabilities, organization, like projects, or services.
Service	Means of delivering value to customers by facilitating outcomes that users want to achieve without the ownership of specific costs and risks.	Typically has three aspects: • Interaction • Offering • Service system ServiceNow provides three base service types: • Business • Technical • Application You can extend the base types to align

Term	Definition	Notes
		with the service types in your organization.
Service catalog	Provides consumable view of available products, services, service commitment options, and offerings.	Helps manage which services a user may have access to. Also, catalogs are the starting point for access to available services. For example, IT services catalog.
Service commitment	Defines the service delivery obligations agreed to between the consumer and the provider.	Often manifested in the form of contracts such as service level agreements, operational level agreements, and underpinning contracts. Service commitments include specific performance characteristics that differentiate one offering from another.
Service instance (called application service before CSDM v5)	A service type that is a logical representation of a deployed application stack.	Examples of service instances are hosting, data backup, and recovery.

Term	Definition	Notes
		Note: Applications and service instances do not have a one-to-one relationship. A service instance can include multiple applications. An application can be included in multiple service instances. See Use Application Services dashboard to monitor health.
Service offering	A stratification of a service into capability, availability, pricing, and packaging options.	Different levels of performance and features for a given service can be made available. For example, ITSM Standard and ITSM Pro.
Technology management service (formerly Technical service)	A service type that is published to service owners and typically underpins a business or service instance.	Can typically be ordered by service owners. Service owners are able to select the desired offering and service commitment levels via the Service Catalog. For example, computers, storage, and networks.

Term	Definition	Notes
Technology management offering	A technology management offering is a service offering with a service classification of "technical service". Technical service offerings may be further divided as follows: • Location and geography • Environment (production or non-production) • Pricing • Availability • Support group (for Incident Management) • Technical approval group (for Change Management) • Packaging options (commitments) The technical service offering comes from the service, based on how the parent serves a specific technical need. Every operational technical service must be associated with at	Note: Not all technology management offerings must be related to applications or infrastructure CIs. Managed Service Providers can provide technical service offerings.

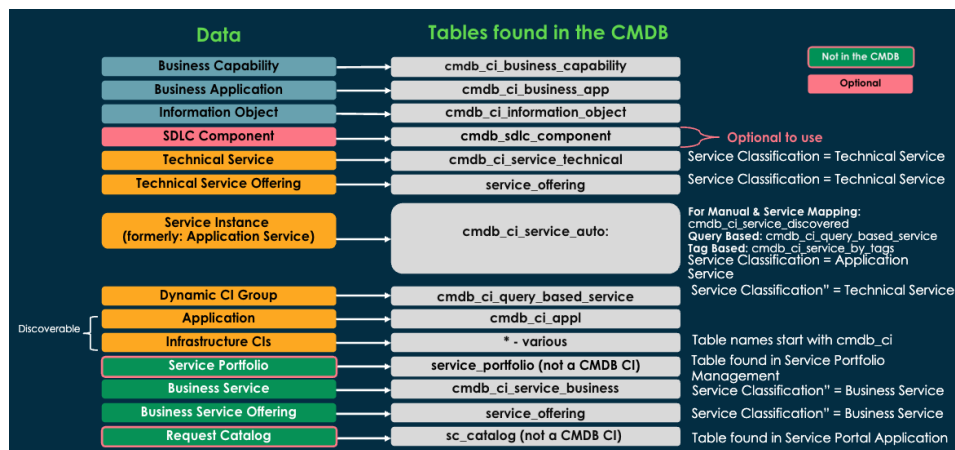
Term	Definition	Notes
	least one technology management offering.	

Related reference

- [CSDM resources](#)

How CSDM concepts map to CMDB tables

The objects in the conceptual CSDM framework must map to the physical model objects (CIs and CI class tables) in the CMDB. You can navigate in the platform to see dictionary attributes.



Business Application data belongs in the cmdb_ci_business_app table, and not in the cmdb_ci_app table.

Note: Business services and technology management services connect to the spm_service_portfolio through the spm_taxonomy_node. See [Service Portfolio Management taxonomy](#).

CSDM videos in the ServiceNow Community

[Playlist of all CSDM videos](#)

Related reference

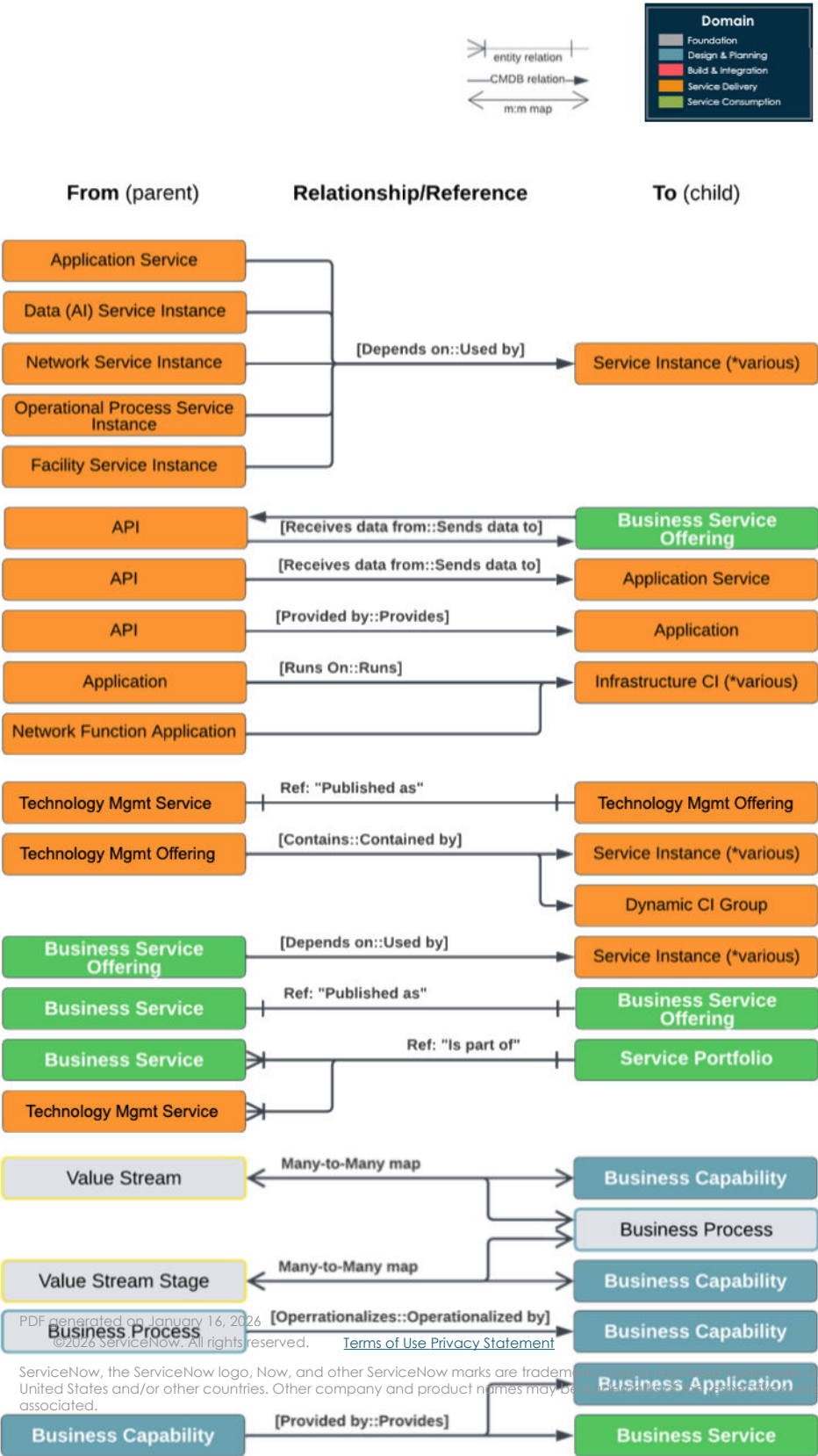
- [CSDM resources](#)

CI relationships in the CSDM

For configuration management to be most effective, establish relationships between the objects and CIs in the conceptual CSDM.

Required CSDM relationships

- Most features and products, such as the Technology Portfolio Management risk assessment and Application Portfolio Management (APM), depend on the relationships.
- The relationships commonly created as part of Service Mapping and Discovery are the standard for infrastructure CIs. If you map the elements manually, be sure to consider how Discovery would have treated them.
- Not all objects in the CSDM conceptual model are CMDB tables, and not all the objects have relationships. You might need to create some of the following required relationships.
- CSDM v5 implements the **Uses::Used By** relationship between Business Applications and Application Services. Earlier CSDM versions used **Consumes::Consumed By**. A Business Application, ultimately, relates to an Application Service table and not any other type of Service Instance.



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Note: Business services and technology management services connect to the `spm_service_portfolio` through the `spm_taxonomy_node`. See [Service Portfolio Management taxonomy](#).

Resources in the ServiceNow Community

For an extended explanation of the business application reference attribute's role in managing platforms from a CSDM perspective, [see this article](#) in the ServiceNow community.

[Playlist of all CSDM videos](#)

Related reference

- [CSDM resources](#)

How life-cycle values for Asset, CI, and IBI are synced

Your organization gains significant value from your ServiceNow AI Platform applications when you take advantage of the option to directly map legacy status values to CSDM life cycle stage and life cycle stage status values and auto-sync the values going forward.

How Asset, CI, and IBI life-cycle values are synced

Any item that is provided as a service or sold to your customer is tracked as an Install Base Item (IBI). The Model category table associates Asset, CI, and IBI for the item.

Sync between asset and IBI

By default, the system syncs life-cycle values between asset and IBI using the CSDM life cycle stage and life cycle stage status life-cycle value pairs.

Sync between asset and CI

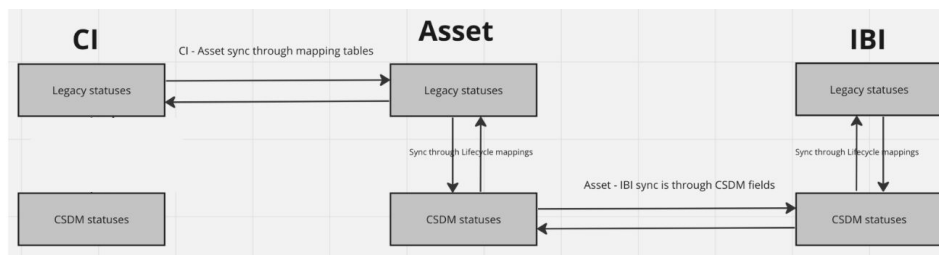
When the CSDM (`csdm.lifecycle.migration.activated`) plugin is activated on your instance, sync between the asset and CI tables depends on the `csdm.lifecycle.sync.between.ci.and.asset.activated` system property setting:

- When the property is set to true, the system syncs the CSDM life cycle stage and life cycle stage status values directly. This option provides significant benefits to your organization.

Important: The property is set to true when you select **Enable life cycle sync** on the Life cycle mappings list view.

- When the property is set to false (default), sync occurs through the legacy **status** and **substatus** values.

Life-cycle sync when the system property is set to false (the default setting)



When the CSDM (csdm.lifecycle.migration.activated) plugin is activated on your instance and the csdm.lifecycle.sync.between.ci.and.asset.activated system property is set to false:

Sync of updates from asset to IBI and CI

Updates made to the **status** or **substatus** value of an asset are synced with the associated IBI and CI. For example, when the **State** value of the asset is updated to **In use**, sync happens as follows:

- The life cycle stage and life cycle stage status values of the asset are set to **Operational** and **In use**, these values are then synced with life cycle stage and life cycle stage status values of IBI asset.
- The **status** and **substatus** values of the CI change to **Installed** and **In use**.

Sync of updates from IBI to asset and CI

Updates made to the life cycle stage and life cycle stage status values of an IBI asset are synced with the legacy **status** and **substatus** values of

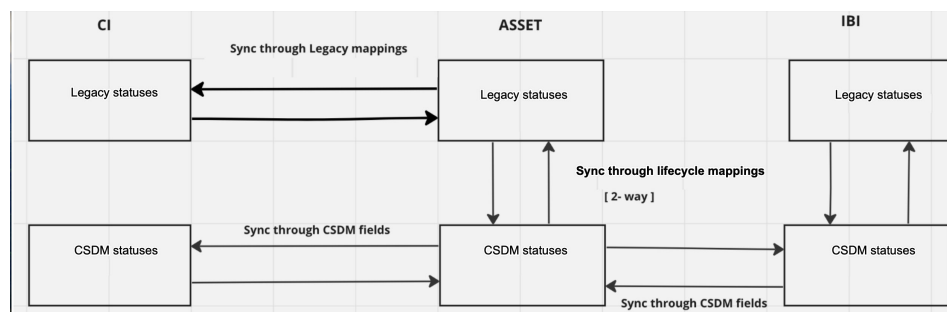
the asset through the CSDM values. The **status** and **substatus** values of the asset are then synced with the associated CI. For example, when the life cycle stage and life cycle stage status values of an IBI asset are updated to **Defective** and **In Stock**, sync happens as follows.

1. The life cycle stage and life cycle stage status values of the asset change to **In Stock** and **Defective**. These values are synced with the **status** and **substatus** values of the asset.
2. The **status** and **substatus** values of the asset are then synced with the associated CI.

Sync of updates from CI to asset and IBI

Updates made to the **status** and **substatus** values of a CI are synced with the associated asset. The **status** and **substatus** values of the asset are then synced with the life cycle stage and life cycle stage status values of the IBI asset through the CSDM values. For example, when the **status** and **substatus** values of a CI are updated to **Retired** and **Donated**, the corresponding values of the asset change. The changes from the asset are synced with the IBI asset and the life cycle stage and life cycle stage status values change to **End of Life** and **Retired**.

Life-cycle sync when the system property is set to true



When the CSDM (csdm.lifecycle.migration.activated) plugin is activated on your instance and the csdm.lifecycle.sync.between.ci.and.asset.activated system property is set to true (or you select the select **Enable life cycle sync** on the Life cycle mapping table list page, as described in [Enabling life-cycle sync from legacy-to-asset](#)):

Sync of updates from asset to CI and IBI

Updates made to the legacy **status** and **substatus** values of the asset are synced between the asset and CI as follows:

- The life cycle stage and life cycle stage status values of the asset are updated with values mapped to the legacy **status** and **substatus** values of the asset.
- The **status** and **substatus** values of the asset are then directly synced with the **status** and **substatus** values of the associated CI.
- The life cycle stage and life cycle stage status values of the asset are then directly synced with the associated CI.

However, sync of updates from asset to IBI is only through the CSDM life-cycle value pairs.

Sync of updates from IBI to asset and CI

Updates made to the life cycle stage and life cycle stage status values of the IBI are synced as follows:

- The legacy **status** and **substatus** values of the asset are synced with the CSDM life-cycle value pairs.
- The **status** and **substatus** values of the asset are synced with the **status** and **substatus** values of the associated CI.
- The life-cycle value pairs of the asset are directly synced with the associated CI.

Sync of updates from CI to asset and IBI

Updates made to the life cycle stage and life cycle stage status values of a CI are synced as follows:

- The life cycle stage and life cycle stage status values of the CI are directly synced with the associated asset.
- The **status** and **substatus** values of the asset are then updated with the values mapped to the CSDM values.
- The **status** and **substatus** values of the asset are then synced with the associated CI.

- The **status** and **substatus** values of IBI are synced with the asset through the CSDM life-cycle value pairs.

Related topics

- [Asset and CI management](#)

Life cycle mapping form

Use the Life Cycle Mapping module to specify how existing legacy status values should be converted to CSDM life-cycle value pairs (life cycle stage and life cycle stage status). You map both asset and CI legacy status values to life-cycle value pairs.

Fields on the Life cycle mapping form

Life cycle mapping form

Field	Description
Mapping for table	Legacy CMDB table and descending tables that this mapping applies to. Applies to a descending table unless there is a mapping configured specifically to the descending table.
Priority	Priority of applying this mapping definition for the table. The Priority value on the life cycle mapping table is used when a record update triggers the "legacy-to-life-cycle" sync, and multiple legacy value states on the record match with mapping rules. Of the multiple values, the highest priority (lowest numerical value) entry is used to set the life-cycle

Field	Description
	value pairs. If the highest priority entry can't be used, the system uses the next record in priority. Note: Priority is not used if Reverse sync is active because there should be a maximum of one reverse-sync choice for each life-cycle value per class.
Active	Denotes whether to apply this mapping definition. Deactivation results in lower-priority mappings being used or setting standard life cycle fields to TBD.
Legacy field name	Legacy field in the specified Mapping for table that is currently being used to store a life cycle stage. The value should be used as the source for the life cycle mapping.
Legacy subfield name	Value to set if the Legacy field name has a subfield. For example, if hardware Status has a substatus, then specify the name of the substatus here.
Legacy field value	Legacy value in the specified Mapping for table that is currently being used store life cycle stage. The value should be used as the source for the life cycle mapping.

Field	Description
Legacy subfield value	Value that is set if the Legacy field value has a subfield. For example, if hardware Status has a substatus, then specify the value of the substatus here.
Life cycle control	Class and life cycle stage and life cycle stage status that are used as the authoritative source of valid combinations for life cycle mapping.
Table	Standard object table that the Life cycle control belongs to.
Life cycle stage	Standard life cycle stage to map the specified Legacy field value to. Note: The Life cycle control setting determines that this particular life cycle stage is appropriate for the mapping. If there is no match in the life_cycle_mapping table, the value is set to TBD.
Life cycle stage status	Standard life cycle stage status to map the specified Legacy field value to. Note: The Life cycle control setting determines that this particular life cycle stage status is appropriate for the mapping.

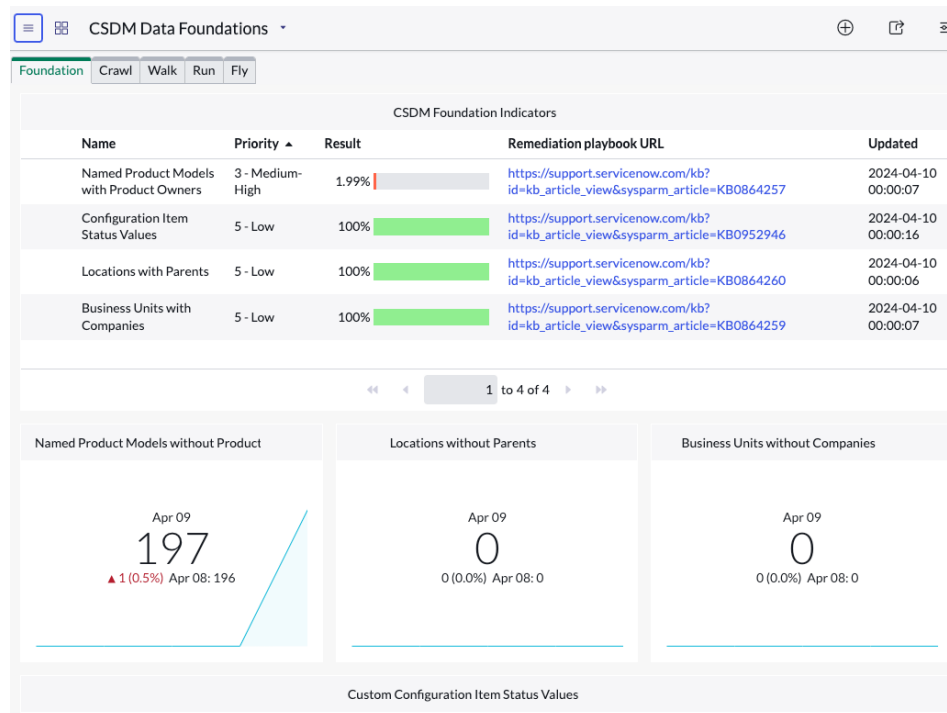
Field	Description
	If there is no match in the life_cycle_mapping table, the value is set to TBD .

'Foundation' stage reports on the CSDM Data Foundations dashboard

The **Foundation** tab on the CSDM Data Foundations dashboard identifies data that might be missing from the CMDB.

Reports on the Foundation tab

In this example, a report on the **Foundation** tab indicates that most named product models don't have associate product owners. Without this data, you might not be able, for example, to identify products that are reaching end-of-life. For more information, see [Products and product models](#).



CSDM Foundation Indicators report

- The **Priority** value is the product of the weight of the metric and the severity of the actual score. Priority ranges from 1 — Critical (the highest priority), to 5 — Low (the lowest priority).

•

The **Result** column displays a color-coded bar showing the percentage of CIs or the measured item that are in compliance for the key foundational metric.

- Red: 0–50% are in compliance.
- Yellow: 50–90% are in compliance.
- Green: More than 90% are in compliance.

To ensure optimum performance, the system stops collecting data for some metrics when they reach a specified number of non-compliant CIs. For information on managing data collection for a metric that isn't

needed or that affects performance of the dashboard, see [Manage performance](#).

- The **Remediation playbook URL** column displays links to knowledge articles in Now Support with instructions for bringing the CIs into compliance. Use your Now Support credentials to access the knowledge article.

To view detailed information on the impact of a metric and for details on working on issues, select the appropriate **Remediation playbook URL**.

Named product models without product

Application, service, and software class instance CIs aren't created through Discovery, so their **Model ID [model_id]** values might not refer to product model records. To help you to migrate to a product-centric management paradigm, each instance of a logical CI should be associated with a product model. For recommendations, see [Auto-generate product models for logical CIs](#).

For more information, see [Products and product models](#).

Locations without parents

You can create a hierarchy of location data using the Parent attribute to match your reporting requirements. Missing parent locations break the structure. See [Location \[cmn_location\] table](#).

Business units without companies

The hierarchy of your business is populated in the Business Unit table with a reference to the parent company. A business unit is a part of your organization that is responsible for specific operations, such as finance, human resources (HR), or IT. A hierarchy within a business unit is common. For large multinational organizations, you may have business units that identify independent regional operations and the specific operations within the region.

- [Auto-generate product models for logical CIs](#)

Use the CSDM Product Model Assignment job to auto-generate a product model record (application model, service model, or software model) for each logical CI that is not yet associated with a product model. Product models are ideal for associating CIs that are parts of a single digital product.

Related concepts

- [CSDM Data Foundations dashboard](#)

Auto-generate product models for logical CIs

Use the CSDM Product Model Assignment job to auto-generate a product model record (application model, service model, or software model) for each logical CI that is not yet associated with a product model. Product models are ideal for associating CIs that are parts of a single digital product.

Before you begin

Users with read access to modified CI records can view the new product models.

Role required: admin

About this task

Application, service, and software class instance CIs are not created through Discovery, so their **Model ID [model_id]** values might not refer to product model records. To assist you in migrating to a product-centric management paradigm, each instance of a logical CI should be associated with a product model. The CSDM Product Model Assignment job operates on the following classes and uses the name of the class instance as the name of the new product model.

- Service Offering
- Technology Management Service (formerly Technical service)
- Service Instance (called application service before CSDM v5): The script adds the version (as it relates to the business application) to the associated software model name. For example, the software model for the MyAppService service instance CI might be **MyAppService - version: 2.1**.
- Business Service
- Business Application

The CSDM Product Model Assignment job calls the CSDMModelUtil script. The script performs these actions for each instance of the supported CSDM classes that does not refer to a product model:

1. Create a new application model, service model, or software model record with the same name as the CI. If the requisite info for generating the name value does not appear in the CI, the script uses the default value that you specified.
2. For the CI, add a reference to the new product model in the **Model ID [model_id]** field.

Operation of the script:

- The script observes the access rules of the CMDB admin that runs the script.
- If an auto-generated product model record for a class instance CI would be identical to an existing record, then the existing record is used for that CI.
- If your data includes CIs with identical names (this is actually an error), then the resulting model_id values might conflict. Validate the resulting model_ids after running the script.

Procedure

1. Specify the default name to use if the name / version values of a CI are insufficient to auto-generate the name for the new product model.
 - a. Navigate to **Configuration > CI Class Manager** and then select **Open Hierarchy**.
 - b. Navigate to each of the supported classes in turn (service instance, business service, and so on).
 - c. On the **Basic Info** page for the class, enter a name in the **Default Product Model** field.

CI Class Manager

Hierarchy Configuration Item Add Child Class

Configuration Item Basic Info

Class Info ^

Basic Info

Attributes

Identification Rule

Reconciliation Rules

Suggested Relationships

All Relationship Rules

Health v

CI List

Basic Info provides the requisite information about the CI class. In order to create a new class, the display name and icon must be completed.

* Display Name Application Service

* Table Name cmdb_ci_service_auto

Description

Max 255 characters

Managed By Group

Default Product Model

Extensible ☒

Principal Class ☐

Icon

Save

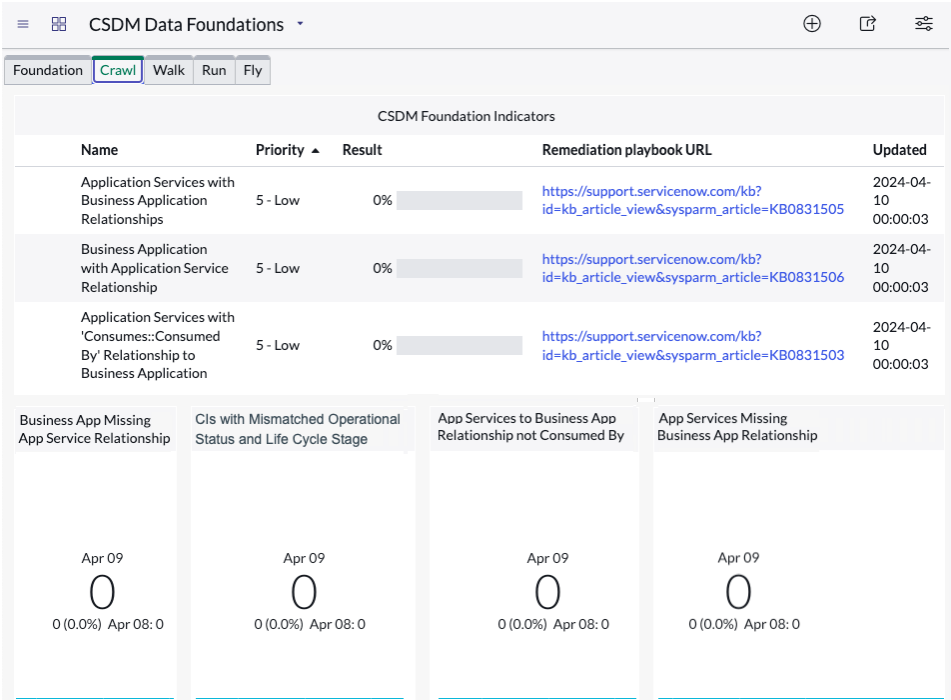
2. Run the CSDM Product Model Assignment job.
 - a. Navigate to **System Scheduler > Scheduled Jobs > Scheduled Jobs**.
 - b. Search for and open the CSDM Product Model Assignment job.
 - c. Select **Execute Now**.
The script runs and generates the product models.
3. Review the created product models to ensure that the new **Model ID [model_id]** values are correct.
Use:
 - **All > Product Catalog > Product Models > Application Models**
 - **All > Product Catalog > Product Models > Service Models**
 - **All > Product Catalog > Product Models > Software Models**
4. Run the job whenever you want to create product model settings for new class instances.

'Crawl' stage reports on the CSDM Data Foundations dashboard

Crawl stage reports on the CSDM Data Foundations dashboard

Reports on the Crawl tab

In this example, a report on the **Crawl** tab indicates that there are service instances that are not associated with a business application.



CSDM Foundation Indicators report

- The **Priority** value is the product of the weight of the metric and the severity of the actual score. Priority ranges from 1 — Critical (the highest priority), to 5 — Low (the lowest priority).
-

The **Result** column displays a color-coded bar showing the percentage of CIs or the measured item that are in compliance for the key foundational metric.

- Red: 0–50% are in compliance.
- Yellow: 50–90% are in compliance.

- Green: More than 90% are in compliance.

To ensure optimum performance, the system stops collecting data for some metrics when they reach a specified number of non-compliant CIs. For information on managing data collection for a metric that isn't needed or that affects performance of the dashboard, see [Manage performance](#).

- The **Remediation playbook URL** column displays links to knowledge articles in Now Support with instructions for bringing the CIs into compliance. Use your Now Support credentials to access the knowledge article.

To view detailed information on the impact of a metric and for details on working on issues, select the appropriate **Remediation playbook URL**.

App services missing business app relationship

Percent of CIs missing an important relationship. The service instance is a base-system CMDB table that should map to the related business application in use. Because the service instance is typically the system that a caller identifies when they report an issue with an application, the relationship is required if you are to identify the affected business application. For more information, see [Use Application Services dashboard to monitor health](#).

CIs with Mismatched Operational Status and Life Cycle Stage

The report identifies life-cycle value mismatches among CMDB, asset, and IBI. This information is helpful when you're aligning existing status data to the CSDM guidelines.

App service to business app not Consumed by

The report identifies incorrect relationships.

Business apps missing app service relationship

Percent of CIs missing an important relationship.

Managing performance of data collection

The system runs scheduled jobs that collect data for three CSDM metrics:

- App services that are missing a business app relationship

- Business apps that are missing an app service relationship
- App services where the relationship to business apps is not Consumed by

For information on managing data collection for a metric that isn't needed or that affects performance of the dashboard, see [Manage performance](#).

Related concepts

- [CSDM Data Foundations dashboard](#)

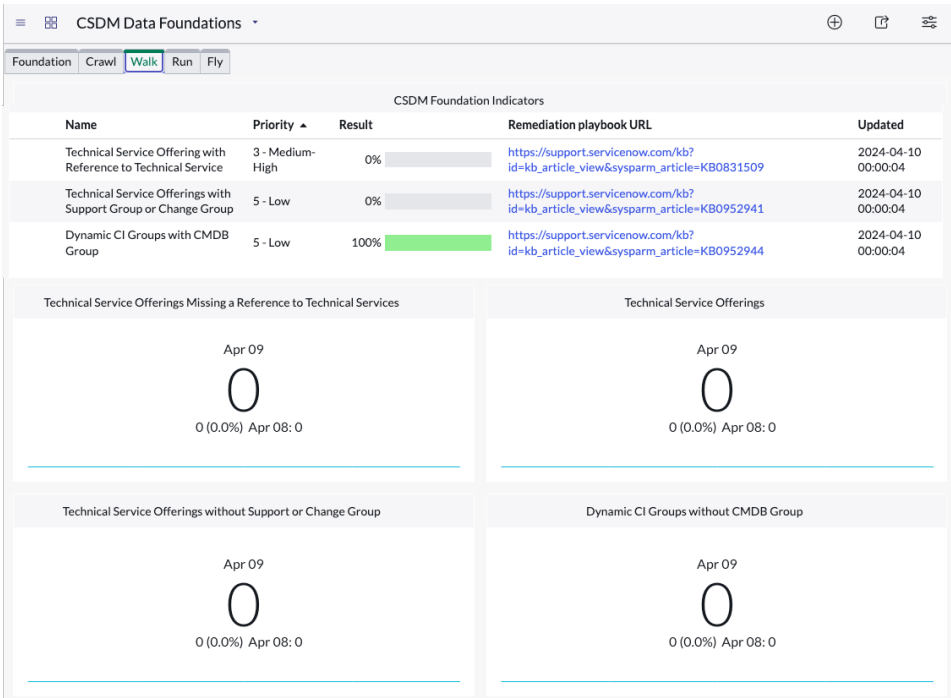
'Walk' stage reports on the CSDM Data Foundations dashboard

Walk stage reports on the CSDM Data Foundations dashboard

Reports on the Walk tab

In this example, a report on the **Walk** tab indicates that there are technology management offerings that are not associated with a support group. This indicates a risk that downtimes will take longer to remedy.

To view detailed information on the impact of a metric and for details on working on issues, select the appropriate **Remediation playbook URL**.



CSDM Foundation Indicators report

- The **Priority** value is the product of the weight of the metric and the severity of the actual score. Priority ranges from 1 — Critical (the highest priority), to 5 — Low (the lowest priority).
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To ensure optimum performance, the system stops collecting data for some metrics when they reach a specified number of non-compliant CIs. For information on managing data collection for a metric that isn't

needed or that affects performance of the dashboard, see [Manage performance](#).

- The **Remediation playbook URL** column displays links to knowledge articles in Now Support with instructions for bringing the CIs into compliance. Use your Now Support credentials to access the knowledge article.

Related concepts

- [CSDM Data Foundations dashboard](#)

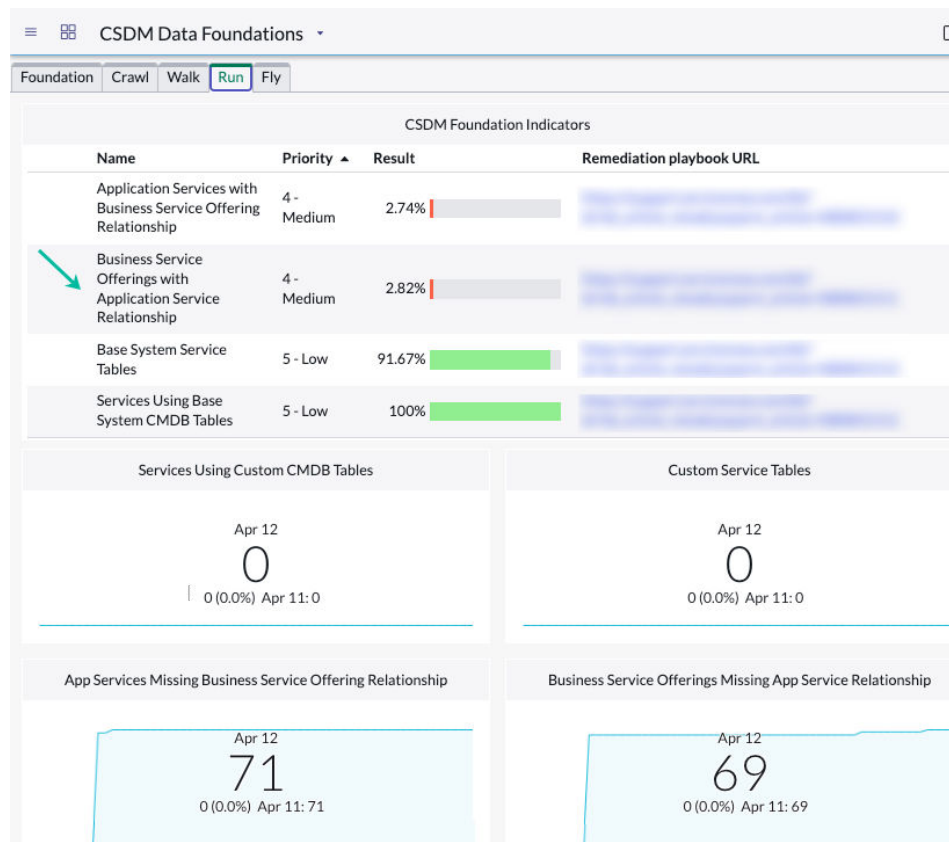
'Run' stage reports on the CSDM Data Foundations dashboard

Run stage reports on the CSDM Data Foundations dashboard

Reports on the Run tab

In this example, a report on the **Run** tab indicates that several business service offerings don't have the required relationships to service instances. With this knowledge, Service owners can add the relationships to ensure that customer service agents get complete information on the upstream impacts of applications that are down.

To view detailed information on the impact of a metric and for details on working on issues, select the appropriate **Remediation playbook URL**.



CSDM Foundation Indicators report

- The **Priority** value is the product of the weight of the metric and the severity of the actual score. Priority ranges from 1 — Critical (the highest priority), to 5 — Low (the lowest priority).

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Managing performance of data collection

The system runs scheduled jobs that collect data for three CSDM metrics:

- App services that are missing a business app relationship
- Business apps that are missing an app service relationship
- App services where the relationship to business apps is not Consumed by

For information on managing data collection for a metric that isn't needed or that affects performance of the dashboard, see [Manage performance](#).

Related concepts

- [CSDM Data Foundations dashboard](#)

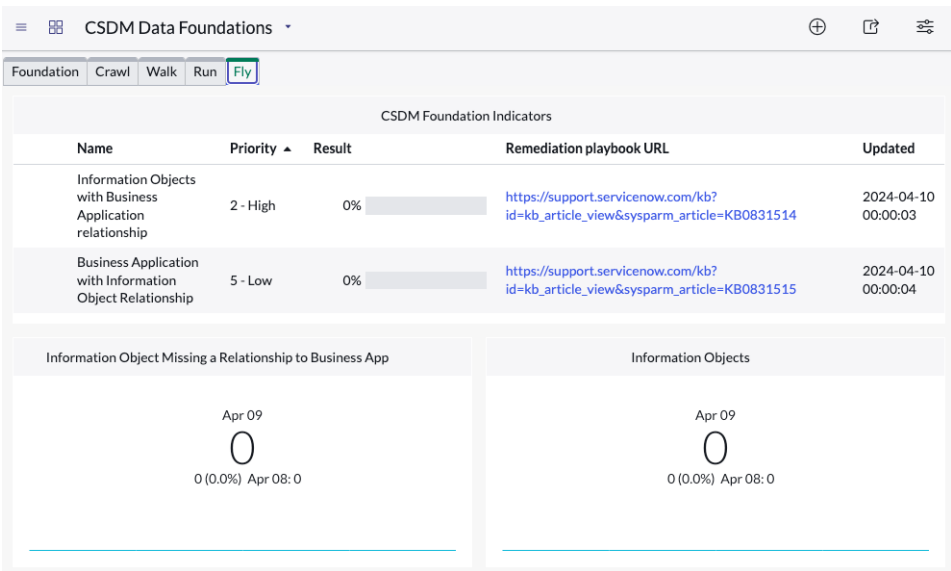
'Fly' stage reports on the CSDM Data Foundations dashboard

Fly stage reports on the CSDM Data Foundations dashboard

Reports on the Fly tab

In this example, a report on the **Fly** tab indicates that there are information objects that are not associated with a business application.

To view detailed information on the impact of a metric and for details on working on issues, select the appropriate **Remediation playbook URL**.



CSDM Foundation Indicators report

- The **Priority** value is the product of the weight of the metric and the severity of the actual score. Priority ranges from 1 — Critical (the highest priority), to 5 — Low (the lowest priority).
-

The **Result** column displays a color-coded bar showing the percentage of CIs or the measured item that are in compliance for the key foundational metric.

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To ensure optimum performance, the system stops collecting data for some metrics when they reach a specified number of non-compliant CIs. For information on managing data collection for a metric that isn't needed or that affects performance of the dashboard, see [Manage performance](#).

- The **Remediation playbook URL** column displays links to knowledge articles in Now Support with instructions for bringing the CIs into compliance. Use your Now Support credentials to access the knowledge article.

Note: Starting with the CSDM Data Foundations dashboard v2.2, the following metrics no longer appear on the **Fly** tab:

- Information objects missing an app service relationship
- Catalog request items related to service offerings

Related concepts

- [CSDM Data Foundations dashboard](#)

CSDM life-cycle terms

Most ServiceNow products and ServiceNow AI Platform applications align closely with the Common Service Data Model. This table defines terms as they are used across the ServiceNow AI Platform.

Terms related to life cycle

CI

Configuration item: Record in the CMDB CI [cmdb_ci] table.

life-cycle value pair

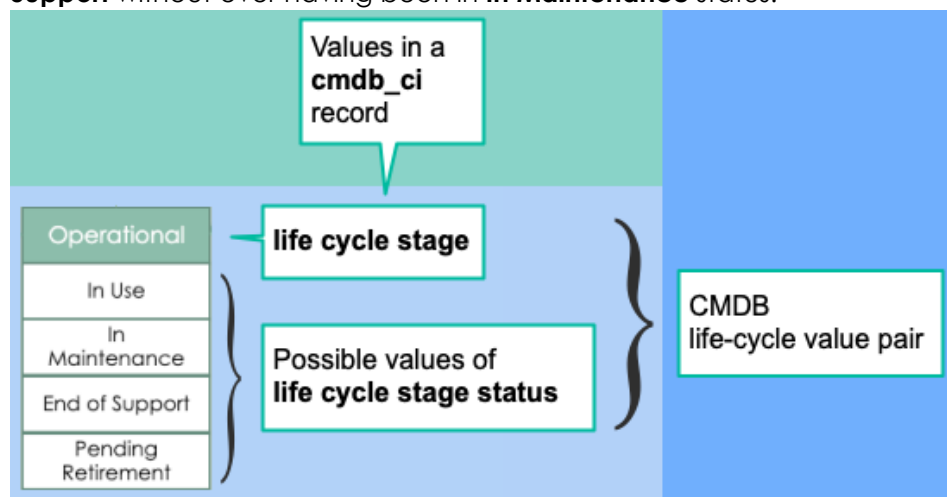
A life-cycle value pair is the combination life cycle stage and life cycle stage status values for a CI, asset, or IBI over the life cycle of a product instance. The pair of values fully describe the life-cycle state of the product instance.

The standard CSDM life-cycle value pair covers all phases of a product instance life cycle.

- A life cycle stage is one of the broad phases that a CI moves through, for example from inception or procurement to operation and then perhaps to end of life.

- life cycle stage status is the particular status of a CI within its current life cycle stage.

For example, a tangible/physical CI in the **Operational** stage might change stage status over time from **In Use** to **In Maintenance** to **End of Support**. A different tangible/physical CI might go from **In Use** to **End of Support** without ever having been in **In Maintenance** status.



CI legacy values

Value types that were in place before CSDM life-cycle values were implemented:

- install status
- operational status
- hardware status
- hardware substatus

asset

Production asset: Record in the asset [alm_asset] table.

install base item (IBI)

Install-Base-Item: Record in the install base item [sn_install_base_item] table. Any item that is provided as a service or sold to your customer is

tracked as an Install Base Item (IBI). The Model category table associates Asset, CI, and IBI for the item.

product instance

A product instance (PI) is comprised of an operational asset (asset), an install-base-item (IBI), and a configuration item (CI). A PI is a logical grouping of the [Asset/CI/IBI] that represents instances of products (goods and services) throughout their life cycles in ServiceNow applications and workflows. Possible formulations:

- Asset+CI+IBI
- Asset+CI
- Asset+IBI

Life cycle mapping table

The base system includes the life cycle mapping [life_cycle_mapping] table. The table is prepopulated with mappings between widely used legacy status values and the equivalent CSDM life cycle stage and life cycle stage status values. Each record in the table (called a mapping rule) specifies how to map a legacy status value, based on its table, to the best-fit CSDM life-cycle value pair.

The table typically contains multiple record entries per class, each entry that maps a specific legacy status to a life-cycle value pair. When there are multiple record entries for a class, the entries are prioritized by importance and likelihood for containing meaningful values for the mapping process. The highest priority rule is used when multiple candidate rules apply.

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Life cycle mapping

cmdb_ci_hardware

...

Update

Delete

* Mapping for table

Hardware [cmdb_ci_hardware]

Application

* Priority

10

Active ☒

Fields

Legacy value

* Legacy field name

Install Status

* Legacy field value

Pending Install

Legacy subfield name

Legacy subfield value

CSDM life cycle values

Life cycle control

Hardware - Dep

Table

Hardware

Life Cycle Stage

Deploy

Life Cycle Stage Status

Test

See [Map legacy status values to CSDM life-cycle values](#) and [How life-cycle values for Asset, CI, and IBI are synced](#).

Life cycle of document and contract entities

The document and contract life-cycle value pairs represent the overall life cycle of document assets (contracts) and CIs (business process documentation) as related to their products. The life-cycle values for the document and contract life-cycle process are visible only in tables related to document entities in Contracts and CMDB.

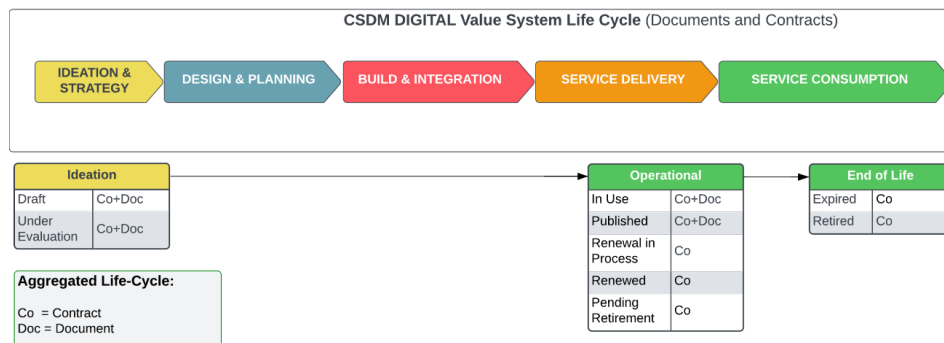
Life-cycle values for document and contract entities

For definitions of the values in the diagram, see [Definitions of life-cycle values for document and contract entities](#)

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The life-cycle value pairs for Customer Success Management documents are visible only in tables related to documents in Contract Management and the CMDB.

Note: The [life_cycle_control] table uses the type of CI (tangible/ physical, document and contract, location and so on) to determine which life cycle stage status values are available for each life cycle stage.

- Definitions of life-cycle values for document and contract entities

The document and contract life-cycle value pairs represent the overall life cycle of document assets (contracts) and CIs (business process documentation) as related to their products. The life-cycle values for the document and contract life-cycle process are visible only in tables related to document entities in Contracts and CMDB.

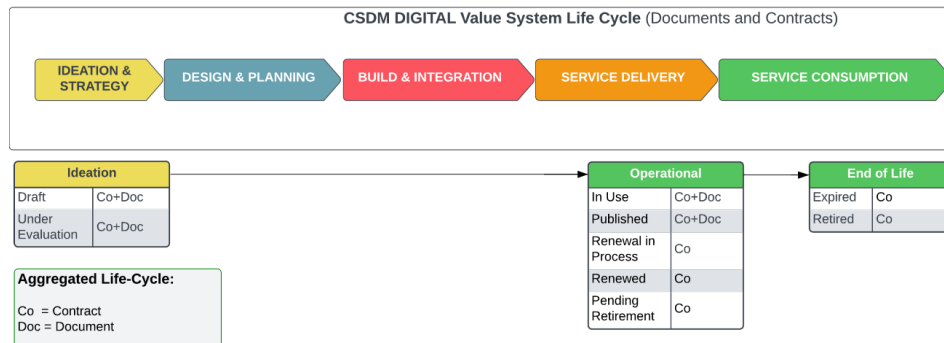
Related reference

- Definitions of life-cycle values for document and contract entities

Definitions of life-cycle values for document and contract entities

The document and contract life-cycle value pairs represent the overall life cycle of document assets (contracts) and CIs (business process documentation) as related to their products. The life-cycle values for the document and contract life-cycle process are visible only in tables related to document entities in Contracts and CMDB.

Definitions of document and contract life cycle stage and life cycle stage status values



Ideation life cycle stage

The materials are in initial draft form.

- **Draft:** The document or contract is in the initial creation and editing process.
- **Under Evaluation:** After creation, the materials are reviewed before activation as operational in use or published.

Operational life cycle stage

The materials are in active use.

- **In Use:** The document or contract is in active operational status before first renewal
- **Published:** After approval for release, publication is the formal act of making the materials visible for consumption.
- **Renewal in Process:** For an existing contract that is set to expire, the effort to renew the contract is occurring.
- **Renewed:** The contract has been renewed.
- **Pending Retirement:** The document, such as contract, has a pending expiration/end date or expiration date.

End of Life life cycle stage

The document or contract is no longer active.

- **Expired:** Date-limited material such as contracts that are past their expiration date.
- **Retired:** Materials that are no longer needed by the organization.

Related concepts

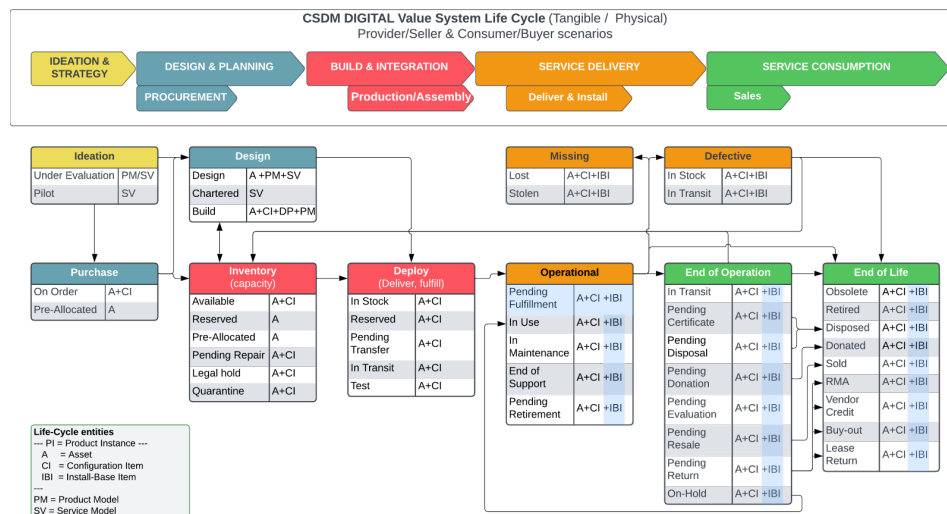
- [Life cycle of document and contract entities](#)

Life cycle of tangible/physical CIs

The tangible/physical life-cycle states represent the overall life cycle of physical assets and CIs as related to their products. Tangible/physical assets are physical items that are stocked, for example computers, monitors, and keyboards. The stages and statuses for the tangible/physical life-cycle process are visible only in hardware-related tables in Asset Management and the CMDB.

Life-cycle values for tangible/physical CIs

For definitions of the values in the diagram, see [Definitions of life-cycle values for tangible/physical CIs](#).



Note: The [life_cycle_control] table uses the type of CI (tangible/physical, document and contract, location and so on) to determine which life cycle stage status values are available for each life cycle stage.

For additional information on how you can benefit from implementing life-cycle value pairs for CMDB entities, see the ['Map existing status values to CSDM life-cycle value pairs'](#) section in the 'Foundation domain' topic.

Examples of tangible/physical classes

View attributes, identification rules, and other important schema structures for the CMDB Computer [cmdb_ci_computer] class in [Hardware \[cmdb_ci_hardware\]](#) class.

- [Definitions of life-cycle values for tangible/physical CIs](#)

The tangible/physical life-cycle states represent the overall life cycle of physical assets and CIs as related to their products. Tangible/physical assets are physical items that are stocked, for example computers, monitors, and keyboards. The stages and statuses for the tangible/physical life-cycle process are visible only in hardware-related tables in Asset Management and the CMDB.

- [Retiring a service instance can affect tangible/physical CIs](#)

Because a service instance might depend on a hardware (tangible/physical) device that other service instances depend on, you must take care when retiring a service instance.

- [Tangible/physical tables in the CMDB](#)

List of tangible/physical tables.

Related reference

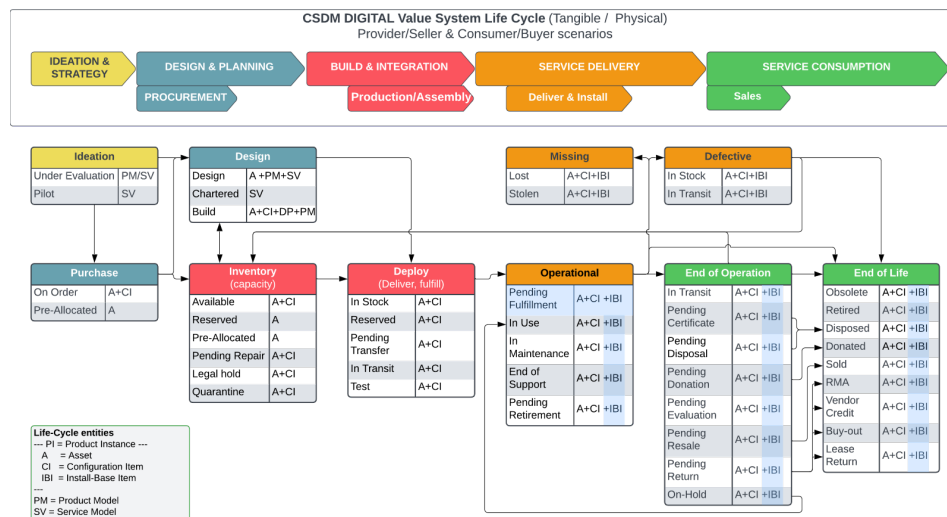
- [Definitions of life-cycle values for tangible/physical CIs](#)
- [Retiring a service instance can affect tangible/physical CIs](#)

Definitions of life-cycle values for tangible/physical CIs

The tangible/physical life-cycle states represent the overall life cycle of physical assets and CIs as related to their products. Tangible/physical assets are physical items that are stocked, for example computers, monitors, and keyboards. The stages and statuses for the tangible/physical life-cycle process are visible only in hardware-related tables in Asset Management and the CMDB.

Definitions of tangible/physical life cycle stage and life cycle stage status values

Note: The CI label tangible/physical includes and supersedes the "Hardware" label that had been used in earlier releases.



Ideation life cycle stage

New tangible or physical models in this stage are being assessed for potential use in the organization.

- **Under Evaluation:** The model is being reviewed for compatibility, functionality, and business fit before approval.
- **Pilot:** The model is in a limited deployment for real-world testing before broader adoption.

Purchase life cycle stage

After a tangible or physical model is approved, the organization proceeds with procurement.

- **On Order:** The asset has been ordered from a vendor but has not yet been received.
- **Preallocated:** The asset is already assigned to a specific purpose or user before its arrival.

Inventory life cycle stage

The asset is received and stored in inventory before deployment.

- **Available:** The asset is in stock and is ready for deployment.
- **Reserved:** The asset is set aside for a specific request but is not yet deployed.
- **Preallocated:** The asset is assigned to a request but is not yet physically allocated.
- **Pending Repair (CI):** The asset is in stock but requires repairs before being usable such as on-site visit from manufacturer or RMA swap.
- **Legal Hold (CI):** The asset cannot be deployed due to legal restrictions or compliance reasons.
- **Quarantine (CI):** The asset is isolated due to potential security risks, failures, or non-compliance.

Deploy life cycle stage

The asset is prepared and moved to operational use.

- **In Stock:** The asset is stored and is designated for deployment, but is in the process of being prepared for deployment.
- **Reserved:** The asset is assigned to a user or department but has not been physically moved.
- **Pending Transfer:** The asset is scheduled for relocation to another facility or user.
- **In Transit:** The asset is currently being moved between locations.

- **Test:** The asset has been identified as part of a deployment activity, has been built, and is ready for testing prior to operational usage .

Operational life cycle stage

The asset is actively in use or is undergoing maintenance.

- **Pending Fulfillment:** The product is put/placed into its operational environment/location but needs to be configured/provisioned before it can be used.
- **In Use:** The asset is deployed and functioning in a live environment.
- **In Maintenance:** The asset is being serviced or is undergoing repairs while still considered operational.
- **End of Support:** The asset has reached the end of vendor or internal support but is still in use.
- **Pending Retirement:** The asset is still operational but is marked for decommissioning.

End of Operation life cycle stage

End of Operation is an intermediate stage between Operational and the next stage. Operation of the product have ended, but the product instance is transitioning to a new life cycle stage and life cycle stage status.

- **Pending Evaluation:** Non-operational intermediate status in which a product is evaluated to determine its next stage and stage status.
- **Pending Return:** Non-operational status before the asset is returned and/or the provider received the returned asset in an end of life stage: **RMA, Vendor Credit, Buyout or Lease Return.**
- **In Transit:** Non-operational, transitional status to support repurposing the asset and/or true end-of-life.
- **Pending Resale:** Intermediate status prior to selling the asset to a third-party entity .
- **Pending Certificate:** Intermediate status prior to receiving a certificate of disposal.

- **Pending Disposal:** Intermediate status prior to the **Disposed** end of life stage.
- **Pending Donation:** Intermediate status prior to the **Donated** end of life stage.
- **On Hold:** Operation of the asset is halted for one of the following reasons:
 - Suspended: service, subscription, and so on.
 - Blocked: credit card, account, and so on.
 - Paused: subscription, lease, and so on.

Defective life cycle stage

Tangible or physical devices that have failed and require replacement or repair.

- **In Stock:** The defective asset is stored and is awaiting disposition.
- **In Transit:** The defective asset is being shipped for return, repair, or disposal.

Missing life cycle stage

The asset is unaccounted for or stolen.

- **Lost:** The asset is missing and cannot be located.
- **Stolen:** The asset has been confirmed as stolen.

End of Life life cycle stage

The asset is being decommissioned or disposed of.

- **Pending Disposal:** The asset is marked for disposal but has not yet been removed.
- **In Transit:** The asset is being shipped to a disposal or return location.
- **Pending Certificate:** The asset awaits official documentation for decommissioning or disposal.
- **Disposed:** The asset has been discarded following proper procedures.

- **Donated:** The asset has been given to a non-profit, employee, or other entity.
- **RMA:** The asset is returned to the vendor for replacement or repair under warranty.
- **Sold:** The asset has been sold as part of asset liquidation.
- **Vendor Credit:** The asset was returned to the vendor for a credit instead of replacement.
- **Buyout:** The asset was leased and has been purchased instead of returned.
- **Lease Return:** The leased asset has been returned to the vendor at the end of its lease period.

Retiring a service instance can affect tangible/physical CIs

Because a service instance might depend on a hardware (tangible/physical) device that other service instances depend on, you must take care when retiring a service instance.

Considerations

A service instance is the logical representation of the underlying tangible/physical and software CIs that work together to implement a business application or system. The service instance represents an instance of the business application or system.

Tangible/physical and software CIs are managed using the tangible/physical life-cycle value pairs. Because a service instance is a logical representation, it is managed as using the intangible/logical life-cycle value pairs. The tangible/physical CIs that are part of the service map under a service instance have their own life cycles, but they are related through the service instance as a specific set of dependencies or decomposition.

Example 1: A tangible/physical CI is not retired when the service instance is retired

When a service instance is retired, the associated tangible/physical device might not be retired. For example, the device might remain idle,

unrelated to any service instance, until it is reallocated for use by a different service instance.

Example 2: A tangible/physical CI is shared by multiple service instances

In the common scenario of a shared database, multiple service instances (each with a unique database schema) share a single database service. The database service runs on a single physical host.

When one of the service instances is retired, the database service and host cannot be retired. All other service instances still depend on the database service that is running on the host.

See [Use Application Services dashboard to monitor health](#).

Tangible/physical tables in the CMDB

List of tangible/physical tables.

Holistic life cycle: CMDB tangible/physical tables (from cmdb_ci)

The tangible/physical life-cycle states represent the overall life cycle of physical assets and CIs as related to their products. Tangible/physical assets are physical items that are stocked, for example computers, monitors, and keyboards. The stages and statuses for the tangible/physical life-cycle process are visible only in hardware-related tables in Asset Management and the CMDB.

Label (display name)	Table name
Accessory	cmdb_ci_acc
Communication Device	cmdb_ci_comm
Computer Peripheral	cmdb_ci_peripheral
Computer Room AC	cmdb_ci_crac
Display Hardware	cmdb_ci_display_hardware
Facility Hardware	cmdb_ci_facility_hardware

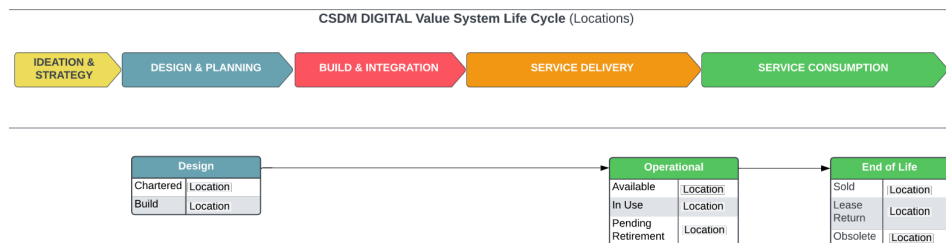
Label (display name)	Table name
Hardware	cmdb_ci_hardware
Imaging Hardware	cmdb_ci_imaging_hardware
IP Device	cmdb_ci_ip_device
Monitoring Equipment	cmdb_ci_monitoring_hardware
Network Adaptor	cmdb_ci_network_adaptor
Printing Hardware	cmdb_ci_printing_hardware
Rack	cmdb_ci_rack
Storage Device	cmdb_ci_storage_device

Life cycle of location entities

The values for the location life-cycle process reflect the locations used by your organization and are visible only in the common data locations table.

Life-cycle values for location entities

For definitions of the values, see [Definitions of life-cycle values for location entities](#).



Note: The [life_cycle_control] table uses the type of CI (tangible/ physical, document and contract, location and so on) to determine which life cycle stage status values are available for each life cycle stage.

For additional information on how you can benefit from implementing life-cycle value pairs for CMDB entities, see the ['Map existing status values to CSDM life-cycle value pairs'](#) section in the 'Foundation domain' topic.

- [Definitions of life-cycle values for location entities](#)

The values for the location life-cycle process reflect the locations used by your organization and are visible only in the common data locations table.

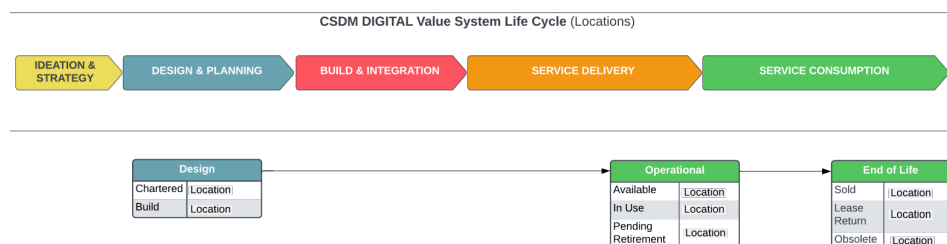
Related reference

- [Definitions of life-cycle values for location entities](#)

Definitions of life-cycle values for location entities

The values for the location life-cycle process reflect the locations used by your organization and are visible only in the common data locations table.

Definitions of location life cycle stage and life cycle stage status values



Ideation life cycle stage

The location is in the initial planning and setup phase.

- **Chartered:** The location is approved for development, but no physical presence has yet been established.
- **Build:** The location is under construction, setup, or is being prepared for operational use.

Operational life cycle stage

The location is active and in use for business operations.

- **Available:** The location is fully built and ready for use but currently unoccupied or not assigned for a specific function.
- **In Use:** The location is actively occupied and is being used for its intended purpose.
- **Pending Retirement:** The location is still operational but is scheduled for decommissioning or repurposing.

End of Life life cycle stage

The location is no longer in active use and is being decommissioned.

- **Sold:** The location has been sold to another entity.
- **Lease Return:** A leased location has been returned to the owner or landlord after the lease period ended.
- **Obsolete:** The location is no longer viable for use due to physical deterioration, business strategy changes, or other constraints.

Note: The [life_cycle_control] table uses the type of CI (tangible/physical, document and contract, location and so on) to determine which life cycle stage status values are available for each life cycle stage.

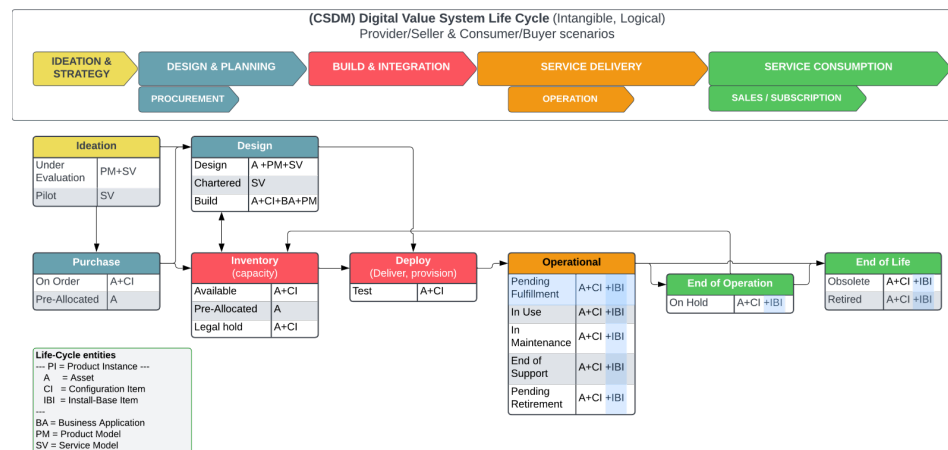
For additional information on how you can benefit from implementing life-cycle value pairs for CMDB entities, see the ['Map existing status values to CSDM life-cycle value pairs' section in the 'Foundation domain' topic](#).

Life cycle of intangible/logical entities

The intangible/logical life-cycle value pairs represent the overall life cycle of logical assets and CIs as related to their products. A logical or software asset includes items like applications, services, and licenses. The life cycle stage and life cycle stage status values of logical items are visible only in tables related to intangible/logical items in Asset Management and the CMDB.

Life-cycle values for intangible/logical CIs, assets, and IBIs

For definitions of the values in the diagram, see [Definitions of life-cycle values for intangible/logical entities](#).



Note: The [life_cycle_control] table uses the type of CI (tangible/physical, document and contract, location and so on) to determine which life cycle stage status values are available for each life cycle stage.

Service instances use intangible/logical life cycles

Because service instances are logical in nature, they should use the Logical life-cycle value pairs. Service instances follow the same life-cycle guidance as any other logical CI.

See [Use Application Services dashboard to monitor health](#).

- Definitions of life-cycle values for intangible/logical entities

The intangible/logical life-cycle value pairs represent the overall life cycle of logical assets and CIs as related to their products. A logical or software asset includes items like applications, services, and licenses. The life cycle stage and life cycle stage status values of logical items are visible only in tables related to intangible/logical items in Asset Management and the CMDB.

- Intangible/logical tables in the CMDB

List of intangible/logical tables.

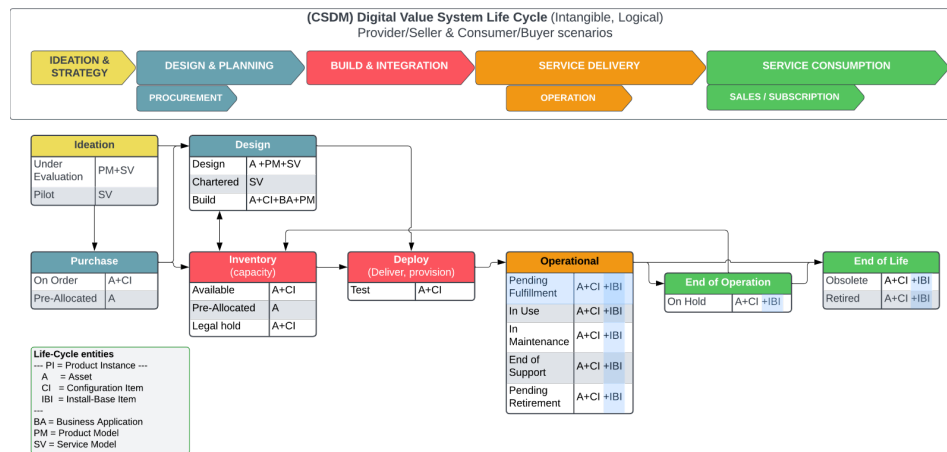
Related reference

- Intangible/logical tables in the CMDB

Definitions of life-cycle values for intangible/logical entities

The intangible/logical life-cycle value pairs represent the overall life cycle of logical assets and CIs as related to their products. A logical or software asset includes items like applications, services, and licenses. The life cycle stage and life cycle stage status values of logical items are visible only in tables related to intangible/logical items in Asset Management and the CMDB.

Definitions



Ideation life cycle stage

In this stage, new software or service products are evaluated and tested before they are approved for use.

- **Under Evaluation:** The software is being assessed for business fit, licensing implications, and technical feasibility.
- **Pilot:** The software is deployed in a limited way to test its functionality and usability before full-scale implementation.

Purchase life cycle stage

After the software is approved, the organization procures the necessary licenses or subscriptions.

- **On Order:** Licenses or software subscriptions have been purchased but are not yet available for use.
- **Preallocated:** Licenses are assigned to users or departments before they are officially deployed.

Inventory life cycle stage

The software licenses are now available for assignment and deployment.

Available: The software licenses are ready for allocation but have not yet been assigned to a specific system or user.

Design life cycle stage

For software development or major configurations, this phase defines the logical and architectural structure.

- **Chartered:** A project is initiated to design or configure software solutions.
- **Design:** The logical architecture and system requirements are being defined.
- **Build:** The software is being developed, customized, or configured for deployment.

Deploy life cycle stage

Software and services are prepared for operational use and are tested before full implementation.

Test: The software is installed in a test environment for validation before production deployment.

Operational life cycle stage

The software is actively in use by the organization.

- **In Use:** The software is deployed and functioning on production systems.
- **End of Support:** The software vendor or internal IT team no longer provide updates or technical support, but the software might still be in use.

Note: gggA CI might be in the **Operational** stage, but might no longer be supported by the vendor or publisher or third-party. That doesn't mean, however, that it can be or should be retired.

- **Pending Retirement:** The software is scheduled for decommissioning but remains operational until a replacement is implemented.

End of Operation life cycle stage

End of Operation is an intermediate stage between Operational and the next stage. Operations of the product have ended, but the product instance might transition to a new life cycle stage and life cycle status.

On Hold: Operation of the asset is halted for one of the following reasons:

- Suspended: service, subscription, and so on.
- Blocked: credit card, account, and so on.
- Paused: subscription, lease, and so on.

End of Life life cycle stage

The software is no longer in active use and is being decommissioned.

- **Retired:** The software is no longer used and its licenses might have been revoked or reassigned.
- **Obsolete:** The software is no longer relevant due to security risks, incompatibility, or technological advancements.

Intangible/logical tables in the CMDB

List of intangible/logical tables.

Intangible/logical tables in the CMDB (from cmdb_ci)

The intangible/logical life-cycle value pairs represent the overall life cycle of logical assets and CIs as related to their products. A logical or software asset includes items like applications, services, and licenses. The life cycle stage and life cycle stage status values of logical items are visible only in tables related to intangible/logical items in Asset Management and the CMDB.

Intangible/logical tables

Table label (display name)	Table name
Application Cluster	cmdb_ci_application_cluster

Table label (display name)	Table name
Batch Job	cmdb_ci_batch_job
Business Application	cmdb_ci_business_app
Business Capability	cmdb_ci_business_capability
Business Process	cmdb_ci_business_process
CIM Profiles	cmdb_CIM_profile
Cloud Key Pair	cmdb_ci_cloud_keu_pair
Cloud Mgmt Subnet	cmdb_ci_subnet
Cloud Resource Base	cmdb ci resource base
Cluster	cmdb_ci_cluster
Cluster Node	cmdb_cil_cluster_node
Cluster Resource	cmdb ci cluster resource
Cluster Virtual IP	cmdb_ci_cluster_vip
Configuration File	cmdb_ci_config_file
Database	cmdb_ci_database
Database Catalog	cmdb_ci_db_catalog
Datastore Disk	cmdb_ci_vcenter_datastore_disk
Disk Partition	cmdb_ci_disk_partition
DNS Name	cmdb_ci_dns_name
DRS VM Config	cmdb_ci_drs_vm_config
Endpoint	cmdb_ci_endpoint

Table label (display name)	Table name
Environment	cmdb_ci_environment
Fiber Channel Port	cmdb_ci_fc_port
Information Object	cmdb_ci_information_object
IP Address	cmdb_ci_ip_address
IP Network	cmdb_ci_ip_network
IP Phone	cmdb_ci_ip_phone
IP Service Instance	cmdb_ci_ip_service
Load Balancer Interface	cmdb_ci_lb_interface
Load Balancer Pool	cmdb_ci_lb_pool
Load Balancer Pool Member	cmdb_ci_lb_pool_member
Load Balancer Service	cmdb_ci_lb_service
Load Balancer VLAN	cmdb_ci_lb_vlan
Logical Partition	cmdb_ci_lpar
Memory Module	cmdb_ci_memory_module
Network Appliance Hostname	cmdb_ci_net_app_host
Network Hostname	cmdb_ci_network_host
Network Traffic	cmdb_ci_net_traffic
Operating System Level Virtualization Container	cmdb_ci_oslv_container
Operating System Level Virtualization Image	cmdb_ci_osiv_image
Operating System Level Virtualization Image TAg	cmdb_ci_oslv_image_tag

Table label (display name)	Table name
Operating System Level Virtualization Local Image	cmdb_ci_oslv_local_image
Package	cmdb_ci_os_packages
Patch	cmdb_ci_patches
Port	cmdb_ci_port
Print Queue	cmdb_ci_print_queue
Qualifier	cmdb_ci_qualifier
SAN Connection	cmdb_ci_san_connection
SAN Endpoint	cmdb_ci_san_endpoint
SAN Fabric	cmdb ci san fabric
SAN Zone	cmdb_ci_san_zone
SAN Zone Alias	cmdb_ci_san_zone_alias
SAN Zone Alias Member	cmdb_ci_san_zone_alias_member
SAN Zone Member	cmdb_ci_san_zone_member
SAN Zone Set	cmdb_ci_san_zone_set
Service	cmdb_ci_service
Storage Area Network	cmdb_ci_san
Storage Controller	cmdb_ci_storage_controller
Storage Export	cmdb_ci_storage_export
Storage File Share	cmdb_ci_storage_fileshare
Storage HBA	cmdb_ci_storage_hba

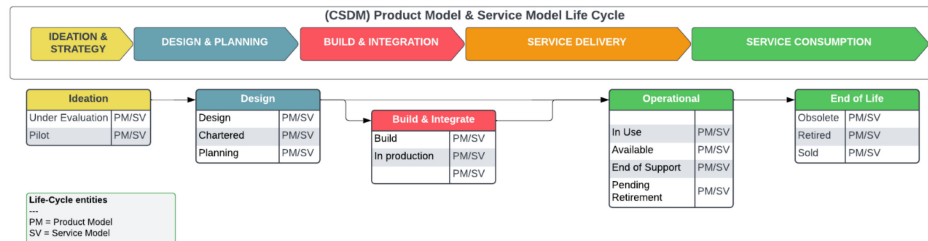
Table label (display name)	Table name
Storage Pool	cmdb_ci_storage_pool
Storage Pool Member	cmdb_ci_storage_pool_member
Storage Volume	cmdb_ci_storage_volume
Tomcat Connector	cmdb_ci_tomcat_connector
UPS Bypass	cmdb_ci_ups_bypass
UPS Alarm	cmdb_ci_ups_alarm
UPS Input	cmdb_ci_ups_input
UPS Output	cmdb_ci_ups_output
Veritas Disk Group	cmdb_ci_veritas_disk_group
Virtual Machine Object	cmdb_ci_vm_object
Virtual Private Cloud	cmdb_ci_vpc
Virtual Private Network	cmdb_ci_vpn
Websphere Cell	cmdb_ci_websphere_cell

Life cycle of product entities

The product life-cycle value pairs represent the overall life cycle of a product model, a specific version, or a product configuration. The life-cycle values for the product life-cycle process are visible only in Product (Models) tables.

Life-cycle value pairs for products

For definitions of the values in the diagram, see [Definitions of life-cycle values for product entities](#).



Note: The [life_cycle_control] table uses the type of CI (tangible/physical, document and contract, location and so on) to determine which life cycle stage status values are available for each life cycle stage.

For additional details on products, see [Products and product models](#).

For additional information on how you can benefit from implementing life-cycle value pairs for CMDB entities, see the 'Map existing status values to CSDM life-cycle value pairs' section in the 'Foundation domain' topic.

- Definitions of life-cycle values for product entities

The product life-cycle value pairs represent the overall life cycle of a product model, a specific version, or a product configuration. The life-cycle values for the product life-cycle process are visible only in Product (Models) tables.

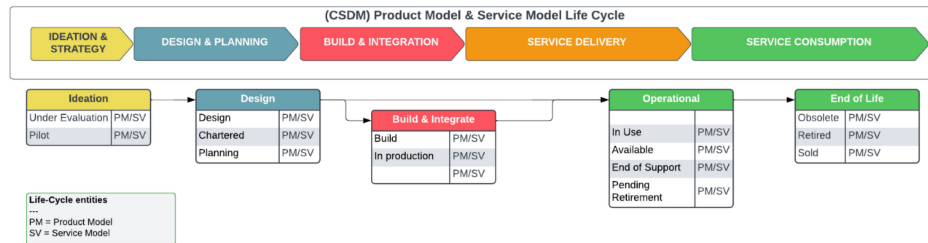
Related reference

- Definitions of life-cycle values for product entities

Definitions of life-cycle values for product entities

The product life-cycle value pairs represent the overall life cycle of a product model, a specific version, or a product configuration. The life-cycle values for the product life-cycle process are visible only in Product (Models) tables.

Definitions of product life cycle stage and life cycle stage status values



Ideation life cycle stage

New product models to determine feasibility and potential adoption.

- **Under Evaluation:** The product is being assessed for business value, functionality, and technical viability.
- **Pilot:** In a small-scale deployment, the product is being tested in a controlled environment before broader adoption.

Design life cycle stage

After a product is approved for development or configuration, its structure and features are defined.

- **Design:** The product is being architected, including defining specifications, features, and configurations.
- **Chartered:** The product model has been formally approved for development or procurement.

Build & Integrate life cycle stage

The product is being developed, tested, and prepared for operational deployment.

- **Build:** The product is being developed, covers the activities from code commit through compiled or packaged deliverables being ready for deployment to a lower (non-production) environment.
- **In Production:** Covers the final promotion and release of changes into the live, customer-facing environment, with all required governance.

Operational life cycle stage

The product is actively available and used within the organization.

- **Available:** The product is fully developed and ready for procurement or deployment.
- **Pending Retirement:** The product is still operational but is scheduled to be phased out.
- **End of Support:** The product is no longer receiving updates, maintenance, or vendor support.

Note: gggA CI might be in the **Operational** stage, but might no longer be supported by the vendor or publisher or third-party. That doesn't mean, however, that it can be or should be retired.

End of Life life cycle stage

The product is no longer in active use and is being retired.

- **Retired:** The product is no longer available for new purchases or deployments but may still exist in limited use.
- **Obsolete:** The product is completely phased out and is no longer supported or used.